Data Technology and Market Efficiency*

Zigang Li†

This version: November 11, 2026; latest version

Abstract

How will advances in data technology affect market equilibrium and informational efficiency when adoption by investors is uneven? I show that data technology development and dissemination improve asset managers' stock-picking ability, especially for those with larger shares of data-science labor. Using U.S. institutional holdings and employment records, I estimate information production across investors and a demand system with heterogeneity in private information and learning from prices. I estimate investors' private information from holdings' forecasting power for future payoffs beyond public information and prices. Data technology developments are captured by increases in data scientist productivity. Estimates show that better-informed investors have more elastic demand, while learning from prices makes demand 61 percent less elastic at the median informative stock. A counterfactual technology advance calibrated to more than a decade of data scientist productivity growth raises price informativeness by 26 percent, more for small stocks. New technology adopters crowd out other investors, increasing market segmentation as stocks' ownership compositions diverge. Aggregate demand becomes more inelastic for 47 percent of stocks.

Keywords: data technology, institutional investors, price informativeness, demand system, ownership composition, demand elasticity, learning from prices
JEL Classification: G11, G12, G14, G23, O33

*I am deeply indebted to my committee Ing-Haw Cheng, Itay Goldstein, Charles Martineau, Stijn Van Nieuwerburgh, and Liyan Yang for their invaluable guidance and support. I am grateful for comments from Alexandre Corhay, Winston Dou, Allen Hu, Raymond Kan, Jincheng Tong, and Renxuan Wang.

†Rotman School of Management, University of Toronto. Email: zigang.li@rotman.utoronto.ca. Website: https://zigangli.github.io.

1 Introduction

Over the past two decades, rapid advances in data technology have changed how institutional investors process information (Goldstein et al., 2021). Asset managers have hired more data scientists and expanded their use of new technologies.[1] These teams use the tools to forecast sales, earnings, and other firm fundamentals that guide investment decisions.[2] The spread of these methods is changing information production across asset management, with market-wide effects on stock prices and the information they convey. Adoption is uneven, however, and investors differ greatly in their capacity to use new tools. As technology improves further, these differences will shape who gains information and how information is transmitted through financial markets. How will a further advance in data technology affect market equilibrium and informational efficiency?

I find moderate and uneven increases in price informativeness, the share of firm payoff uncertainty resolved by prices.[3] A data technology advance equal to the estimated data scientist productivity growth over more than one decade raises average price informativeness by 3.1 percentage points from 11.8 percent. Price informativeness increases more for small stocks: the increase averages 4.4 points in the smallest quintile and 2.3 points in the largest. Adopters' improved private signals strengthen their demand and crowd out other institutions, increasing market segmentation. Adopters' positions rise by 15.4 percent, while other institutions' positions fall by 1.85 percent. The average difference between stocks' ownership compositions rises from 53.12 to 53.60 percent. This difference measures the share of ownership that must change hands for two stocks' ownership compositions to match. The technology advance also makes aggregate demand less elastic for 46.5 percent of stocks.

To quantify these effects, I construct a framework in which investors differ in their capac-

[1] Data technology consists of the algorithms and computing infrastructure that let investors process a wider range of data at lower cost to produce information and signals for investing. Abis and Veldkamp (2024) document growth in the use of machine learning (ML) and AI by data analysts in investment management. Industry reports describe adoption of new data technologies (CFA Institute, *AI Pioneers in Investment Management*, 2019; AIMA, *Casting the Net*, 2020). Investor reports, prospectuses, Form ADV brochures and other communications describe the use of ML/NLP and alternative data (*Two Sigma, 2026; Citadel, 2026; BlackRock, 2024; AQR, 2025, 2026; etc.*).

[2] Applications to fundamental forecasting are described in the CFA Institute's *Handbook of Artificial Intelligence and Big Data Applications in Investments* (2023) and investor communications (*AllianceBernstein, 2019; T. Rowe Price, 2019; Allspring, 2024; Fidelity Investments, 2025; etc.*). Data technology also helps investors use trading data to extract information from prices (Farboodi and Veldkamp, 2020). I focus on technology used for forecasting firm fundamentals. Section 2.1, Appendix A.3, and Online Appendix OA2 give further examples.

[3] I focus on how well prices forecast future asset payoffs (Bond et al., 2012). Following Dávila and Parlatore (2026), I measure price informativeness as the proportional reduction in forecast-error variance for firm payoffs when prices are added to other public information. Related work also regresses future cash flows on current valuations and public-information controls (Bai et al., 2016; Farboodi et al., 2022).

ity to process data into private forecasts of firm fundamentals and learn from prices when choosing portfolios. I measure investors' private information from the additional forecasting power of an investor's holdings for future firm payoffs, beyond public information and information contained in prices. Based on the model, I develop a demand system in which private information and price informativeness shape investor demand elasticity. I estimate information production and the demand function using detailed institutional holdings in the U.S. stock market and employment data, then calculate counterfactuals to quantify the effects of a further data technology advance on market equilibrium and informational efficiency.

Two facts motivate the analysis. First, data scientist employment has grown rapidly and unevenly across asset managers. Matching institutional holdings to employment histories, I find that data-science labor grew 8.5-fold, compared with 2.3-fold growth in investment analysts and portfolio managers. At the end of 2024, asset managers employing data scientists held 80.4 percent of all asset managers' equity assets but accounted for only 11.7 percent of those managers. Data scientist employment is concentrated among larger managers. Only 4.4 percent of managers in the bottom half by AUM employed data scientists, compared with 86.5 percent in the top one percent.

Second, asset managers with larger shares of data-science labor show greater increases in stock-picking ability around technology breakthroughs. Stock-picking ability is higher when a manager overweights firms relative to market weights before favorable earnings surprises. I focus on natural language processing (NLP) and computer vision (CV), which expand investors' ability to extract fundamental information from text and images. Extensive corporate disclosures give NLP models more material from which to extract information about firm fundamentals, while images of store-based retailers reveal customer traffic and can predict sales and earnings.[4] For each breakthrough, I compare changes in stock-picking ability on these stocks across managers with different pre-breakthrough shares of data-science labor in total employment. A one-standard-deviation higher pre-breakthrough share of data-science labor is associated with an additional increase in stock-picking ability of 0.91×10^{-3} after NLP breakthroughs and 0.55×10^{-3} after CV breakthroughs.

I develop a model in which investors with heterogeneous data scientist teams produce private information about firms, learn from equilibrium prices, and choose portfolios. Data scientists turn data into signals about firms' idiosyncratic payoffs. Investors combine these signals with the information conveyed by prices when evaluating expected payoffs and risk. Their trades reveal some of their information through market prices. A technology advance

[4]Textual analysis of corporate filings extracts information about future earnings, profitability, and cash flows (Li, 2010; Cohen et al., 2020; Azimi and Agrawal, 2021). Satellite images of store parking lots provide timely signals of retailers' sales and earnings (Zhu, 2019; Kang et al., 2021; Katona et al., 2025).

raises information productivity at investors with data scientist teams, giving them more precise private signals.

Who gains information, by how much, and which stocks they hold determine the effects of a technology advance. Data technology adopters' better forecasts raise their demand, and their demand makes prices more informative. This erodes other investors' informational advantage and reduces the expected payoff from trading on their existing private signals. Competition raises prices and lowers expected excess payoffs. In the counterfactual, other institutions are crowded out most where adopters contribute more information. They shift portfolio weight toward stocks with less information from adopters, increasing market segmentation. At the same time, learning from prices can reduce their uncertainty and partly cushion their aggregate ownership loss.

Uneven adoption of data technology also shapes aggregate demand elasticity. Adopters' better private information reduces payoff uncertainty and, by itself, makes their demand more elastic. By contrast, greater price informativeness strengthens learning from prices, making all investors' demand less elastic. Other investors receive no private-information gain. A price increase signals better fundamentals, so they raise their payoff forecasts and reduce their demand less, weakening their responses to price changes. Demand elasticity is therefore an endogenous equilibrium outcome. For each stock, the distribution of ownership across investors with different data scientist teams shapes the gains in private information and how informative prices become. The model jointly determines demand elasticity and price informativeness. In the counterfactual, learning from prices dominates the direct effect of adopters' better private information for many stocks, making aggregate demand less elastic. Less elastic aggregate demand can amplify the price impact of non-fundamental shocks, increasing price fluctuations and fragility (Goldstein et al., 2025).

To measure private information, I separate stock-picking ability from the management of exposure to common risk factors. Investors choose stocks for their expected payoffs relative to those of stocks with similar risk and for their contribution to factor exposures, following the stock-picking and factor-management logic of Haddad et al. (2026). The model decomposes holdings into demand based on public information, private signals, and factor exposure adjustments. I focus on private information about firms' idiosyncratic payoffs, measured using subsequent earnings conditional on stock characteristics, analyst forecasts, and prices. Investors who systematically hold larger positions in firms with favorable idiosyncratic earnings surprises have a positive holdings–surprise covariance. Correcting this covariance for information already revealed by prices and factor exposure management yields an investor-level information measure scaled by risk tolerance.[5] The measure captures hold-

[5] Kacperczyk et al. (2016) measure fundamental stock-picking ability using market-relative portfolio

ings' additional forecasting power for future payoffs beyond public information and prices. I derive the information estimator's sampling variance and asymptotic normal distribution. The variance depends on the amount and concentration of information.

Measured information is positive at asset managers, persistent within managers, and concentrated in a small group of managers. Asset managers in the highest quintile of past information still have more information than those in the lowest quintile over the following three years, after a one-year gap. Information is concentrated. In the average quarter, the top decile of institutions accounts for 84 percent of positive measured information. These findings support a model in which persistent differences in data processing capacity generate substantial heterogeneity in the information investors have.

I then estimate how data scientist teams produce information for active asset managers. Following Abis and Veldkamp (2024), the production function has two layers: data-management workers process raw data into a stock of clean data, and data-science teams analyze that data to produce information. Investor fixed effects account for persistent data processing efficiency and risk tolerance, while year fixed effects measure common changes in productivity. Estimated productivity in 2024 is 3.28 times its 2011 level. This increase in data scientist productivity captures the effects of data technology developments on information production. The estimated data-science labor exponent is 0.143: at given productivity and other staffing, doubling data scientist employment raises information production by about 10.5 percent.

The demand system estimates how private information and price informativeness affect investors' demand elasticity. I extend the granular instrumental variable (GIV) framework of Gabaix and Koijen (2024) by using measured information to account for correlated trading on payoff news. Investors receive signals about the same firm's fundamentals, so favorable news can make several investors buy together. The model determines how strongly their demand responds to this shared news and to prices. Within each stock-quarter, I project other investors' demand on these model-implied sensitivities, excluding the focal investor. The residual demand, weighted by the investors' assets under management, supplies the instrument. These residual demand shocks generate price variation that identifies the focal investor's demand elasticity. I use them in GMM estimation, building on the asset-market applications of Chaudhry and Li (2026) and Chaudhary et al. (2026).

The demand estimates support both information channels. Investors with more predicted private information respond more strongly to price changes, while investors respond less strongly where prices are more informative. Institutional demand is inelastic at the me-

weights and subsequent earnings shocks. I use positions and earnings news conditional on public information and prices and correct for learning from prices.

dian, with an implied elasticity of -0.51. At the median informative stock, an uninformed investor's demand is 60.6 percent less elastic when she learns from prices. Learning from prices therefore accounts for a substantial part of the estimated demand elasticity.

I construct the counterfactual by combining the estimated information production function and demand system in the market of 2024Q4. I raise data scientist productivity by an amount equal to the estimated increase from 2011 to 2024. The technology advance applies to active managers employing data scientist teams, whom I call adopters. Each adopter gains 69.5 percent more private information at fixed data scientist employment, distributed across stocks according to her initial information allocation. The model aggregates these information gains stock by stock, updates the information conveyed by prices, and determines investors' demand. Market clearing gives counterfactual prices. Average price informativeness rises by 3.1 percentage points from 11.8 percent. Price informativeness increases by 4.4 points for small stocks, compared with 2.3 for large stocks. Because adopters supply a larger share of small stocks' information, the same proportional technology advance brings these stocks a larger information gain.

Uneven information gains crowd out other institutions and increase market segmentation. Adopters' positions rise by 15.4 percent, while other institutions' positions fall by 1.9 percent. A decomposition separates the contribution of private-information gains from learning from prices. Holding information in prices fixed, adopters' private-information gains reduce other institutions' positions by 1.95 percent. Allowing prices to become more informative offsets 0.11 percentage points of that loss, as other holders learn more from prices. Crowding out is strongest where adopters' private information accounts for a larger share of information in the price. Other institutions' positions fall by 2.3 percent in the highest stock quintile of this share and 0.7 percent in the lowest. This crowding out makes stocks' ownership compositions more distinct. Their average difference rises from 53.12 to 53.60 percent. Larger counterfactual technology advances further increase this separation.

The data technology advance also changes aggregate demand elasticity. Holding information in prices fixed, the private-information gains make aggregate demand 4.5 percent more elastic at the median among stocks with informative prices. Once investors learn from the more informative prices, aggregate demand is 2.5 percent less elastic at the median. It becomes less elastic for 46.5 percent of all stocks.

Related literature. This paper contributes to the economics of data by estimating how data technology developments and investors' information production shape market equilibrium. Research on data and technology explains how scale, data availability, and processing shape learning (Begenau et al., 2018; Farboodi and Veldkamp, 2020; Dugast and Foucault,

2018, 2025). Empirical studies show that information production responds to easier access to corporate disclosures and to changes in publicly supplied information (Gao and Huang, 2020; Chen et al., 2020). Abis and Veldkamp (2024) estimate a two-layer production function and analyze how the growth of knowledge production affects price informativeness and the cost of capital. I build on their technology, measure investor-level information output from holdings, and estimate the productivity of data scientist teams. Combining these estimates with heterogeneous asset demand, I study how a common technology advance affects investors' demand and price informativeness. This also contributes to market macrostructure, which studies how institutions' organization and characteristics determine asset prices (Haddad and Muir, 2025). In related work, Bonelli and Foucault (2026) document ownership reallocation from long-term investors toward hedge funds as alternative-data coverage expands. In addition, data technology can also help firms by improving their forecasts of consumer demand, raise firms' markups, and affect firm dynamics (Corhay et al., 2025). In this paper, I focus on how data technology affects investors' information production and market equilibrium through investors' demand.

A second strand studies information acquisition and portfolio choice. Studies with rational expectations equilibrium models explain how heterogeneous private signals determine portfolio choices and become partly revealed through prices (Grossman and Stiglitz, 1980; Hellwig, 1980; Admati, 1985). Van Nieuwerburgh and Veldkamp (2010) jointly determine learning and portfolio choice, showing how information acquisition can lead investors to specialize. Kacperczyk et al. (2016) connect heterogeneous processing capacity to mutual funds' attention allocation between fundamental stock-picking and market-timing abilities. Institutional incentives also shape information acquisition through delegation and benchmarking (Gârleanu and Pedersen, 2018; Breugem and Buss, 2019). Goldstein and Yang (2015) show that investors learning about different components of a payoff can complement one another: information conveyed by prices reduces uncertainty about the other component and encourages trading and acquisition. Kacperczyk et al. (2025) show how investor size and market power affect both learning and the transmission of private information into prices. I estimate the information produced by data-science labor and evaluate a productivity increase at observed levels of data scientist employment.

The measurement contribution is to recover investor-level information from holdings and future firm payoffs. Forecasting price efficiency evaluates the information incorporated into market prices by how accurately those prices predict firms' future fundamentals (Bond et al., 2012). Farboodi et al. (2022) recover aggregate data precision from prices and fundamentals, separating private information from information conveyed by prices. Farboodi et al. (2025) value observed financial data from their incremental predictive content for returns, account-

ing for information revealed through prices. Stock-level measures quantify how much prices reveal about fundamentals (Bai et al., 2016; Dávila and Parlatore, 2026). My estimator builds on the portfolio-covariance logic of Kacperczyk et al. (2016). It attributes private information to individual investors by conditioning earnings news on public information and prices and correcting for information contained in prices and factor management. I derive its sampling variance and asymptotic distribution.

I build on the literature on asset demand systems (Koijen and Yogo, 2019, 2026; Huebner, 2023; Haddad et al., 2025; Jansen et al., 2024; Koijen et al., 2024; Li et al., 2026; Bretscher et al., 2026; Chaudhry and Li, 2026; Chaudhary et al., 2026). Koijen et al. (2024) use demand to measure investor informativeness and study how capital reallocations affect prices and price informativeness. My framework provides a microfoundation for measuring investors' private information when they learn from prices and links this information to data-science productivity. I also develop counterfactuals that map heterogeneous information gains into changes in price informativeness and investors' demand. Haddad et al. (2026) show how homogeneous substitution conditional on observables supports identification of relative demand elasticities when investors substitute across many assets. For each investor, my model extends the assumption by requiring homogeneous substitution conditional on observables, private information, and price informativeness, and I estimate how those information channels shape demand elasticities. Dou et al. (forthcoming) show that active funds tilt toward stocks with low exposure to common fund flows to hedge fund-size risk.

The analysis also builds on work explaining why demand elasticities vary. Haddad et al. (2025) show that investors' responses to one another matter for aggregate elasticity and that information acquisition can be one of the forces driving these responses. Chaudhry and Li (2026) document variation in elasticity with the size of demand shocks and accumulated price dislocations. Davis (2025) shows that quantitative investment strategies need not generate highly elastic demand. Bastianello and Fontanier (2025) show how mistaken inference from prices can make demand less elastic. Goldstein et al. (2025) show that learning from prices can amplify the price impact of non-fundamental shocks and increase financial market fragility. I estimate how private information and rational learning from prices affect investor demand. Combining these estimates with heterogeneous productivity increases quantifies when a technology advance makes aggregate demand for a stock less elastic.

The paper connects asset managers' skill to its sources and market-wide effects. Holdings and trades reveal skill among some mutual fund managers (Kacperczyk et al., 2005; Kacperczyk and Seru, 2007; Kacperczyk et al., 2008; Cremers and Petajisto, 2009; Baker et al., 2010), and Kacperczyk et al. (2014) relate variation in skill to information processing and attention allocation. Recent evidence connects AI skills and data scientist staffing to

investment performance (Cen et al., 2025; Aragon et al., 2026). Sheng et al. (forthcoming) find larger increases in performance after ChatGPT's publication at hedge funds with pre-existing AI staff. Bonelli and Foucault (forthcoming) show that alternative data erode traditional experts' advantage in stock-picking ability and induce reallocation toward uncovered stocks. I estimate the productivity of data scientist teams from the earnings-based information measure and quantify how a data technology advance changes ownership and informational efficiency.

Consdering the real effects of financial markets, the feedback effect, could extend this paper's effects to firms' decisions. Bond et al. (2012) and Goldstein (2023) review this feedback, and Chen et al. (2007) test whether managers learn from prices. Goldstein et al. (2023) find that wider dissemination of corporate filings through EDGAR raised stock liquidity, lowered the cost of equity capital, and increased equity financing and investment. Firms hold more cash and invest less in response to greater stock price fragility, which reflects their exposure to non-fundamental demand shocks arising from the composition of equity ownership (Friberg et al., 2024). The counterfactual in this paper holds firm characteristics and fundamentals fixed. In a setting with the feedback effect, the quantified changes in prices and information could also affect firms' decisions.

Section 2 presents the data and motivating evidence. Section 3 develops the model. Sections 4–6 measure information and estimate production and demand. Section 7 quantifies the technology advance.

2 Data and Motivating Facts

Institutional holdings and employment histories establish two facts that motivate the model. Data-science employment has grown rapidly and unevenly across asset managers. Managers with larger shares of data-science labor also show greater increases in stock-picking ability around breakthroughs in data technology.

2.1 Data Technology in Asset Management

Asset managers use data technology to turn data about firm fundamentals into forecasts for investment decisions. They combine alternative data, drawn from sources other than financial statements and market prices, with statistical and machine-learning methods to forecast sales, earnings, and other fundamentals (Green and Zhang, 2024). Natural-language processing extracts information from written and spoken language. Investors apply it to corporate filings and earnings-call transcripts to interpret disclosures and forecast funda-

mentals. Forward-looking tone and changes in financial reports predict future earnings (Li, 2010; Cohen et al., 2020). Neural-network measures of annual-report sentiment predict future profitability and operating cash flows (Azimi and Agrawal, 2021). FinBERT, a version of Google's BERT language model adapted to financial text, improves information extraction from earnings calls (Huang et al., 2023). Goldman Sachs's quantitative equity funds began using natural-language processing on these calls in 2014.[6] Longer or more complex filings give these tools more text to process.

Computer vision, which identifies objects and patterns in images, provides one example. A model can count cars in satellite images of retailers' parking lots and turn those images into measures of customer traffic before firms report sales (Kang et al., 2021). Growth in parking-lot fill rates predicts same-store sales and returns around earnings announcements (Katona et al., 2025).

Data-science teams develop and apply these methods, combining the tools with datasets about firm fundamentals to produce information for investment decisions. In an Alternative Investment Management Association survey, 53 percent of hedge-fund managers use alternative data, and 85 percent of those using it for more than five years are looking to hire data scientists.[7] Newly published language and computer-vision models expand these teams' forecasting tools. These applications motivate the stock-picking ability measure in Section 2.4 and the information-production model in Section 3. Appendix A.3 and Online Appendix OA2 give further institutional evidence.

2.2 Data

I combine quarterly institutional holdings from Thomson Reuters, 1980–2024, with CRSP prices and Compustat fundamentals. The holdings are quarter-end long positions in U.S.-listed equities reported on Form 13F by investors with discretion over at least $100 million in reportable securities. A 13F institution is the unit of observation, and its AUM is the market value of its reported equity holdings.

Each 13F institution receives a fixed investor type describing the business behind its reported portfolio (Appendix A.1). Asset managers are the subgroup whose primary activity is investment: hedge funds, mutual funds, investment advisers, and public pension and sovereign funds. Banks, insurers, and broker-dealers form a separate comparison group because their equity demand also reflects financial intermediation and insurance objectives. Banks and broker-dealers may hold shares as market-making inventories or as hedges against

[6] Goldman Sachs Trust, *Tax-Advantaged Equity Funds Annual Report* (2014), pp. 20, 27.
[7] Noble and Balint, *Casting the Net* (AIMA, 2020), a survey of 100 hedge-fund managers.

exposures from customer trades. Insurers' equity allocations are shaped by policyholder liabilities and regulatory capital requirements.

Stock characteristics come from the WRDS data library of Jensen et al. (2023), which constructs them from CRSP and Compustat. Size is log market capitalization, and book-to-market is book equity divided by market equity. The demand system uses the five nonprice characteristics in Koijen and Yogo (2019): log book equity, operating profitability, investment, dividends to book equity, and market beta. Profitability is earnings before interest, taxes, depreciation, and amortization, less interest expense, divided by book equity. Investment is the annual proportional change in total assets. The dividend ratio is total dividends over the preceding twelve months divided by book equity. Market beta is estimated from a 60-month rolling regression of excess stock returns on excess market returns, requiring at least 36 observations. Filing length comes from the Loughran–McDonald summaries of firms' 10-K and 10-Q filings. The measures of stock-picking ability and information use quarterly EBIT from Compustat (Section 4). Online Appendix OA1 gives the variable definitions.

I/B/E/S provides revisions to median quarterly earnings-per-share forecasts, forecast dispersion, and analyst coverage. Along with stock characteristics, these variables control for public information when I estimate investors' private information.

Individual employment histories from Revelio Labs cover 2009–2024 and draw on publicly available professional profiles, including LinkedIn.[8] I match 13F institutions to the companies with employment records in the Revelio data, allowing the match to change when an institution's name changes. Matches use company identifiers and standardized names, with addresses, websites and corporate affiliations used to resolve ambiguous cases. Data-science (DS) labor is the weighted headcount of data scientists and quantitative researchers in the investment process. It includes data-science work in statistical and machine-learning forecasting, text, computer vision, and alternative-data signals, as well as all quantitative-research titles. This measure captures computational research capacity. Appendix A.2 summarizes the classification, and Online Appendix OA1.2 gives the details of the classification.

The share of DS labor is DS employment divided by total employment, reported as a percentage. A DS employer has positive DS employment in the quarter. When 13F filers share an employer, I allocate its workforce in proportion to their 13F assets, or equally if none has assets, so each employee counts once. The employment panel includes all 13F institutions matched to Revelio in the corresponding quarter.

I combine institutional holdings with stock characteristics and analyst forecasts to construct a quarterly panel of 13F institutions. I use this full 13F panel to measure investors'

[8]Revelio's sampling weights use government employment statistics to adjust for differences in profile coverage across occupations and locations. See Revelio Labs, *Data Dictionary*, "Sampling Weights".

Table 1: Descriptive Statistics

(a) Quarterly Investor Panel, with Employment Data

	N	Mean	SD	25%	50%	75%
1(DS Labor > 0) (%)	263,987	12.0	32.4	0.0	0.0	0.0
# DS Labor	263,987	1.8	21.0	0.0	0.0	0.0
Share of DS Labor (%)	263,987	0.67	4.15	0.00	0.00	0.00
# Investment Staff	263,987	41.8	410.5	1.0	3.1	9.6
# Data Engineering	263,987	2.7	34.6	0.0	0.0	0.0
# Admin. and Operations	263,987	29.7	321.7	0.0	1.1	4.1
# Finance and Accounting	263,987	4.7	32.5	0.0	0.0	2.1
# Human Resources	263,987	6.7	71.6	0.0	0.0	0.0
# IT Support	263,987	10.4	118.1	0.0	0.0	0.0
# Employees	263,987	316.5	3,283.5	5.1	12.5	37.6
AUM ($bn)	263,987	4.59	56.88	0.07	0.20	0.81
# Stocks Held	263,987	174	371	20	59	143
Years since First 13F	263,987	9.89	9.35	2.50	6.75	14.75

(b) Quarterly Investor Panel, Whole Sample

	N	Mean	SD	25%	50%	75%
AUM ($bn)	449,561	3.84	44.96	0.08	0.23	0.87
# Stocks Held	449,561	186	390	25	68	159
Years since First 13F	449,561	8.70	8.42	2.25	6.00	12.50

Table 1: Descriptive Statistics

(c) Quarterly Stock Panel

	N	Mean	SD	25%	50%	75%
Size (ln ME)	910,743	5.19	2.32	3.50	5.05	6.76
B/M	910,743	0.79	0.93	0.29	0.56	0.95
Op. Profitability	910,743	0.04	0.76	−0.01	0.17	0.29
Asset Growth	910,743	0.27	1.08	−0.03	0.07	0.23
Market Beta	910,743	1.11	0.74	0.61	1.03	1.50
Idio. Vol. (Daily, %)	910,743	3.56	2.47	1.86	2.87	4.49
13F Ownership (%)	910,743	37.9	31.7	8.5	31.0	64.7
# 13F Holders	910,743	100	208	8	31	111
# Analysts	910,743	3.7	5.4	0.0	1.0	5.0
Earnings Surprise (SUE)	910,743	0.83	8.88	−1.80	0.73	3.46

Notes. Panel (a) describes institutions matched to Revelio employment records, and panels (b) and (c) describe the full investor and stock panels defined in Section 2.2. DS Labor counts data scientists and quantitative researchers under the definitions in Section 2.2. Its share is DS labor per 100 employees. Employee counts use Revelio's sampling weights. Investment staff includes investment analysts and portfolio managers. AUM is the market value of holdings in the U.S. equity panel, in billions of dollars. Investor age is years since the first 13F report in the data. ME is market equity in millions of dollars, and B/M is book-to-market equity. Operating profitability is operating profits over book equity. Asset growth is annual total-asset growth. Market beta uses 60 months. Idiosyncratic volatility is the annual-window daily standard deviation of market-model residuals, in percent. Ownership statistics use stock-quarters covered by the 13F data. Ownership is institutions' percentage of outstanding shares. SUE is the standardized earnings surprise defined in Online Appendix OA1.3. Online Appendix OA1 gives the construction details.

information and estimate stock demand.

I match the 13F panel to the employment data to estimate how research teams produce information. Table 1 reports summary statistics for the matched employment panel, the full 13F panel, and the quarterly stock panel. Online Appendix OA1.5 describes the construction of the estimation samples.

2.3 Fact 1: DS Employment Has Grown Rapidly and Unevenly

DS employment at asset managers grew far faster than employment of investment analysts and portfolio managers. From 2009 to 2024, aggregate DS labor rose 8.5-fold, from 788 to 6,732 (Figure 1(a)), while employment of investment analysts and portfolio managers grew 2.3-fold. The DS share of investment staff rose from 2.39 to 8.41 percent (Figure 1(b)). The DS share of investment staff divides DS labor by DS labor plus investment analysts and portfolio managers. Growth also occurred within managers: among those present in 2009Q1, DS labor grew 5.3-fold by 2024Q4.

DS employers are a small share of managers and hold most assets. At the end of the employment panel, 11.7 percent of managers employed DS labor and held 80.4 percent of AUM, up from 57.3 percent in 2009Q4. The AUM share reached 92.6 percent among mutual funds and 68.5 percent among hedge funds (Online Appendix Table OA1).

DS employment is concentrated among larger asset managers. Only 4.4 percent of managers in the bottom half by AUM had DS staff, compared with 13.0 percent between the 50th and 90th percentiles, 38.4 percent between the 90th and 99th, and 86.5 percent in the top one percent (Figure 1(c)). Employment also differs across manager types. Using each type's aggregate employment, the DS labor share was 7.45 percent at hedge funds, 1.26 percent at investment advisers, 0.90 percent at mutual funds, and 0.36 percent at public pension and sovereign funds.

2.4 Fact 2: DS Labor and Increases in Stock-Picking Ability after Technology Breakthroughs

I study asset managers' stock-picking ability around major breakthroughs in natural language processing (NLP) and computer vision (CV).[9] Each technology helps forecast the fundamentals of particular firms, whose stocks I call exposed stocks. For NLP, exposed stocks are text-heavy stocks, the top decile of filing length before a breakthrough after ad-

[9]The NLP events focus on information extraction from disclosures and exclude recent generative large language models (LLM), because their applications extend beyond textual analysis to coding and autonomous agent workflows.

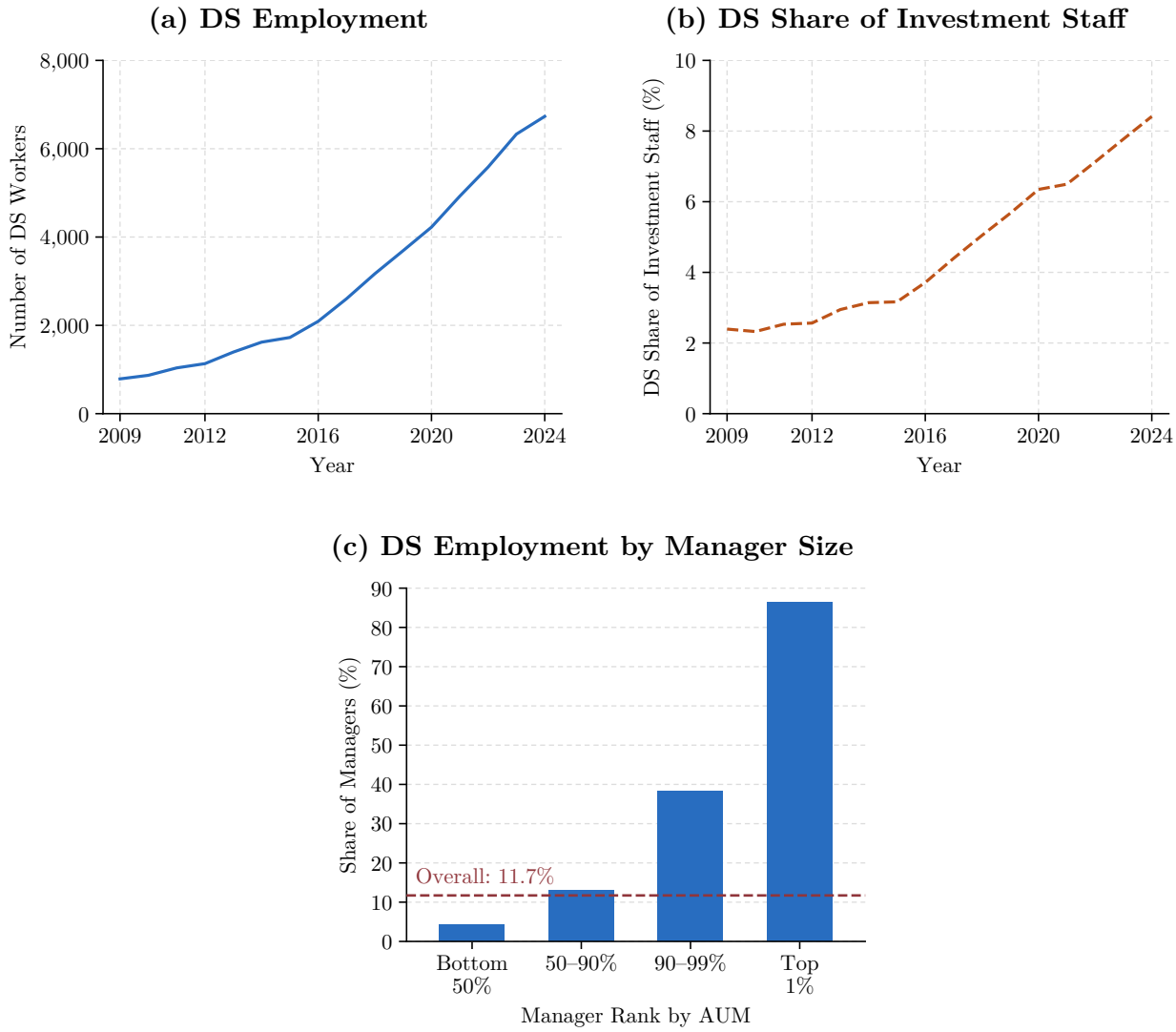


Figure 1: DS Labor at U.S. Asset Managers

Notes. Asset managers and DS labor follow Section 2.2. Headcounts use Revelio's sampling weights. Panels (a) and (b) plot fourth-quarter DS employment and its share of investment staff, respectively. The share uses the orange dashed line. The denominator comprises DS labor, investment analysts and portfolio managers. The share is the ratio of aggregate headcounts. Panel (c) reports the percentage of managers with positive DS employment in each AUM percentile group in 2024Q4. Every manager has equal weight. The red dashed line gives the overall percentage calculated across all managers.

justing for size and industry. Their longer reports give the tools more material to process. For CV, they are store-based retailers and restaurants: satellite images of their parking lots reveal customer traffic and can predict sales and earnings (Katona et al., 2025; Bonelli and Foucault, forthcoming). The classification uses retail SIC codes, including restaurants and excluding nonstore retailers (Appendix A). If DS teams help asset managers use new tools, stock-picking ability on exposed stocks should improve more at managers with larger pre-event shares of DS labor.

Following Kacperczyk et al. (2016), I measure the fundamental stock-picking ability of asset manager i on the exposed stocks in quarter t as

$$\mathrm{SP}_{i,t} = N_{i,t}^{-1} \sum_{n \in E_{i,t}} \left(w_{i,n,t} - w_{n,t}^{m} \right) \mathrm{SUE}_{n,t+4}, \tag{1}$$

where $E_{i,t}$ is the set of exposed stocks the asset manager holds with an observed earnings surprise, and $N_{i,t}$ is the number of those stocks. The portfolio weight $w_{i,n,t}$ uses the asset manager's whole portfolio, and $w_{n,t}^{m}$ is the stock's market weight. The earnings surprise $\mathrm{SUE}_{n,t+4}$ is operating earnings over the next four quarters minus operating earnings over the past four, scaled by the standard deviation of the eight preceding year-over-year changes in quarterly earnings. The measure is high when the asset manager overweights firms before favorable earnings surprises. It is defined when the asset manager holds at least ten exposed stocks with an observed surprise.

The breakthroughs come from a list of new NLP and CV algorithms and models that appear in the Epoch AI database and rank in the top 0.01 percent of their field and year by citations (Appendix A; Online Appendix OA2.5). I group nearby developments into waves. Each event falls in the first quarter ending at least two months after the highest-ranked model's code or trained models became public, allowing time for use before holdings are observed. The NLP events are Word2Vec and BERT. The CV events are AlexNet, ResNet, Mask R-CNN, ViT, and SAM (Appendix Table A1).

I build a separate dataset around each event and stack them for estimation (Cengiz et al., 2019). Each dataset follows asset managers' stock-picking ability on exposed stocks from six quarters before an event to six after.[10] DS employers have DS staff in any of the four quarters before the event. Comparison asset managers have none during those quarters.

The pre-event share of DS labor, $\mathrm{DS}_{i,s}$, is asset manager i's mean DS labor per 100 employees over the four quarters before event s, standardized across asset managers in the event's dataset. Asset managers without DS staff have a zero share before standardization.

[10] Managers must meet the employment requirements in Online Appendix OA1.5 in at least one of the four quarters before the event.

The event study is

$$\mathrm{SP}_{i,s,t} = \sum_{k \neq -1} \beta_k \, \mathbf{1}\{t - t_s = k\} \times \mathrm{DS}_{i,s} + \alpha_{i,s} + \lambda_{s,t} + u_{i,s,t}, \tag{2}$$

where t_s is the event quarter, and $k = -1$, the quarter before the event, is the reference. Asset-manager-by-event effects $\alpha_{i,s}$ absorb each asset manager's average stock-picking ability within the window, and event-by-quarter effects $\lambda_{s,t}$ absorb common movements. Thus, β_k measures the change from quarter -1 to quarter k per standard deviation of the pre-event share of DS labor.

Managers with larger shares of DS labor show greater increases in stock-picking ability on exposed stocks. Figure 2 plots the NLP coefficients in panel (a) and CV coefficients in panel (b). Both increase at the event quarter. The NLP increase peaks two quarters after the event and is smaller from quarter four onward. This pattern may reflect other asset managers' gradual adoption of the algorithms, narrowing the initial advantage of managers with larger DS shares. The pooled NLP increase per standard deviation is 0.91×10^{-3}. For CV, the increases remain positive through quarter six and are largest in quarters five and six. This persistence may reflect slower catch-up in using CV algorithms among managers with smaller DS shares. The pooled increase per standard deviation is 0.55×10^{-3}. Appendix Tables A2 and A3 report pooled estimates, specifications adding shares of six other types of employment, a specification with an indicator of whether managers employ DS staff, and results for alternative definitions of exposed stocks, i.e., complex-text stocks measured by the top decile of the share of Loughran-McDonald complexity words in 10-K filings for NLP, and retailers covered by the satellite data provider in Katona et al. (2025) for CV. Online Appendix Table OA2 gives all staffing coefficients from column (2). Appendix B describes the pooled specification, weights, and pre-event tests.

2.5 From Facts to Model

The facts connect data scientist employment to investment outcomes. DS labor has grown rapidly and remains concentrated, while managers with larger DS employment shares show greater increases in stock-picking ability around breakthroughs. The model gives research teams a role in producing private signals about firm payoffs and lets their trades convey some of that information to others through prices. The production function in Section 5 estimates how much more information a given team produces as productivity improves. This estimated change sets the size of the counterfactual technology advance.

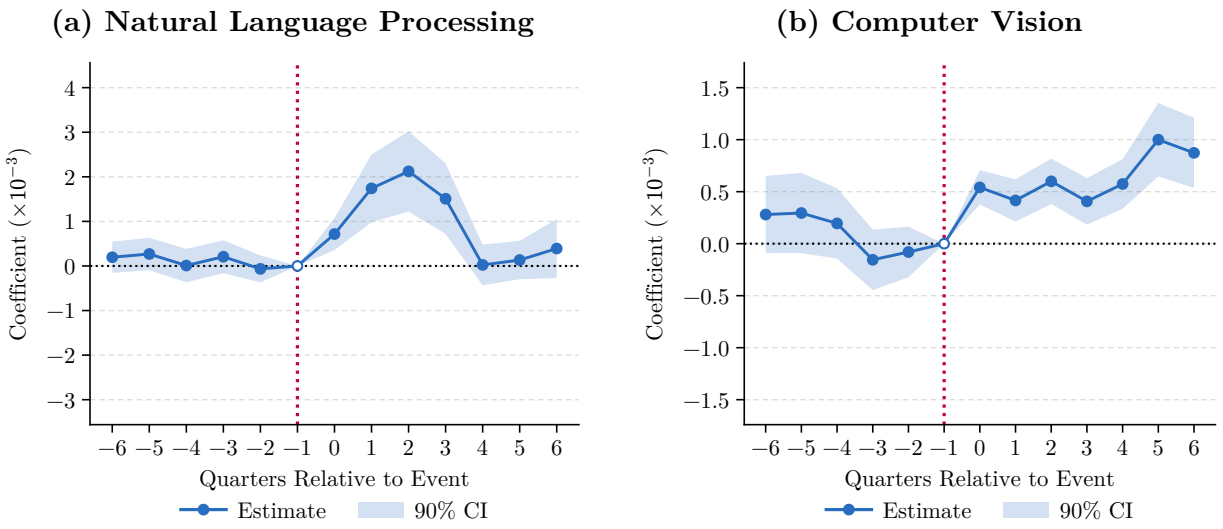


Figure 2: Stock-Picking Ability around Data-Technology Breakthroughs

Notes. Changes in stock-picking ability around the two language events in panel (a) and five computer-vision events in panel (b), listed in Appendix Table A1. The specification follows equation (2). Points are the event-time coefficients, β_k, in equation (2), multiplied by 10^3. Bands give 90 percent confidence intervals. Each coefficient measures the change per standard deviation of the pre-event DS share of total employment, relative to quarter −1. Stock-picking ability, defined in equation (1), uses text-heavy stocks in panel (a) and store-based retailers in panel (b). The hollow marker and vertical line mark the omitted quarter. DS employers have weight one, and comparison investors are reweighted to give each event the same ratio of comparison investors to DS employers (Wing et al., 2024). Standard errors are clustered by investor.

3 Model

Data technology improves the information of investors with data scientist teams. As these investors trade, they compete with other holders for shares and convey some of their information through prices. I study these effects in a noisy rational-expectations economy with heterogeneous investors and many stocks (Grossman and Stiglitz, 1980; Hellwig, 1980; Admati, 1985). The model links an investor's information to its holdings and explains how learning from prices transmits an uneven technology advance across investors. Appendix C gives the proofs.

3.1 Setup

Timing. There are three dates. At date 0, data scientist teams use their existing data stock to produce signals and allocate them across stocks. At date 1, investors observe their signals and choose portfolios. At date 2, payoffs are realized. Data scientist teams and data stocks are in place at the beginning of date 0.

Assets. There are N stocks, each with supply normalized to one share, and a riskless asset with gross return R_f. Stock n trades at price $p(n)$, and investor i holds the fraction $q_i(n)$ of its shares. I assume that payoffs follow a one-factor model.[11] The payoff of stock n is

$$f(n) = \mu(n) + \beta(n)\, F + \varepsilon(n), \qquad F \sim N(0,1), \qquad \varepsilon(n) \sim N\Big(0, 1/\tau_\varepsilon(n)\Big), \tag{3}$$

where $\mu(n)$ is the expected payoff, $\beta(n)$ the loading on the common factor F, and $\varepsilon(n)$ the idiosyncratic payoff innovation. The factor and the innovations are mutually independent. The payoff precision $\tau_\varepsilon(n)$ is the inverse of the innovation's prior variance. Following Koijen et al. (2024), I allow expected payoffs and factor loadings to depend on observable characteristics $x(n)$:

$$\mu(n) = \Phi_\mu^\top x(n) + \xi_\mu(n), \qquad \beta(n) = \Phi_\beta^\top x(n) + \xi_\beta(n). \tag{4}$$

Φ_μ and Φ_β are vectors of coefficients on firm characteristics. Known residuals $\xi_\mu(n)$ and $\xi_\beta(n)$ allow expected payoffs and factor loadings to differ among stocks with the same characteristics.

Investors. There is a continuum of investors, indexed by i.[12] Investor i has initial wealth

[11] With multiple factors and no mandates, Proposition C.2 gives the corresponding factor-management term in demand. The information estimator keeps its form, and price informativeness is unchanged in the benchmark economy.

[12] Appendix H derives the information moment with finitely many investors. Under the benchmark inference rule and the restrictions on price impact stated there, demand also keeps its form with effective risk

W_{1i} and chooses a portfolio q_i, whose elements are $q_i(n)$, to maximize U_i subject to her budget constraint:

$$U_i = \gamma_i \,\mathbb{E}[W_{2i} \mid \mathcal{I}_i] - \frac{\gamma_i^2}{2} \operatorname{Var}(W_{2i} \mid \mathcal{I}_i) + \frac{\delta_i^\phi}{2}\Big(\phi^\top q_i\Big)^2, \qquad W_{2i} = R_f W_{1i} + q_i^\top\Big(f - R_f p\Big), \quad (5)$$

where W_{2i} is wealth at date 2, f and p are vectors of payoffs and prices, $\mathcal{I}_i$ is her information set, and $\gamma_i > 0$ is absolute risk aversion.[13] Risk tolerance is γ_i^{-1}. Investment mandates and priorities also shape portfolio choice (Haddad et al., 2026). Some institutions must keep a prescribed mix of stocks. Others follow broader objectives stated in their prospectuses. The last term captures a soft investment mandate or priorities that the investor can trade off against expected return and risk: $\phi(n) \geq 0$ is a stock attribute, such as carbon emissions, and $\phi^\top q_i$ is total portfolio exposure to it. A negative coefficient δ_i^ϕ penalizes the magnitude of this exposure, and a positive one rewards it.[14] Without a mandate, $\delta_i^\phi = 0$. Because the objective depends on total exposure, changing one position affects the incentive to hold other stocks with the same attribute.

Information. Investors use data to learn about payoffs. I measure their information in units of signal precision: one data point on stock n is a signal about its payoff innovation that is exactly as informative as the prior. The count therefore measures how much the data teach the investor about payoffs.[15] Investor i's jth data point on the stock is

$$\eta_i^j(n) = \varepsilon(n) + u_i^j(n), \qquad u_i^j(n) \sim N\Big(0, 1/\tau_\varepsilon(n)\Big), \qquad j = 1, \ldots, k_i(n), \quad (6)$$

where $k_i(n) \geq 0$ counts her data points on the stock. I assume that the noise $u_i^j(n)$ of a data point is normal, independent across data points, stocks and investors, and independent of the factor and the innovations. The average of her data points, $\eta_i(n) = k_i(n)^{-1} \sum_j \eta_i^j(n)$, is a sufficient statistic for them, and I call it her signal on the stock. Given $\varepsilon(n)$, it is normal with mean $\varepsilon(n)$ and precision $k_i(n)\tau_\varepsilon(n)$, so her private precision about the stock and her total private information are

$$\tau_i(n) = k_i(n)\,\tau_\varepsilon(n), \qquad K_i = \sum_n k_i(n). \quad (7)$$

tolerance. The appendix gives the conditions required for production estimation and counterfactuals.

[13] Under normality, the first two terms equal $-\ln \mathbb{E}[\exp(-\gamma_i W_{2i}) \mid \mathcal{I}_i]$ and give the CARA portfolio (Veldkamp, 2011; Kacperczyk et al., 2016).

[14] A hard mandate fixes exposure at $\phi^\top q_i = \bar{c}_i$, as with a required allocation to an asset category. Online Appendix Proposition OA3.1 derives optimal demand under this constraint.

[15] This count measures a dataset's information in precision units. Raw records may have different information content. It can take noninteger values because signal precision is continuous.

She can also learn from prices p, so her information set is $\mathcal{I}_i = \{\eta_i(1), \ldots, \eta_i(N), p\}$. Signals reveal information about idiosyncratic payoffs, and investors keep their prior about the common factor.[16]

The count $k_i(n)$ measures private precision relative to payoff precision. It is unchanged by the units of payoffs or shares, and the first data point halves the prior variance. Thus K_i sums information in a common unit across stocks.[17]

Allocation of data. A predetermined rule allocates an investor's total information across stocks:

$$k_i(n) = K_i \, \Psi_i\Big(x(n)\Big), \qquad \Psi_i\Big(x(n)\Big) \geq 0, \qquad \sum_{n \in \mathcal{N}_i} \Psi_i\Big(x(n)\Big) = 1, \tag{8}$$

where $x(n)$ is a vector of observable stock characteristics known at date 0, and $\mathcal{N}_i$ is the investor's stock universe. Ψ_i is zero outside it. She holds no stocks outside this universe. The predetermined rule depends on stocks through these variables and separates scale from mix: a proportional rise in K_i raises every $k_i(n)$ in proportion. It can reflect the coverage of a data source. Since the variance of the payoff innovation can be included in $x(n)$, this allocation rule can also summarize an information choice, including specialization in stocks on which information is most valuable (Van Nieuwerburgh and Veldkamp, 2009, 2010).

Noise traders. Besides investors, noise traders trade every stock, buying $z(n)$ shares of stock n. $z(n)$ is normal with mean zero and variance $\sigma_z^2(n)$, independent across stocks and of every other shock. Together, the factor, the payoff innovations, the noise of every data point and the noise trades are therefore jointly normal and mutually independent.[18]

3.2 Equilibrium

Investors choose optimal portfolios given their information sets, and prices clear markets.

Definition 1 (Equilibrium)**.** Given the data counts $k_i(n)$, an equilibrium is a set of functions for prices p, and portfolio choices $\{q_i\}$, such that three conditions hold. (i) Given $\mathcal{I}_i$, each q_i maximizes the objective (5). (ii) Markets clear,

$$\int q_i \, di + z = \mathbf{1}, \tag{9}$$

where z stacks the noise trades and $\mathbf{1}$ is the vector of supplies. (iii) Each investor rationally

[16] This stock-level information complements the attention allocation between aggregate and stock-specific shocks studied by Kacperczyk et al. (2016).

[17] A budget on these counts is a scale-free precision budget. An entropy budget instead charges the same for each halving of remaining variance(Van Nieuwerburgh and Veldkamp, 2010; Veldkamp, 2011). Here each successive halving requires twice as many new data points.

[18] Assumption C.1 states the boundedness and factor-pervasiveness conditions used in the proofs.

updates her beliefs by Bayes' rule given the price function.

Beliefs. The price conveys part of what other investors know. I conjecture, and verify in Proposition 2, that when $\kappa(n) > 0$, stock n's price reveals the signal

$$\eta_p(n) = \varepsilon(n) + \frac{z(n)}{\kappa(n)}, \qquad \kappa(n) = \int \gamma_i^{-1}\tau_i(n)\, di, \qquad \tau_p(n) = \frac{\kappa(n)^2}{\sigma_z^2(n)}, \tag{10}$$

where $\kappa(n)$ is aggregate information weighted by risk tolerance and $\tau_p(n)$ is price-signal precision. Demand responds to each private signal with coefficient $\gamma_i^{-1}\tau_i(n)$. Adding these coefficients gives $\kappa(n)$. Noise trading prevents full revelation. Doubling $\kappa(n)$ divides price-signal noise by two and multiplies its precision by four. When $\kappa(n) = 0$, investors learn nothing from the price and omit its signal from their beliefs.

I write $\tilde{\tau}(n) = \tau_\varepsilon(n) + \tau_p(n)$ for the public precision of stock n and $\hat{\tau}_i(n) = \tilde{\tau}(n) + \tau_i(n)$ for investor i's posterior precision.

Lemma 1 (Beliefs)**.** *Under the conjecture, investor i's beliefs about the payoff innovations are normal and independent across stocks, with*

$$\mathbb{E}[\varepsilon(n) \mid \mathcal{I}_i] = \frac{\tau_i(n)\,\eta_i(n) + \tau_p(n)\,\eta_p(n)}{\hat{\tau}_i(n)}, \qquad \mathrm{Var}\big(\varepsilon(n) \mid \mathcal{I}_i\big) = \frac{1}{\hat{\tau}_i(n)}. \tag{11}$$

Her beliefs about the factor are its prior.

Conditional independence makes precisions add. The price contributes the same precision to every investor, while private signals distinguish their beliefs. No signal reveals the factor. Price informativeness, $\tau^{\mathrm{R}}(n) = \tau_p(n)/\tilde{\tau}(n)$, is the share of prior uncertainty about the innovation that the price resolves (Dávila and Parlatore, 2026).

Optimal demand. At a given expected excess payoff, a better-informed investor takes a larger position because she faces less uncertainty per share. Buying the stock may also increase her portfolio's factor exposure or move it away from a mandate. She adjusts other positions to account for these portfolio effects.

Write $m_i(n) = \mathbb{E}[f(n) \mid \mathcal{I}_i] - R_f p(n)$ for expected excess payoff, and $M_i = \hat{\Omega}_i + \beta\beta^\top - \delta_i^\phi \gamma_i^{-2}\phi\phi^\top$ for the curvature matrix of the objective. Here $\hat{\Omega}_i$ is the diagonal posterior covariance matrix of idiosyncratic innovations, with entries $1/\hat{\tau}_i(n)$. I assume M_i is positive definite, which makes the objective strictly concave and the optimal portfolio unique. This holds automatically for $\delta_i^\phi \leq 0$.

In the demand rule below, $\gamma_i^{-1}\hat{\tau}_i(n)m_i(n)$ is the position the investor would choose based on the stock's expected excess payoff and idiosyncratic uncertainty. The other two terms

adjust this position for factor and mandate exposures, following the portfolio logic of Haddad et al. (2026). Their shadow prices, g_i^β and g_i^ϕ, are common to all stocks in her portfolio. Each stock's loadings and posterior precision determine how much its position changes.

Proposition 1 (Optimal demand)**.** *Investor i's problem has the unique solution $q_i = \gamma_i^{-1} M_i^{-1} m_i$. Stock by stock,*

$$q_i(n) = \gamma_i^{-1} \hat{\tau}_i(n) \Big[m_i(n) - \beta(n) g_i^\beta - \phi(n) g_i^\phi \Big], \tag{12}$$

where the factor and mandate adjustments g_i^β and g_i^ϕ solve the two-by-two system

$$\begin{pmatrix} g_i^\beta \\ g_i^\phi \end{pmatrix} = G_i^{-1} \begin{pmatrix} \beta^\top \hat{\Omega}_i^{-1} m_i \\ \phi^\top \hat{\Omega}_i^{-1} m_i \end{pmatrix}, \qquad G_i = \begin{pmatrix} 1 + \beta^\top \hat{\Omega}_i^{-1} \beta & \beta^\top \hat{\Omega}_i^{-1} \phi \\ \beta^\top \hat{\Omega}_i^{-1} \phi & -\gamma_i^2 / \delta_i^\phi + \phi^\top \hat{\Omega}_i^{-1} \phi \end{pmatrix}.$$

Without a mandate, $\delta_i^\phi = 0$, the system reduces to its first equation, so that $g_i^\phi = 0$ and $g_i^\beta = \beta^\top \hat{\Omega}_i^{-1} m_i / (1 + \beta^\top \hat{\Omega}_i^{-1} \beta)$. The portfolio's exposures are $\beta^\top q_i = \gamma_i^{-1} g_i^\beta$ and, for $\delta_i^\phi \neq 0$, $\phi^\top q_i = -\gamma_i g_i^\phi / \delta_i^\phi$.

A favorable signal can induce the investor to buy one stock and adjust other positions to manage the resulting exposure. The two-by-two system makes these choices consistent across her portfolio. Its shadow prices depend on the signals and prices observed when trading, even though preferences and loadings are fixed.

Equilibrium price function. Aggregating these demands and imposing market clearing gives the following price function. The benchmark replaces realized factor and mandate adjustments by their means in aggregate demand. Individual portfolios keep their realized adjustments.[19]

Proposition 2 (Equilibrium price function)**.** *Under the benchmark market-clearing approximation, stock n has the equilibrium price function*

$$R_f \, p(n) = \mu(n) - \frac{1 + \beta(n) H_\beta(n) + \phi(n) H_\phi(n)}{\mathcal{T}(n)} + \frac{\kappa(n) \, \varepsilon(n) + z(n)}{\mathcal{T}_p(n)}. \tag{13}$$

The coefficient $\mathcal{T}(n)$ determines the expected excess payoff needed to absorb supply, and $\mathcal{T}_p(n)$ determines the price impact of trading. The terms $H_\beta(n)$ and $H_\phi(n)$ aggregate the mean factor and mandate adjustments. These quantities are defined below. If $\kappa(n) > 0$, the price is a strictly increasing affine function of $\eta_p(n)$, which verifies the conjecture. If $\kappa(n) = 0$, the price reveals nothing and moves with $z(n)$ alone.

[19] Under the diversification and market-depth conditions in Appendix C, the market-clearing error shrinks as investors' stock universes expand. The same discussion gives bounds for finite portfolios.

Consider first a stock with $\beta(n) = \phi(n) = 0$. Before signals and noise trades are realized, its expected excess payoff is $1/\mathcal{T}(n)$, the compensation investors require to hold its unit supply. Greater risk tolerance or more precise information lowers this compensation and raises the expected price. Let $\bar{\gamma}^{-1} = \int \gamma_i^{-1}\, di$ be the total risk tolerance of investors who can hold the stock, suppressing its dependence on n. Then

$$\mathcal{T}(n) = \int \gamma_i^{-1}\hat{\tau}_i(n)\, di = \bar{\gamma}^{-1}\Big[\tau_\varepsilon(n) + \tau_p(n)\Big] + \kappa(n).$$

Holding payoff beliefs fixed, a purchase of one share from these investors raises the price by $1/[R_f\mathcal{T}(n)]$. When the purchase comes from noise traders, investors also learn from the resulting price rise. They interpret part of it as favorable payoff news and require a larger price increase to sell the same number of shares.

For an investor with no private signal, the demand slope based on the prior alone is $-R_f\gamma_i^{-1}\tau_\varepsilon(n)$. Learning from prices multiplies this slope by

$$d_p(n) = \frac{\kappa(n)}{\kappa(n) + \bar{\gamma}^{-1}\tau_p(n)} = \frac{1}{1 + \bar{\gamma}^{-1}\kappa(n)/\sigma_z^2(n)} \in (0, 1], \tag{14}$$

with $d_p(n) = 1$ at $\kappa(n) = 0$. As aggregate private information $\kappa(n)$ rises or noise-trading variance $\sigma_z^2(n)$ falls, prices convey more information and $d_p(n)$ falls: the investor sells less in response to a price increase.

For an investor with a private signal, private precision adds $-R_f\gamma_i^{-1}\tau_i(n)$ to this slope. Aggregating across investors gives the demand slope $-R_f\mathcal{T}_p(n)$, where

$$\mathcal{T}_p(n) = \kappa(n) + \bar{\gamma}^{-1}\tau_\varepsilon(n)\, d_p(n) = \mathcal{T}(n)\, d_p(n). \tag{15}$$

A noise-trader purchase changes the price by $1/[R_f\mathcal{T}_p(n)]$ per share, $1/d_p(n)$ times the impact with payoff beliefs fixed. At a given price, a unit payoff innovation raises informed demand by $\kappa(n)$. The equilibrium price rises by $\kappa(n)/[R_f\mathcal{T}_p(n)]$. Independent private-signal errors average out, leaving the payoff innovation and noise trading as the two sources of price fluctuations.[20]

Factor risk and mandates enter expected excess payoffs through $\beta(n)H_\beta(n)/\mathcal{T}(n)$ and $\phi(n)H_\phi(n)/\mathcal{T}(n)$. The first term compensates holders for factor exposure, and the second reflects how holding the stock serves their investment objectives. The aggregates are

$$H_\beta(n) = \int \gamma_i^{-1}\hat{\tau}_i(n)\, \bar{g}_i^\beta\, di, \qquad H_\phi(n) = \int \gamma_i^{-1}\hat{\tau}_i(n)\, \bar{g}_i^\phi\, di.$$

[20] With finitely many investors, individual signal errors also enter prices. An investor then learns from the part of the price not accounted for by her own signal. Appendix H.2 gives the inference rule.

Here $\bar{g}_i^a = \mathbb{E}_0[g_i^a]$ for $a \in \{\beta, \phi\}$ is the mean adjustment given date-0 primitives and information allocations.

Relative demand slopes. Price changes also induce trades to adjust factor and mandate exposures. These adjustments cancel from the difference in holdings of two stocks whose positions respond equally to changes in either shadow price. For distinct stocks $n, n' \in \mathcal{N}_i$, equation (12) gives the conditions

$$\hat{\tau}_i(n')\beta(n') = \hat{\tau}_i(n)\beta(n), \qquad \hat{\tau}_i(n')\phi(n') = \hat{\tau}_i(n)\phi(n).$$

Precision enters these conditions because a given change in either shadow price induces a larger adjustment when the investor has a more precise payoff forecast. Because posterior precision varies across investors, the matching conditions are investor specific.[21]

Proposition 3 (Relative demand slopes)**.** *Under the benchmark equilibrium price function, hold information allocations, private signals and all other prices fixed. For the two stocks defined above,*

$$\frac{\partial\left[q_i(n) - q_i(n')\right]}{\partial p(n)} = -R_f\,\gamma_i^{-1}\Big[\tau_i(n) + \tau_\varepsilon(n)\,d_p(n)\Big]. \tag{16}$$

The coefficients on the right-hand side sum to $-R_f\mathcal{T}_p(n)$ across the stock's eligible holders.

A more precise private signal makes the demand more elastic. Learning from prices makes the demand less elastic through $d_p(n)$. Section 6 estimates these two information effects. Slopes are measured in shares per dollar of price.

3.3 Data Processing and Information Production

Following Abis and Veldkamp (2024), I separate data preparation from analysis. Data managers acquire and organize raw records into usable datasets. Data scientist teams analyze these data to produce information about firms' payoffs. For an investor i with a data scientist team, the two stages in quarter t are

$$D_{it} = (1-\delta_D)D_{i,t-1} + \left(L^{\mathrm{DM}}_{i,t-1}\right)^{\rho},$$
$$K_{it} = \tilde{a}_i A_t D_{it}^{\nu}\left(L^{\mathrm{DS}}_{i,t-1}\right)^{1-\nu}.$$

Here K_{it} is total private information, D_{it} is the usable data stock, L^{DM} is data-management labor, and L^{DS} includes data scientists and quantitative researchers. Both labor inputs

[21] Corollary C.1 gives the own-price response, including the portfolio-exposure adjustment.

are measured at the previous quarter-end, before signals arrive. The depreciation rate is $\delta_D \in [0,1]$. The exponents $0 < \rho, \nu < 1$ describe diminishing returns to data-management labor and to data in analysis. Investor efficiency $\tilde{a}_i$ and common productivity A_t determine how much information a team produces with given inputs.

The information measure uses quarterly earnings innovations. I treat usable data as specific to forecasting the next earnings announcement and assume full depreciation within one quarter, $\delta_D = 1$. Each announcement reveals the outcome that these data help forecast. Because the sample has limited coverage of data-management labor, I also assume $L^{\mathrm{DM}}_{i,t-1} = c_i L^{\mathrm{DS}}_{i,t-1}$, where $c_i > 0$ is constant over time. Substituting these assumptions gives

$$K_{it} = a_i A_t \left(L^{\mathrm{DS}}_{i,t-1}\right)^{\alpha}, \tag{17}$$

where $\alpha = 1 - \nu + \rho\nu$ and $a_i = \tilde{a}_i c_i^{\rho\nu}$. The labor elasticity α combines the information gains from preparing more data and from analyzing them as the team expands. Section 5 estimates this relationship using the information measured from investors' holdings.

3.4 What a Data-Technology Advance Does

A data technology advance raises information productivity at investors with DS teams. Holding employment fixed, each adopter's private information grows in proportion to the increase in A_t. The information gains reach stocks unevenly because these investors hold different portfolios and contribute different amounts of information. Their demand affects other investors through more informative prices and competition for the fixed supply of stocks.

Price informativeness. The effect differs across stocks because adopters account for different shares of aggregate information. For $\kappa(n) > 0$, write $\Delta_\kappa(n) = \kappa'(n)/\kappa(n) - 1$ for the proportional increase in aggregate information, where a prime denotes the economy after the data technology advance. This increase weights each adopter's information gain by her initial contribution to $\kappa(n)$. Price signal precision then rises by the factor $[1+\Delta_\kappa(n)]^2$. For a given proportional information gain, price informativeness rises most at an intermediate initial level: nearly uninformative prices convey little new knowledge, while highly informative prices leave little uncertainty to resolve.[22]

[22]The increase is largest at initial price informativeness $1/[2 + \Delta_\kappa(n)]$, just below one half for a small technology advance (Corollary C.4).

Crowding out. For a clean example, take a stock with $\beta(n) = \phi(n) = 0$, so it has no factor or mandate exposure. The expected trading payoff attributable to investor i's private information is

$$\Pi_i(n) = \gamma_i^{-1} \frac{\tau_i(n)}{\tilde{\tau}(n)} = \gamma_i^{-1} k_i(n)[1 - \tau^{\mathrm{R}}(n)]. \tag{18}$$

A more informative price reduces this informational rent from a given private signal.[23] Bonelli and Foucault (forthcoming) show how this erosion of informational advantage lowers the trading returns of investors whose expertise improves less with data technology.

Expected ownership depends on the investor's willingness to bear risk relative to that of other holders. Even an investor with no private signal holds shares in expectation because the stock compensates her for bearing payoff risk. For the same stock, the expected excess payoff $1/\mathcal{T}(n)$ clears the unit supply, giving

$$\mathbb{E}[q_i(n)] = \frac{\gamma_i^{-1} \hat{\tau}_i(n)}{\mathcal{T}(n)}. \tag{19}$$

Investors with more precise forecasts or greater risk tolerance demand more. A technology advance raises posterior precision but lowers the expected excess payoff as investors compete for the fixed supply. At unchanged risk tolerance,

$$\frac{\mathbb{E}[q_i'(n)]}{\mathbb{E}[q_i(n)]} = \frac{\hat{\tau}_i'(n)}{\hat{\tau}_i(n)} \frac{\mathcal{T}(n)}{\mathcal{T}'(n)}. \tag{20}$$

An investor's expected demand falls when her precision grows less proportionally than holders' average precision, weighted by risk tolerance. Learning from prices can improve her forecast even as better-informed adopters crowd her out.[24]

Proposition 4 (Crowding out)**.** *Let $\beta(n) = \phi(n) = 0$, and let some investors' private precision on stock n rise, holding $\sigma_z(n)$, $\tau_\varepsilon(n)$, $\bar{\gamma}^{-1}$ and all other investors' private precisions fixed. If $\tau^{\mathrm{R}}(n) < 1/2$, a small increase in aggregate private information $\kappa(n)$ lowers the expected demand of every investor whose private precision is unchanged. A finite increase in $\kappa(n)$ by the proportion $\Delta_\kappa(n) > 0$ lowers their expected demand whenever $\tau^{\mathrm{R}}(n) < 1/[2 + \Delta_\kappa(n)]$.*

Investors whose private precision is unchanged gain information through more informative prices, while adopters' stronger demand lowers the stock's expected excess payoff. For a small

[23] For nonzero factor loadings, Proposition C.1 gives this rent for the stock-selection position after removing the realized factor adjustment.

[24] With zero mandate penalties, the two propositions extend to the conditional stock-selection demand of stocks with nonzero factor loadings.

increase in $\kappa(n)$ with initial $\tau^{\mathrm{R}}(n) < 1/2$, this payoff falls proportionally more than any of these investors' posterior precisions rise, so all of their expected holdings fall.[25]

Proposition 5 (Crowding out across investors)**.** *Under the conditions for crowding out in Proposition 4, consider investors i and j whose private precisions on stock n remain unchanged. If $\tau_i(n) > \tau_j(n)$ and the price becomes more informative, investor i experiences a larger percentage decline in expected holdings:*

$$\frac{\mathbb{E}[q_i(n)] - \mathbb{E}[q_i'(n)]}{\mathbb{E}[q_i(n)]} > \frac{\mathbb{E}[q_j(n)] - \mathbb{E}[q_j'(n)]}{\mathbb{E}[q_j(n)]}.$$

More informative prices add the same precision to both investors' forecasts. This is a larger proportional improvement for the investor with less private information. Her expected holdings therefore fall by a smaller percentage when competition from adopters lowers the stock's expected excess payoff.

The effect of learning from prices on expected demand depends on the investor's private precision. Holding private precisions fixed, greater public precision raises expected demand among investors whose private precision is below the risk-tolerance-weighted average, $\kappa(n)/\bar{\gamma}^{-1}$, and lowers it among those above (Proposition C.1). It cushions crowding out for relatively uninformed investors and reinforces it for better-informed investors whose private information is unchanged.

Aggregate demand elasticity. A technology advance affects aggregate demand elasticity through better private information and learning from prices. For a stock with $\beta(n) = \phi(n) = 0$, the magnitude of aggregate demand elasticity at a fixed positive price and quantity is proportional to $\mathcal{T}_p(n)$. A fall in $\mathcal{T}_p(n)$ therefore makes aggregate demand less elastic.

Proposition 6 (Decrease in Aggregate Elasticity)**.** *For a stock with $\beta(n) = \phi(n) = 0$, fix $\sigma_z(n)$, $\tau_\varepsilon(n)$ and $\bar{\gamma}^{-1}$. At $\kappa(n) > 0$, a small rise in $\kappa(n)$ lowers the magnitude of aggregate demand elasticity, evaluated at a fixed positive price and quantity, if and only if*

$$\kappa(n) < \bar{\gamma}^{-1}\tau_\varepsilon(n)d_p(n)[1 - d_p(n)].$$

Better private information reduces adopters' uncertainty about payoffs and makes their demand more elastic. Their informed trading also makes prices more informative, strengthening learning from prices for all investors. A higher price then leads investors to revise

[25] Easley and O'Hara (2004) find lower average holdings of uninformed investors when a larger share of fixed total information is private. Here some investors' private information increases. Corollary C.2 gives the finite-change condition.

their payoff forecasts upward, making their demand less elastic. Since aggregate demand is the sum of individual demand curves, the change in aggregate demand elasticity depends on which force dominates.

Under the condition in Proposition 6, learning from prices dominates. To interpret the condition, consider a small percentage increase in aggregate private information. In equation (15), the private-information component of the magnitude of aggregate demand elasticity is proportional to $\kappa(n)$ and grows by that percentage. The component based on investors' common prior is proportional to $\bar{\gamma}^{-1}\tau_\varepsilon(n)d_p(n)$. Stronger learning from prices reduces it by $1-d_p(n)$ times that percentage. The condition holds when the component based on the common prior is large enough that its fall exceeds the rise in the private-information component in absolute size.[26] Section 7 measures how these forces change prices, demand elasticity and portfolios when information gains differ across stocks.

4 Measuring Investors' Information

Private information gives investors' holdings forecasting power for future payoffs beyond public information and prices. Following the portfolio-covariance logic of Kacperczyk et al. (2016), I measure information by asking whether an investor holds more of stocks with subsequently favorable payoff news. The model supplies corrections for information already revealed by prices and for portfolio adjustments that offset the investor's stock-level demand.

Consider an investor's position in a stock before its next earnings announcement. I remove the earnings news predicted by public variables and the price, then ask whether the investor's position predicts the remaining earnings surprise. A holding above the amount predicted by public controls contributes positively when the earnings surprise is favorable. A holding below that amount contributes positively when the earnings surprise is unfavorable. The investor may have learned from public information, such as analyst forecasts. What matters is whether her forecast contains information beyond that already in public variables and prices.

4.1 From Holdings to Information

Let $\tilde{\varepsilon}(n) = \varepsilon(n) - \mathbb{E}[\varepsilon(n) \mid p]$ denote payoff news left unrevealed by prices, measured empirically by the residual earnings surprise. The investor's risk-tolerance-weighted information is $\theta_{it} = \gamma_{it}^{-1}K_{it}$. Risk tolerance matters because an investor willing to bear more risk takes

[26] From equation (14), $\partial \log d_p(n)/\partial \log \kappa(n) = -[1 - d_p(n)]$. Appendix C proves the local condition, and Corollary C.3 gives its finite-change counterpart.

larger positions on the same signal.

Two adjustments recover information from the covariance of demand with residual payoff news. First, prices reveal the fraction $\tau^{\mathrm{R}}(n)$ of payoff uncertainty, reducing the covariance even when private information is unchanged. Second, buying on favorable private information changes the portfolio's factor and mandate exposures. The exposure adjustment multiplies the stock's response to its private signal by $1 - h_i(n)$. Dividing by this factor recovers the information behind that response.[27]

Proposition 7 (Information identity)**.** *The covariance of investor i's demand for stock n with its residual payoff news is $\mathbb{E}[q_i(n)\tilde{\varepsilon}(n)] = \gamma_i^{-1} k_i(n)[1 - \tau^{\mathrm{R}}(n)][1 - h_i(n)]$. Hence*

$$\gamma_i^{-1} K_i = \sum_n \frac{\mathbb{E}\Big[q_i(n)\,\tilde{\varepsilon}(n)\Big]}{\Big[1 - \tau^{\mathrm{R}}(n)\Big]\Big[1 - h_i(n)\Big]}. \tag{21}$$

Demand based on public information and trades on private signal noise have zero covariance with residual payoff news. With full positions and known model objects, realized products in equation (21) therefore give an unbiased estimate of risk-tolerance-weighted information.[28]

An investor without private information has a zero population moment. The identity holds for raw demand: removing a public component leaves its population covariance unchanged (Corollary D.1). Public-information trades nevertheless add sampling noise because they can coincide with realized payoff news by chance. I residualize positions to reduce this noise and correct below for the effect of fitting the position controls in the same sample.

The stock-picking ability measure in equation (1) uses portfolio weights and earnings surprises. The information measure instead uses positions as fractions of shares outstanding, removes the news predicted by public information and prices, and corrects for the two sources of attenuation.[29] I implement it in four steps. Appendix D.2 specifies the forecasting regressions and estimator corrections.

(1) I use operating earnings announced in the quarter after holdings are observed, subtract earnings for the same fiscal quarter a year earlier, and divide the difference by book

[27] Appendix D derives $h_i(n) = \hat{\tau}_i(n) v_n^\top G_i^{-1} v_n$, with $v_n = (\beta(n), \phi(n))^\top$. For $\delta_i^\phi \leq 0$, $0 \leq h_i(n) < 1$. If $\phi(n)$ is spanned by the included characteristics, controls containing $\tilde{\tau}(n)\phi(n)$ remove the public-precision component of the mandate adjustment. The private-precision component remains. Online Appendix Proposition OA3.1 covers hard mandates, and Proposition C.2 covers multiple factors.

[28] With finitely many investors, the moment measures incremental precision beyond the price, weighted by effective risk tolerance when price impact is present (Appendix H).

[29] The forecasting stage adapts Dávila and Parlatore (2026). Attributing risk-tolerance-weighted information to investors complements the latent-demand attribution of Koijen et al. (2024) and the aggregate price-informativeness trends in Bai et al. (2016).

equity at the holdings date. This comparison accounts for recurring seasonal patterns. I forecast this earnings change using stock characteristics and analyst-forecast revisions, coverage and dispersion. The augmented forecast adds price, its interactions with characteristics, and industry effects. Its error, $\tilde{\varepsilon}_{t+1}(n)$, is the residual earnings surprise.

(2) The inverse conditional variances of the two forecast errors give prior and public precisions, $\hat{\tau}_\varepsilon(n)$ and $\hat{\tau}_\varepsilon(n)+\hat{\tau}_p(n)$. The baseline variance divided by the augmented variance gives the correction $1/[1-\hat{\tau}^{\mathrm{R}}(n)]$. I anchor this correction to variance ratios within size and market-to-book bins and bound it between one and 100. The lower bound assigns zero measured price informativeness to stocks whose fitted variance does not fall.

(3) I measure positions in book-equity dollars: $Q_{it}(n)$ is the fraction of outstanding shares held times the firm's book equity. Residualizing these positions on public controls, including every forecasting regressor, gives $\tilde{Q}_{it}(n)$. The estimation universe $\mathcal{N}_{it}$ contains stocks held in any of the previous four quarters. A current position is zero if the stock is no longer held. Since 13F reports long positions only, the measure captures information reflected in those positions.[30]

(4) I sum the corrected products of residual positions and earnings surprises:

$$\hat{\theta}_{it} = \sum_{n\in\mathcal{N}_{it}} \frac{\tilde{Q}_{it}(n)\,\tilde{\varepsilon}_{t+1}(n)}{\big[1-\hat{\tau}^{\mathrm{R}}(n)\big]\big[1-\hat{h}(n)\big]\big[1-P_{nn}\big]}. \tag{22}$$

To estimate $h_i(n)$, I weight each stock's squared estimated market beta by its estimated public precision and divide by the sum across the investor's stock universe. Market betas are estimated using 60 months of monthly returns. This feasible $\hat{h}(n)$ replaces unobserved investor-specific precisions with public precision (Corollary D.2). The diagonal P_{nn} of the position regression's projection matrix is the weight on stock n's own position in its fitted value. Subtracting this fitted value removes part of the information covariance, and dividing by $1-P_{nn}$ restores that part. Investor and quarter indices are suppressed.[31]

[30] Long positions yield a lower bound in expectation on risk-tolerance-weighted information in the Gaussian benchmark, up to the exposure and projection approximations. Censoring multiplies each stock's expected information contribution by its probability of being held, which is at most one (Lemma D.2).

[31] I cap $\hat{h}(n)$ at 0.5 and floor $1-P_{nn}$ at 0.05. An observed mandate attribute outside the span of market beta must also enter the exposure correction. Appendix D.2 gives the specification.

4.2 Statistical Properties

The estimator's precision depends on how much information an investor has and how she allocates it across stocks. The results condition on public information and the predetermined information allocation. Write V_{it} for the leading sampling variance. Appendix D.3 states the regularity conditions and gives the proofs.

Proposition 8 (Bias and precision)**.** *Under the conditions in Appendix D.3,*

$$\mathbb{E}[\hat{\theta}_{it}] - \theta_{it} = O(\theta_{it}/|\mathcal{N}_{it}|), \qquad \mathrm{Var}(\hat{\theta}_{it}) = V_{it}\Big[1 + O(|\mathcal{N}_{it}|^{-1})\Big].$$

For $K_{it} > 0$, the leading relative variance is

$$\frac{V_{it}}{\theta_{it}^2} = 2\,\mathrm{HHI}_{it} + \frac{\bar{w}_{it}}{K_{it}} + \frac{\sigma^2_{\mathrm{mis},it}}{\theta_{it}^2},$$

where $\mathrm{HHI}_{it} = \sum_n [k_{it}(n)/K_{it}]^2$ measures information concentration, $\bar{w}_{it} = \sum_n \frac{k_{it}(n)}{K_{it}[1-\tau^{\mathrm{R}}(n)]}$ averages the price-informativeness correction, and $\sigma^2_{\mathrm{mis},it}$ is the variance contributed by public demand left in residual positions.

Relative bias vanishes as the stock universe grows. Spreading information across stocks reduces dependence on payoff news from a few stocks, lowering the concentration term. Reducing the signal-noise term $\bar{w}_{it}/K_{it}$ also requires more information, at given allocation shares and price informativeness. More informative prices amplify sampling noise because a larger correction is needed to recover information already revealed by prices. Public demand left after residualization adds noise without contributing information.

Proposition 9 (Asymptotic normality)**.** *Under the conditions of Proposition 8, let $V_{it}(n)$ denote stock n's contribution to $V_{it} > 0$. If $\max_n V_{it}(n)/V_{it} \to 0$ as $|\mathcal{N}_{it}| \to \infty$, then*

$$\frac{\hat{\theta}_{it} - \theta_{it}}{\sqrt{V_{it}}} \xrightarrow{d} N(0,1).$$

The normal approximation applies when no stock accounts for a nonvanishing share of the estimator's sampling variance. If information also becomes diffuse, K_{it} grows and residual public-demand variance becomes negligible relative to θ_{it}^2, the estimate recovers the investor's information with a vanishing proportional error.

I report two normalizations. *Information per dollar* divides $\hat{\theta}_{it}$ by lagged AUM, the value of the entire 13F portfolio at the previous quarter-end, and is reported per billion dollars.

This normalization removes the size effect on the estimated total information.[32] *Standardized information* divides by an industry-based noise scale. Multiplying every position by the same positive factor scales the estimate and this noise scale proportionally. Standardization therefore removes the proportional effect of position size and risk tolerance (Lemma D.1). In the data, I sum the squared information contributions of twelve industries to measure sampling uncertainty, allowing stocks within each industry to move together.[33]

4.3 Properties of Measured Information

Measured information is higher in the asset-management group than in the comparison group of banks, insurers and broker-dealers (Table 2; Section 2.2). Mean standardized information is 0.059 for the asset-management group and 0.010 for the comparison group. The difference is 0.049, and the difference in information per dollar is 0.123. Both are significant. The comparison group's mean is statistically indistinguishable from zero.

Hedge funds rank first on both measures.[34] Their standardized information of 0.205 is statistically significant. The median hedge fund's information per dollar, averaged over quarters, is 0.951 per billion dollars of lagged AUM, compared with 0.169 for mutual funds and 0.108 for investment advisers.

Measured information is persistent. Each quarter, I sort investors within each group into quintiles by their mean standardized information over the previous twelve quarters, leave a four-quarter gap, and follow them for twelve quarters. In the asset-management group, the top quintile exceeds the bottom by 0.114, with monotonically increasing quintile means (Table 2, Panel (b)). The comparison group's spread is 0.003 and statistically indistinguishable from zero. The gap prevents the formation and evaluation windows from using the same quarterly earnings observation. The persistence is not driven by investor size. After controlling for investor size and the number of stocks held, the spread is 0.107. Sorting these investors on size instead gives an imprecisely estimated spread of −0.033 (Online Appendix Table OA4).

Information is also concentrated. In the average quarter, the top decile of institutions accounts for 84 percent of positive measured information. Figure 3 compares the distribution of investors' standardized information. For investors with at least twelve quarterly estimates,

[32] If risk tolerance is proportional to assets, the ratio measures information reflected in long positions times relative risk tolerance, $1/\gamma_{it}^{R}$ (Assumption F.1).

[33] Online Appendix OA4.1 relates this scale to sampling variance and shows that standardized information is bounded in absolute value by $\sqrt{12}$. Standard errors for group statistics use their quarterly time series.

[34] Using publicly disclosed stock holdings, Griffin and Xu (2009) find a small stock-picking advantage for hedge funds over mutual funds that depends on the weighting and benchmark. Agarwal et al. (2013) find that hedge funds' holdings initially kept confidential outperform their publicly disclosed holdings.

Table 2: Measured Information of Institutional Investors

(a) By Institution Type

	N(Inst.)	13F Share (%)	Info. per $bn	Information (Standardized)
Asset Managers	3,549	71.1	0.167***	0.0595***
			(0.042)	(0.0120)
Hedge Funds	261	2.4	0.951***	0.2049***
			(0.347)	(0.0423)
Mutual Funds	562	50.5	0.169***	0.0632***
			(0.050)	(0.0159)
Inv. Advisers	2,665	13.2	0.108**	0.0388***
			(0.050)	(0.0118)
Public Pension & Sovereign	61	5.0	0.041	0.0285
			(0.108)	(0.0494)
Banks, Insurers & Broker-Dealers	724	28.0	0.044	0.0103
			(0.065)	(0.0216)
Difference			0.123***	0.0492***
			(0.036)	(0.0154)

(b) Persistence: Quintile Means Relative to the Group's Mean

	Low	2	3	4	High	High − Low
Asset Managers	−0.0465	−0.0177	−0.0086	0.0051	0.0677	0.1142***
						(0.0187)
Banks, Insurers & Broker-Dealers	0.0081	−0.0142	−0.0164	0.0112	0.0112	0.0031
						(0.0275)

Notes. Information estimates follow Section 4. The sample and institution groups are defined in Section 2.2. Panel (a) reports each group's share of the market value of positions, summed over quarters. Info. per $bn is the time average of each group's quarterly median information per billion dollars of lagged AUM. Standardized information divides the estimate by its industry-based noise scale. The column averages quarterly group means. Difference is asset managers minus banks, insurers and broker-dealers. Panel (b) sorts institutions by mean standardized information over the preceding twelve quarters, leaves a four-quarter gap, and reports means over the next twelve quarters relative to the mean of the five quintiles. Newey–West standard errors use 8 lags in panel (a) and 24 in panel (b). *, **, and *** denote significance at the 10, 5, and 1 percent levels.

I average their quarterly information estimates in levels using inverse squared industry noise scales as weights, then divide the weighted mean by its implied noise scale. The asset-manager distribution has a fatter right tail than the distribution for banks, insurers and broker-dealers. The orange area marks the excess density in this tail.

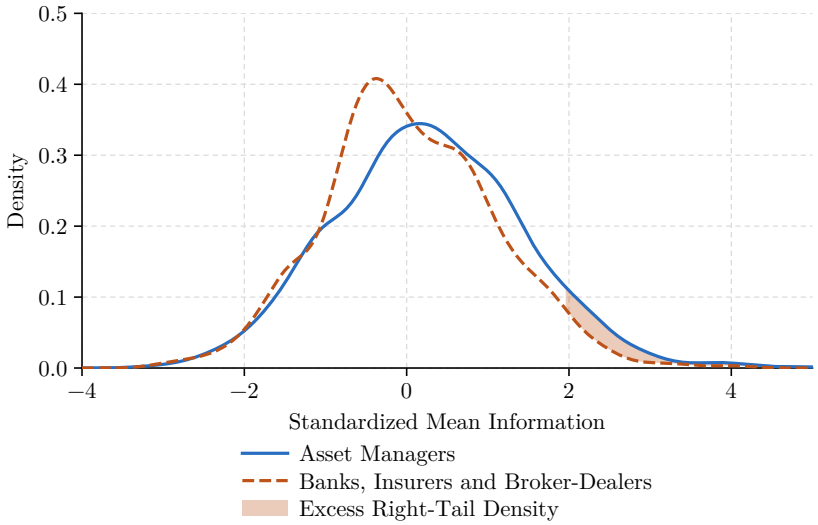


Figure 3: The Distribution of Measured Information across Investors

Notes. Distributions of standardized mean information for institutions with at least twelve quarterly estimates. Every institution has equal weight within its group. The solid blue and dashed dark-orange curves are Gaussian kernel densities for asset managers and for banks, insurers and broker-dealers, using a common pooled Silverman bandwidth. Orange shading marks the excess density in the asset managers' fatter right tail.

5 The Information Production Function

I estimate how data scientist teams process data and produce information about firm fundamentals. The output of the information production function is the measured total information of investors from the previous section. Higher productivity allows a given data scientist team to generate more precise signals about firm payoffs.

5.1 Specification

Following the model in Section 3.3, I estimate the information production function in logs, allowing productivity to vary by calendar year and controlling for employment in other occupations:

$$\log \hat{\theta}_{it} = \omega_i + \omega_{y(t)} + \alpha \log L^{\mathrm{DS}}_{i,t-1} + X^{\top}_{i,t-1} b + u_{it}. \tag{23}$$

Here $\hat{\theta}_{it}$ is investor i's measured total information in quarter t, and $L^{\mathrm{DS}}_{i,t-1}$ is DS employment. The two controls in $X_{i,t-1}$ are log combined employment of investment analysts and portfolio managers, and log employment in all remaining occupations, excluding DS. They account for the size of the investor's other investment staff and the broader organization. All employment inputs use headcounts from the preceding quarter. I assume that risk tolerance is constant within each investor. Investor fixed effects ω_i then absorb these factors along with persistent differences in efficiency. The calendar-year effects $\omega_{y(t)}$ measure common changes in information productivity at given employment. The disturbance u_{it} includes unobserved changes in inputs and productivity, together with sampling error in measured information.

The main sample consists of active asset managers. Active share is half the absolute distance between portfolio and market weights (Cremers and Petajisto, 2009). Managers above the 20th percentile of full-sample mean active share form the active group. They employ 60 percent of DS labor and their holdings are more likely to reflect stock-level research than those of managers close to the market portfolio. I require positive DS employment in at least four of the preceding eight quarters to focus on investors with an established DS workforce. The log specification uses observations with positive measured information and positive employment in each input.

I estimate the coefficients jointly by least squares. Changes in DS employment within investors identify the labor elasticity α, controlling for other employment and common annual changes. Appendix E gives the formal conditions and estimation details.

5.2 Annual Information Productivity

I normalize the 2011 year effect to zero and measure productivity relative to that year:

$$\frac{\widehat{A}_y}{\widehat{A}_{2011}} = \exp\bigl(\hat{\omega}_y - \hat{\omega}_{2011}\bigr), \qquad \hat{\omega}_{2011} = 0. \tag{24}$$

Holding DS and other employment fixed, the estimated production function yields 3.28 times as much information in 2024 as in 2011 (Figure 4). Productivity reaches 1.62 times its initial level in 2016 and 2.53 in 2021, and reaches its highest estimated value in 2024.

The estimated DS labor elasticity is 0.143. At unchanged productivity and other employment, doubling DS employment raises information production by 10.5 percent. With data-management labor proportional to DS employment, this elasticity combines the information gains from preparing more data and from analyzing them. This estimate is consistent with Abis and Veldkamp (2024).

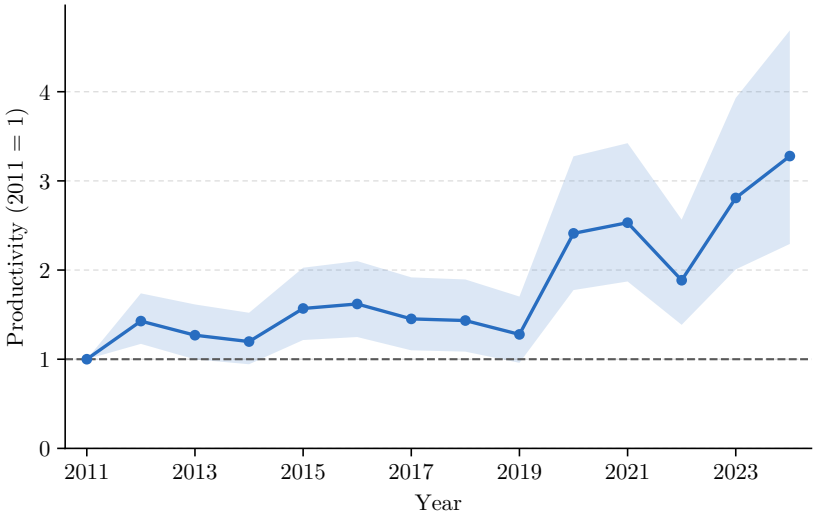


Figure 4: Annual Information Productivity

Notes. Annual productivity from equation (23), normalized to one in 2011. The active-manager sample follows Section 2.2. The line plots $\exp(\hat{\omega}_y - \hat{\omega}_{2011})$. Shading gives pointwise 90 percent confidence intervals based on standard errors clustered by investor.

6 Demand Estimation

I estimate a demand system in which private information and learning from prices shape investors' portfolio choices. The estimation uses quarterly changes in 13F holdings over 1984Q3–2024Q4.

6.1 The Demand System

Private information and learning from prices affect demand through investors' uncertainty about firm payoffs and their interpretation of price changes. More precise private signals and more informative prices both reduce posterior uncertainty. Holding the payoff forecast fixed, this reduction makes demand more sensitive to price. Learning from prices also changes the forecast itself: a higher price signals a higher payoff, offsetting part of the demand response to its higher cost. The estimated price slope combines these effects.

I estimate the relative demand slope to separate stock selection from factor management and adjustments to mandate exposures. Equation (16) compares two stocks for which managing these exposures induces equal position adjustments when one stock's price changes. The difference between their demand responses removes this common adjustment. Following Haddad et al. (2026), I use comparisons across stocks within an investor's portfolio, controlling for stock characteristics and their interactions with information. These interactions matter because stocks with the same factor loading can require different adjustments

to factor exposure when their posterior precisions differ.

I measure demand in book-equity dollars, $Q_{it}(n) = \mathrm{BE}_t(n) q_{it}(n)$, where $q_{it}(n)$ is investor i's fraction of outstanding shares and $\mathrm{BE}_t(n)$ is the firm's book equity (Koijen et al., 2024). I construct the quarterly demand flow $\Delta Q_{it}(n)$ from the change in split-adjusted shares, valued at the previous quarter's book equity per share.[35] Valuing both quarters' shares at the same book equity per share separates trading from changes in valuation. Price is the market-to-book ratio. Its change in the same fixed book-equity units is $\Delta \mathrm{MB}_t(n) = \mathrm{MB}_{t-1}(n) r_t(n)$, where $r_t(n)$ is the ex-dividend return.

Taking first differences of the model's demand schedule gives a price-change term whose coefficient depends on private information and learning from prices. The remaining terms guide the controls used to isolate these information effects. Appendix F gives the derivation. I estimate

$$\begin{aligned} \Delta Q_{it}(n) &= \zeta_{it}(n) \tau_\varepsilon(n) \Delta \mathrm{MB}_t(n) + \Theta_{it}^\top X_{it}(n) + e_{it}(n), \\ \zeta_{it}(n) &= \zeta_0 + \zeta_{\mathrm{type}(i)} + \zeta_{\mathrm{age}} \ln(1 + \mathrm{age}_{i,t-1}) \\ &\quad + \underbrace{\zeta_k \bar{k}_{i,t-1}(n)}_{\text{private information}} + \underbrace{\zeta_\ell \ell_{t-1}(n)}_{\text{learning from prices}} \\ &\quad + \zeta_b \bar{\ell}_{t-1} \mathbf{1}\{\tau^{\mathrm{R}}_{t-1}(n) > 0\} + \zeta_c \mathbf{1}\{\tau^{\mathrm{R}}_{t-1}(n) = 0\}. \end{aligned} \tag{25}$$

The first term is the demand response to a price change. $\tau_\varepsilon(n)$ is the inverse of prior payoff variance. The payoff-precision term increases the magnitude of the price response for stocks with less payoff risk. The coefficient $\zeta_{it}(n)$ captures variation in the slope, with the two underbraced terms measuring the effects of private information and learning from prices. The baseline ζ_0 and the type and age terms account for other heterogeneity across investors. The remaining terms control for changes in relative price signal precision over time and stocks with uninformative prices. Our estimation uses the cross-sectional variation in price informativeness. The controls $\Theta_{it}^\top X_{it}(n)$ account for other demand shifts, and $e_{it}(n)$ is residual demand.

The model predicts a negative coefficient on private information and a positive coefficient on learning from prices, $\zeta_k < 0$ and $\zeta_\ell > 0$. Better private information reduces payoff uncertainty. Investors then buy more when prices fall and sell more when prices rise. Learning from prices also reduces uncertainty, but a higher price raises investors' payoff forecasts. This revision outweighs the effect of lower uncertainty, so investors sell less as prices rise. These information effects extend the characteristics-based demand system of Koijen and

[35] Here $\Delta Q_{it}(n)$ denotes the dollar trading flow divided by lagged assets under management (AUM). Assumption F.1 sets $\gamma_i^{-1} = \mathrm{AUM}_i / \gamma_i^{\mathrm{R}}$, matching the local risk tolerance of CRRA preferences evaluated at AUM. Dividing by AUM removes the component of the relative slope proportional to investor size. The relative-risk-aversion scale γ_i^{R} is evaluated at lagged AUM for each quarterly comparison and may differ across investors.

Yogo (2019); Koijen et al. (2024).

For the private-information interaction in demand, I assume that investors allocate information evenly across the $N_{i,t-1}$ stocks they hold in the sample used to estimate information.[36] Thus $\bar{k}_{i,t-1}(n) = \bar{k}_{i,t-1}/N_{i,t-1}$, where $\bar{k}_{i,t-1}$ is the investor's predicted information. Because both the information estimate and demand come from holdings, I predict information using investor type, lagged size and a historical information score. This captures persistent differences in information and reduces the mechanical link with current trading. The score divides average past information by its estimated standard error and uses estimates through $t-5$. The persistence documented in Section 4 supports this use of past information. I fit the prediction coefficients using other investors' information estimates to reduce overfitting to the investor's own measurement noise. I standardize $\bar{k}_{i,t-1}(n)$ across investor–stock observations, so its coefficient measures the slope difference associated with a one-standard-deviation increase in predicted information per stock.

To measure learning from prices, I use the ratio of aggregate private information to noise-trading variance. I construct this ratio from estimated price informativeness and information aggregated across the stock's holders. A higher ratio makes demand less responsive to price in the model, holding other stock and investor characteristics fixed. For informative prices, $\ell_{t-1}(n)$ is the log of the previous quarter's ratio minus that quarter's median across investor–stock observations. This is the Relative Price Signal Precision measure in Table 3. The term $\bar{\ell}_{t-1}$ measures the quarterly median's deviation from the median over the full sample, so the coefficient on $\ell_{t-1}(n)$ compares stocks within a quarter. I also allow a separate slope for stocks with uninformative prices. Appendix F gives the construction details.

The controls $X_{it}(n)$ account for demand differences associated with observable stock characteristics, factor management and mandate exposures. The investor-quarter intercept makes the comparison across stocks held by the same investor in the same quarter. Stock characteristics and their changes control for differences in expected payoffs. The characteristics are log book equity, profitability, investment, dividends to book equity and market beta. I also include changes in the public earnings forecast from Section 4, after removing the component explained by stock prices. The coefficients on stock characteristics and changes in earnings forecasts vary across investor-quarters.[37] Interactions of factor loadings with information allow exposure adjustments to depend on the precision of payoff forecasts, with coefficients that vary across investor-quarters. Appendix F derives their connection to the relative demand slope.

[36] In an alternative specification, I assume a parametric form for investors' information allocation across stocks, based on the interaction of investor features and stock characteristics. Estimating that demand system gives similar results but with lower power.

[37] Appendix F gives the complete control set.

6.2 Identification with a GIV

Prices and investors' holdings are jointly determined in equilibrium, making prices endogenous in the estimation of demand elasticities. The granular instrumental variable (GIV) uses residual demand from other investors to isolate price variation (Gabaix and Koijen, 2024). The model supplies a natural source of this variation: investor-specific noise in private signals. These signal errors are independent across investors, and, with finitely many investors, they need not cancel in aggregate demand. Noise in other investors' signals can therefore move the market price while the focal investor's private signal stays fixed.

Constructing the instrument removes two components of observed demand. First, I account for investors' heterogeneous price elasticities of demand. This separates changes along their demand schedules from residual demand, which includes trading on private signals. Second, I remove the payoff innovation shared by those signals. Each private signal contains the same firm-level payoff innovation plus investor-specific noise. Responses to the shared innovation are stronger for investors with more private information. Removing this component isolates the variation associated with their signal noise.

I remove both components in a single projection of flows within each stock-quarter. The projection includes the investor variables governing price responses and the sensitivities to common payoff news implied by predicted information and payoff precision. Because flows difference two quarterly positions, the projection includes payoff-news sensitivities at both dates. The projection weights flows by inverse residual variance estimated from other quarters.

The heterogeneous price response can be removed before its coefficients are known. Within a stock-quarter, $\tau_\varepsilon(n)\Delta\mathrm{MB}_t(n)$ is common to all investors, so the price-response term in equation (25) is a linear combination of the observed variables in $\zeta_{it}(n)$. Projecting on these variables removes that combination for any values of its coefficients.[38] The instrument can therefore be constructed before estimating the demand coefficients, and the GMM moments are linear in those coefficients. This procedure follows Gabaix and Koijen (2024) and builds on the asset-market applications of Chaudhary et al. (2026); Chaudhry and Li (2026).

For investor i, I exclude her own flow from the projection and aggregate the other investors' residual flows:

$$Z_{it}(n) = \sum_{j \neq i} \mathrm{AUM}_{j,t-1} \Delta Q_{jt}^{\perp,(-i)}(n). \tag{26}$$

[38] For example, if the price coefficient varies only by investor type, subtracting each type's mean flow within that stock-quarter removes its price-response component without knowing the type coefficients. Projecting on all the variables in $\zeta_{it}(n)$ extends this argument to the specified heterogeneity in demand slopes.

Here $\Delta Q_{jt}^{\perp,(-i)}(n)$ is investor j's residual flow from the projection excluding i. Multiplication by lagged AUM converts it back to book-equity dollars, giving larger investors' demand shocks more weight. Excluding investor i throughout the construction keeps her own signal noise out of the instrument. The resulting price variation identifies her demand response, including learning from prices (Haddad et al., 2026).

The GIV instruments the baseline price term. To estimate heterogeneity, I interact it with each variable in $\zeta_{it}(n)$, including type indicators, log age, predicted information and the price-informativeness terms. Let $\widetilde{Z}_{l,it}(n)$ denote each instrument after removing the included controls, where l indexes the baseline and its interactions. I estimate the coefficients by linear GMM using

$$\mathbb{E}\Big[\widetilde{Z}_{l,it}(n)e_{it}(n)\Big] = 0 \qquad \text{for every slope term } l. \tag{27}$$

Controls serve as their own instruments, and observations receive equal weight in the estimating moments. I calculate standard errors following Chaudhary et al. (2026) to account for each investor's shock entering other investors' instruments. Appendix F derives the instrument and estimating moments.

6.3 Estimates

The sample follows positions held in consecutive quarters and includes stocks with at least sixty investors and investors with at least twenty stocks.

Table 3 shows that investor demand is inelastic at the median, with an implied elasticity of -0.51.[39] Hedge funds have a median elasticity of -1.10, banks and broker-dealers -0.91, investment advisers -0.51, and mutual funds -0.39. This ordering accords with Koijen and Yogo (2019); Koijen et al. (2024).

Figure 5 shows how fitted elasticity varies with private information and price informativeness for a representative position. Its three curves compare the 10th, 50th and 90th percentiles of predicted information per stock among investors with positive estimated private information. The horizontal axis is price informativeness, $\tau_{t-1}^{\mathrm{R}}(n)$: the fraction of payoff uncertainty resolved by prices. Other stock inputs, investor age and position size are fixed at their medians, with type fixed at investment adviser. Price and quantity stay fixed along each curve.

Better-informed investors have more negative demand slopes. A one-standard-deviation increase in predicted information per stock changes the slope coefficient by -0.05, about a

[39] For each position, I calculate elasticity by multiplying the fitted demand slope by lagged market-to-book and dividing by lagged book-equity holdings per dollar of AUM. I average these elasticities within each investor-quarter using book-equity weights, then report medians across investor-quarters.

Table 3: Demand Estimates

	(1) Main Specification
Price (ζ_0)	−0.51
	(0.06)
Price × Information (ζ_k)	−0.05
	(0.01)
Price × Relative Price Signal Precision (ζ_ℓ)	0.25
	(0.04)
Implied Elasticity, Median	
All Institutions	−0.51
Hedge Funds	−1.10
Mutual Funds	−0.39
Inv. Advisers	−0.51
Banks & Broker-Dealers	−0.91
Pension Funds	0.21
Insurers	−0.21
Other	−1.27
Controls	Y
Price × Type, Age	Y
N(Inst.-Quarters)	106,144
N(Stock-Quarters)	197,128

Notes. Section 6 describes estimation. Information standardizes predicted private information per stock, $\bar{k}_{i,t-1}(n)$. Relative Price Signal Precision is the log ratio of aggregate private information to noise-trading variance, centered at the quarterly median. Both measures are lagged one quarter. Elasticities equal fitted slopes times lagged market-to-book divided by lagged book-equity holdings per dollar of AUM. I average them within each investor-quarter using book-equity weights, then report medians across investor-quarters for all institutions and within each type. All price coefficients and their standard errors are reported on a common scale of the median investor-quarter elasticity. This sets the reported coefficient ζ_0 to −0.51 and reports the interactions as slope differences on a common scale. Standard errors appear in parentheses.

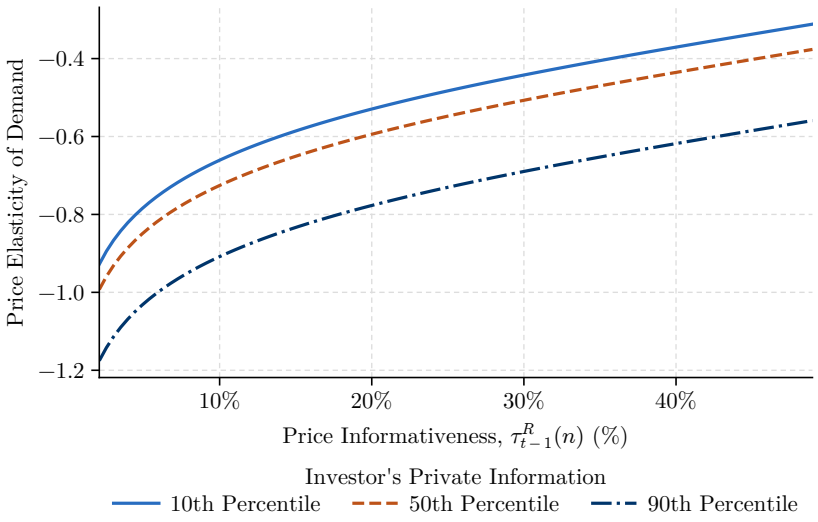


Figure 5: Demand Elasticity and Information

Notes. Fitted demand elasticities for a representative position at the 10th, 50th and 90th percentiles of predicted private information per stock, $\bar{k}_{i,t-1}(n)$. Price informativeness, $\tau^{\mathrm{R}}_{t-1}(n)$, is the percentage of idiosyncratic payoff uncertainty resolved by prices. The axis spans its 5th–95th percentiles across distinct informative stock-quarters. Age uses the investor-quarter median, and type is fixed at investment adviser. Holdings use the median across investor-quarters of each portfolio's median informative holding. Reference prices and quantities stay fixed along the curves. More negative values mean more elastic demand. Appendix F gives the calculation.

tenth of the baseline coefficient. This estimate agrees with the model: more precise private signals reduce uncertainty and increase the willingness to trade against a price change. At the median price informativeness of 19.0 percent, Figure 5's elasticities at the 10th, 50th and 90th percentiles of private information are -0.54, -0.61 and -0.79. Investors with more private information have a larger percentage demand response to a one percent price change.

Learning from prices makes demand less elastic. A one-log-unit increase in $\ell_t(n)$, the ratio of aggregate private information to noise-trading variance within a quarter, raises the slope coefficient by 0.25. Holding other stock inputs fixed, a more informative price raises this ratio. Each curve in Figure 5 therefore rises with price informativeness: the elasticity becomes less negative as investors learn more from price changes.

At the median informative stock, learning from prices makes an uninformed investor's demand elasticity less negative, reducing its magnitude by 60.6 percent. Setting private information to zero at the median informative stock gives an AUM-weighted average slope coefficient of -0.42. Combined with the coefficient on learning from prices, Lemma F.1 gives a discount of $1 + 0.25/(-0.42) \simeq 0.394$.

7 Counterfactual: A Data-Technology Advance

I combine the estimated information production function and demand system to study the effects of future advances in data technology. The experiment takes 2024Q4, the final quarter of the sample, as its baseline and repeats the historical increase in data-science productivity. I analyze the effects on price informativeness, stocks' ownership compositions and market segmentation, and aggregate demand elasticity.

7.1 The Experiment

I combine 2024Q4 13F holdings and price informativeness with lagged estimated private information of investors. Employment of data scientists, investors' risk preferences and mandates, and the set and characteristics of stocks available to each investor keep fixed. Each stock's supply to investors in the sample equals their initial aggregate holdings. The outside sector keeps its demand fixed and supplies no private information.

The data technology advance raises private information at active asset managers with DS employment, whom I call adopters. They initially hold 10.7 percent of the sample's book-equity positions. The productivity increases estimated for active managers motivate this choice. Private information is unchanged for other investors, including managers with DS staff but low active share. The technology advance affects them through competition from adopters and learning from prices.

I repeat the estimated increase in annual productivity from 2011 to 2024. For adopter i, the proportional gain in the baseline predicted information input $\theta_{i,t}$ is

$$\frac{\Delta\theta_{i,t}}{\theta_{i,t}} = \delta\frac{\widehat{A}_{2024} - \widehat{A}_{2011}}{\widehat{A}_{2024}} = \delta\left(1 - \frac{\widehat{A}_{2011}}{\widehat{A}_{2024}}\right). \tag{28}$$

The parameter δ measures the technology advance in multiples of the historical productivity level increase, $\widehat{A}_{2024} - \widehat{A}_{2011}$. The estimated productivity ratio is 3.28 (Figure 4), so $\delta = 1$ raises each adopter's private information by 69.5 percent. The experiments with $\delta = 2$ and 3 double and triple this increase. The production function is multiplicative. At fixed DS employment, its common productivity factor scales each adopter's information in the same proportion. The increase in data-science productivity raises every stock's private signal precision proportionally within an adopter.[40] Appendix G gives details of the calculation of

[40] Under the assumption of an even allocation of information across stocks, this increase gives every initially held stock the same amount of information in units of lagged prior precision. Private-signal precision is the inverse variance of signal noise. Each data point has the stock's prior precision $\tau_\varepsilon(n)$, so $k_i(n)$ data points give signal precision $\tau_i(n) = k_i(n)\tau_\varepsilon(n)$.

information gains and the restriction on factor management.

The technology advance changes price informativeness in general equilibrium. Adopters' stronger demand raises prices, moving other investors along their demand curves. As adopters trade on more precise signals, prices also reveal more information about payoffs. This changes how all investors learn from prices. The resulting changes in payoff uncertainty and in how investors' demand responds to price signals shift investors' demand curves and change their demand slopes.

Aggregating adopters' information gains determines how much additional payoff information enters each stock's price. Investors combine this more precise price signal with their private signals. First, I calculate the counterfactual price informativeness for each stock. Then, investors' counterfactual demand and counterfactual prices are jointly solved based on market clearing conditions. Reported 90 percent confidence intervals are the 5th and 95th percentiles of 200 joint draws of demand coefficients using their estimated covariance, conditional on the production estimates, productivity increase and other inputs.

7.2 Price Informativeness

Table 4 shows that average price informativeness rises by 3.1 percentage points from 11.8 percent. Doubling and tripling the technology advance raise the all-stock average by 6.1 and 8.9 points.

Table 4: Price Informativeness after a Data-Technology Advance

Data Scientist Productivity Increase:	(1) $\delta = 1$	(2) $\delta = 2$	(3) $\delta = 3$
Initial Price Informativeness (%)	11.8	11.8	11.8
Final Price Informativeness (%)	14.9	17.9	20.7
Increase (p.p.)	3.1	6.1	8.9

Notes. Equal-weighted price informativeness across stocks in the counterfactual. Price informativeness is the fraction of idiosyncratic payoff uncertainty resolved by prices. Initial Price Informativeness and Final Price Informativeness are percentages. Increase is in percentage points. The productivity increase, δ, is measured in multiples of the historical data scientist productivity increase. The columns repeat, double and triple that increase at fixed research employment. Section 7.1 defines the technology advance and adopters.

The increases in price informativeness depend on how much new information reaches each stock and how informative its price is initially. Let $\Delta_\kappa(n)$ be the proportional rise in aggregate information. This increase weights adopters' information gains by their initial contributions to the stock. Price-signal precision then rises by the factor $[1 + \Delta_\kappa(n)]^2$: stronger informed demand reduces the noise in the information conveyed by prices. When a price is initially very noisy, a proportional improvement reveals little additional information.

When it is already highly informative, little uncertainty remains to resolve. The increase in price informativeness is therefore largest at an intermediate initial level. In the cross-section, increases rise up to initial price informativeness of about 45 percent and then decline (Figure 6; Corollary C.4). Larger technology advances move this peak toward lower initial price informativeness.

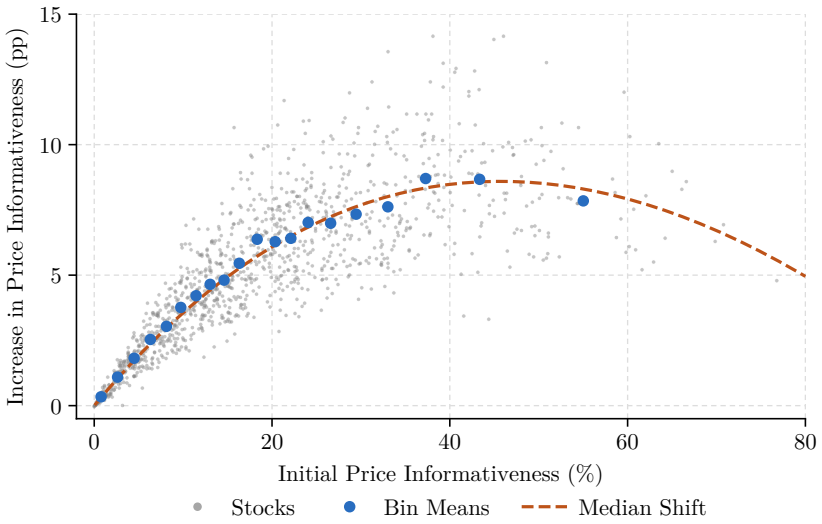


Figure 6: The Increase in Price Informativeness across Stocks

Notes. Increases in price informativeness after a productivity increase equal to its 2011–2024 amount ($\delta = 1$), for stocks with positive price informativeness in 2024Q4. Price informativeness is the fraction of idiosyncratic payoff uncertainty resolved by prices. The horizontal axis is its initial level in percent. The vertical axis is its change in percentage points. Gray points are stocks. Bin Means averages twenty equal-count bins of initial price informativeness. Section 7.1 defines the technology advance and adopters.

Price informativeness increases more for small stocks. The average increase is 4.4 points in the smallest size quintile, compared with 2.3 in the largest. Adopters supply 32.8 percent of small firms' aggregate private information and 20.5 percent of large firms', so the technology advance adds proportionally more information to small firms. Table 5 relates the increases to stock characteristics. Conditional on the other characteristics, increases are larger for stocks with greater idiosyncratic volatility, analyst coverage and market beta.

7.3 Crowding Out and Market Segmentation

7.3.1 Portfolio Reallocation

I measure portfolio reallocation by the percentage change in each group's total positions. A position is the fraction of a firm's outstanding shares held by an institution, measured in units of the firm's initial book equity.

Table 5: Price Informativeness and Crowding Out across Stock Characteristics

	(1) Change in Price Informativeness (p.p.)	(2) Crowding Out (%)
Size (ln ME)	−0.10	0.08***
	(0.14)	(0.02)
B/M (ln)	0.33***	−0.09***
	(0.09)	(0.02)
Op. Profitability	−0.39***	0.09***
	(0.09)	(0.02)
Asset Growth	0.23***	−0.11***
	(0.06)	(0.01)
Dividends / BE	−0.27***	0.00
	(0.09)	(0.02)
Market Beta	1.16***	−0.18***
	(0.08)	(0.01)
Idio. Vol.	1.25***	−0.25***
	(0.10)	(0.02)
Turnover (ln)	−0.29***	0.16***
	(0.09)	(0.01)
Firm Age (ln)	−0.01	0.00
	(0.08)	(0.01)
Analyst Coverage (ln)	1.39***	−0.03*
	(0.12)	(0.02)
Mean of Dependent Variable	5.25	1.53
Adjusted R^2	0.49	0.43
Stocks	1147	1933

Notes. Cross-sectional regressions after a productivity increase equal to the historical increase ($\delta = 1$). Column (1) uses stocks with positive initial price informativeness. The outcome is the increase in price informativeness in percentage points. Column (2) uses the full stock sample. The outcome is the percentage decline in other institutions' total positions. A positive coefficient indicates a larger percentage reduction in positions. Characteristics follow Table 1. B/M and six-month average daily turnover enter in logs. Firm age in months and analyst coverage enter as log one plus their values. Dividends / BE is twelve-month dividends over book equity. Regressors are winsorized at the 1st and 99th percentiles and standardized across the full stock sample. Parentheses report standard errors. *, **, and *** denote significance at the 10, 5, and 1 percent levels.

Data technology adopters expand their positions by 15.4 percent, while other institutions reduce theirs by 1.85 percent (Table 6). Adopters' share of total positions rises from 10.7 to 12.3 percent.

Table 6: Crowding Out of Other Institutions

Panel (a) Position Changes (%)

		Adopters' Coverage					
	All	Low	2	3	4	High	High − Low
Adopters	15.4	4.9	9.4	12.9	13.7	20.2	15.4
	[11.3, 19.9]						[11.5, 19.3]
Other Institutions	−1.8	−0.7	−1.2	−1.5	−1.8	−2.3	−1.5
	[−2.4, −1.4]						[−1.9, −1.1]

Panel (b) Portfolio-Weight Changes (p.p.)

		Adopters' Coverage					
	All	Low	2	3	4	High	High − Low
Other Institutions		0.02	0.03	0.06	0.07	−0.17	−0.19
							[−0.23, −0.15]
N(Stocks)	1933	387	386	387	386	387	

Notes. Changes after a productivity increase equal to the historical increase ($\delta = 1$). Adopters are active asset managers with DS employment. Other Institutions are the remaining institutions. Within each coverage quintile, panel (a) reports each institution group's percentage change in total positions. Panel (b) reports changes in each stock quintile's share of the group's portfolio, in percentage points, valuing holdings at initial prices. Adopters' coverage is the proportion of investors' total information about a stock that comes from adopters' private information. Stocks are sorted into five bins. All pools the five groups. High minus Low compares the extreme quintiles. Brackets give 90 percent confidence intervals.

Competition from adopters' private-information gains drives this reallocation. Lower payoff uncertainty raises their demand. Prices rise as investors compete for the fixed supply, reducing the expected excess payoff available to every investor. An investor is crowded out when the improvement in its forecast precision is too small to offset this decline (equation (20)).

To separate the two sources of information gains, I first hold price informativeness at its initial value. Adopters' new private information then reduces other institutions' positions by 1.95 percent. Allowing prices to become more informative offsets 0.11 percentage points of that decline through learning from prices. Adopters' positions increase by 16.3 percent in the first step. Learning from prices reduces this increase by 0.90 points (Table 7, Panel (a)). Prices adjust to clear the fixed supply in both steps. Appendix G gives the decomposition, and Online Appendix Table OA5 reports alternative orders and larger technology advances.

Learning from prices adds the same signal precision to every investor's forecast, so it gives a larger proportional improvement to investors with less private information. It also

Table 7: Private Information, Learning from Prices, and Institutional Demand

	(1) Private Information Gains	(2) More Informative Prices	(3) Total
Panel (a) Position Changes (%)			
Adopters	16.30	−0.90	15.41
	[11.94, 20.97]	[−1.13, −0.68]	[11.27, 19.86]
Other Institutions	−1.95	0.11	−1.85
	[−2.51, −1.43]	[0.08, 0.14]	[−2.38, −1.35]
Panel (b) Change in Elasticity Magnitude (%)			
Median across Informative Stocks	4.52	−7.01	−2.49
	[4.10, 4.96]	[−7.96, −6.15]	[−3.85, −1.19]

Notes. This table decomposes changes in positions and demand elasticity. Panel (a) reports percentage position changes. Private-signal precision increases first at unchanged price informativeness, followed by the effect of more informative prices. Panel (b) reports median percentage changes in demand-elasticity magnitude across informative stocks, evaluated at initial prices and holdings. In the Private Information Gains column of Panel (b), the effect of learning from prices on the demand slope is held fixed. The contribution of learning from prices is the difference between the median of total changes and the median of changes caused by private information alone (column (1)). Negative values mean less elastic demand. Brackets give 90 percent confidence intervals.

erodes the trading value of existing private information as prices reveal more to competitors (Bonelli and Foucault, forthcoming). For other institutions as a group, the benefit is small relative to the stronger competing demand from adopters. Every institution type in this group loses positions in aggregate (Table 8).

Among adopters, hedge funds expand their positions by 35.3 percent, investment advisers by 12.0 percent and mutual funds by 6.5 percent. Hedge funds begin with more information, so the proportional productivity increase adds more to their private signal precision.

Table 8: Crowding Out of Other Institutions by Institution Type

	Change in Positions (%)						
	Hedge Funds	Inv. Advisers	Mutual Funds	Pension Funds	Banks & Br.-Dealers	Insurers	Other
Adopters	35.3	12.0	6.5	–	–	–	–
Other Institutions	−7.2	−3.9	−1.2	−0.5	−2.2	−0.9	−2.8

Notes. Percentage changes in positions by institution type after a productivity increase equal to the historical increase ($\delta = 1$). Adopters, other institutions and the position adjustment follow Table 6. Each entry divides the indicated group's change in total positions across stocks by its initial total positions, multiplied by 100. Institution types follow Table 3. Other Types includes the remaining institutions. A dash indicates no adopters in that type.

Crowding out is stronger where adopters contribute more of the information of that stock. Adopters' coverage is the proportion of investors' total information about the stock

that comes from adopters' private information. Total information includes public prior information, private information and information learned from prices. Across coverage quintiles, adopters' positions rise by 4.9 percent in the least-covered stocks and 20.2 percent in the most-covered. Other institutions' positions fall by 0.7 and 2.3 percent, respectively (Figure 7(a)). Other institutions also shift portfolio weight away from the most-covered stocks. Valuing holdings at initial prices, their weight in the highest-coverage quintile falls by 0.17 percentage points, while their weight in the lowest rises by 0.02 points (Table 6, Panel (b)).

More-informed institutions receiving no private-information gain have larger losses. Figure 7(b) sorts these institutions by their private information per stock, $\bar{k}_{i,t-1}(n)$. The median decline in total positions rises from 1.8 percent in the lowest information quintile to 14.5 percent in the highest. The ordering is consistent with Proposition 5: for the same stock, a better-informed investor has a smaller proportional increase in forecast precision from learning from prices and suffers a larger percentage loss when crowded out.

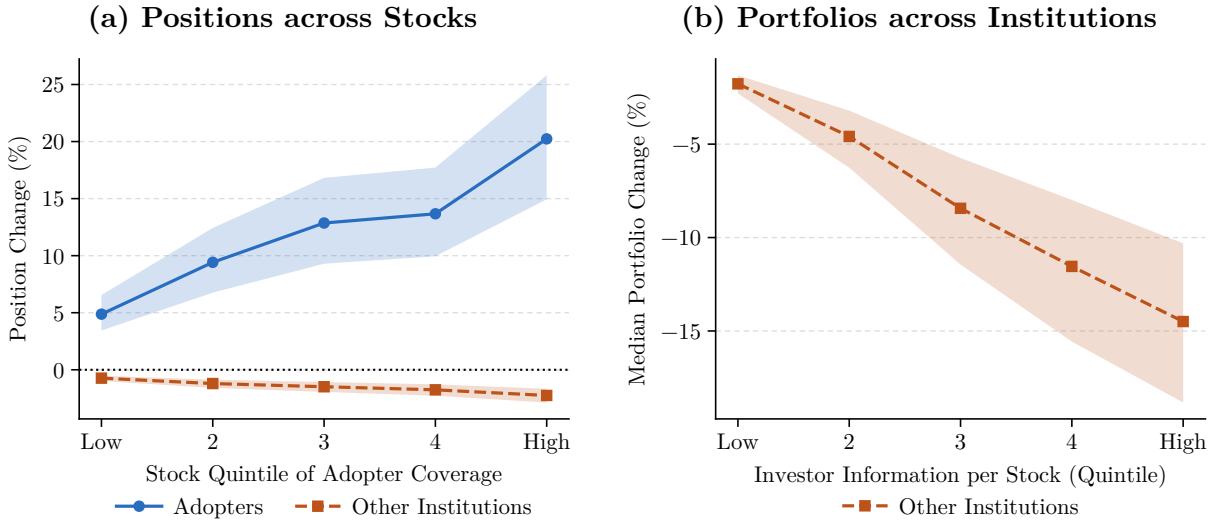


Figure 7: Crowding Out across Stocks and Institutions

Notes. Percentage changes in positions after a productivity increase equal to its 2011–2024 increase ($\delta = 1$). Panel (a) sorts stocks into five equal-count groups by adopters' coverage and reports each institution group's percentage change in total positions within that stock group. Coverage and institution groups follow Table 6. Panel (b) sorts non-adopters by their private information per stock, $\bar{k}_{i,t-1}(n)$. For each investor, I divide the change in total positions across stocks by its initial total positions. Points give medians within information quintiles. Shading gives 90 percent confidence intervals.

Crowding out also varies with stock characteristics (Table 5). A one-standard-deviation increase in log firm size is associated with a 0.08 percentage-point greater reduction in other institutions' positions, and higher turnover with a 0.16-point greater reduction. For most characteristics, the coefficients on crowding out and the change in price informativeness have opposite signs. Stocks with greater idiosyncratic volatility, market beta and analyst coverage

have larger increases in price informativeness and smaller declines in other institutions' positions, conditional on the other characteristics. For these stocks, learning from prices offsets more of the decline. More informative prices have two opposite effects: they can improve investors' forecasts by providing more information, while reducing the value of their existing private information. The balance depends on other institutions' private information relative to the average. The learning from prices effect dominates for relatively less-informed institutions.

7.3.2 Ownership Composition across Stocks

A stock's ownership composition is the distribution of its holdings across institutions in the sample. I measure the difference between two ownership compositions as half the sum of the absolute differences in investors' ownership shares. It is the share of ownership that must change hands for the two compositions to match. This is the dissimilarity index of Duncan and Duncan (1955), analogous to active share (Cremers and Petajisto, 2009).

The average difference rises from 53.12 to 53.60 percent, an increase of 0.48 percentage points (Table 9). Adopters' higher demand crowds other institutions out of certain stocks, making ownership compositions more distinct across stocks. Doubling and tripling the technology advance raise the difference by 0.90 and 1.28 points. The difference between the average ownership compositions of the highest- and lowest-coverage quintiles rises by 0.12 points from 22.63 percent, and by 0.41 and 0.75 points under the larger technology advances.

Table 9: Market Segmentation after the Technology Advance

	(1)	(2)	(3)	(4)
	2024Q4 Level (%)	Change from 2024Q4 (p.p.)		
Data Scientist Productivity Increase:		$\delta = 1$	$\delta = 2$	$\delta = 3$
Panel (a) All Stocks				
Average Difference in Ownership Composition	53.12	0.48 [0.35, 0.62]	0.90 [0.66, 1.15]	1.28 [0.94, 1.62]
Panel (b) Most- and Least-Covered Stock Quintiles				
Difference between Average Ownership Compositions	22.63	0.12 [0.08, 0.17]	0.41 [0.22, 0.60]	0.75 [0.47, 1.07]

Notes. For each stock, ownership shares sum to one across the 13F institutions in the sample. The difference between two ownership compositions is half the sum of the absolute differences in these shares. Panel (a) gives every distinct stock pair equal weight. Panel (b) first averages each institution's ownership share within each quintile, weighting stocks equally, then compares the two average ownership compositions. Coverage is the share of a stock's total information in 2024Q4 that comes from adopters' private information (Table 6). Brackets give 90 percent confidence intervals for the changes.

7.4 Aggregate Demand and Equity Prices

Aggregate demand becomes less elastic for 46.5 percent of stocks (Table 10). These stocks represent 78.3 percent of those with informative prices and account for 19.3 percent of total book equity. Among informative stocks, aggregate elasticity magnitude falls by 2.5 percent at the median. The share of all stocks becoming less elastic falls to 43.3 percent under twice the technology advance and 40.3 percent under three times the technology advance. I evaluate elasticities at initial prices and quantities to measure changes in the demand slope.

Table 10: Demand Elasticity and Equity Prices after the Technology Advance

	(1)	(2)	(3)
	Change after the Technology Advance		
Data Scientist Productivity Increase:	$\delta = 1$	$\delta = 2$	$\delta = 3$
Panel (a) Demand Elasticity			
Aggregate, Median Change (%)	−2.5	−3.7	−3.9
	[−3.9, −1.2]	[−6.2, −1.3]	[−7.1, −0.5]
Less Elastic, Share of Stocks (%)	46.5	43.3	40.3
	[38.4, 52.3]	[33.9, 49.9]	[30.7, 46.8]
Less Elastic, Share of BE (%)	19.3	17.5	15.8
	[9.3, 36.9]	[8.0, 33.4]	[7.0, 30.2]
Panel (b) Equity Prices			
Price Change, VW (%)	2.9	5.5	7.9
	[2.0, 4.0]	[3.8, 7.7]	[5.5, 11.0]
Price Change, EW (%)	1.4	2.8	4.1
	[1.0, 2.1]	[1.9, 4.0]	[2.8, 5.8]

Notes. Changes in elasticity magnitudes and equity prices in the counterfactual. The productivity increase, δ, is a multiple of its historical increase. Aggregate refers to total institutional demand. Median Change is the median percentage change in elasticity magnitude across informative stocks, evaluated at initial prices and holdings. Negative values mean less elastic demand. Less Elastic, Share of Stocks is the percentage of all stocks with a decline in aggregate elasticity magnitude. Share of BE weights this indicator by book equity. VW and EW are market-equity-weighted and equal-weighted means across stocks. Brackets give 90 percent confidence intervals.

The decline reflects that the learning from prices effect dominates the direct effect of better private information on the demand elasticities. Adopters' better private signals lower their uncertainty and make them more willing to trade against a price change. This makes their demand more elastic. More informative prices also reduce uncertainty for all investors, but they also make a price increase a stronger signal of better payoff news. This strengthens investors' learning from prices and makes them decrease their demand less when prices rise. Aggregate demand becomes less elastic when the learning from prices effect dominates.

Holding fixed the contribution of learning from prices to the demand elasticities, new private information raises aggregate elasticity magnitude by 4.5 percent at the median across

informative stocks. Allowing prices to become more informative lowers this median change by 7.0 percentage points, giving a net decline of 2.5 percent (Table 7, Panel (b)). As the technology advance grows, the direct private-information gain dominates for more stocks, reducing the share that become less elastic (Corollary C.3).

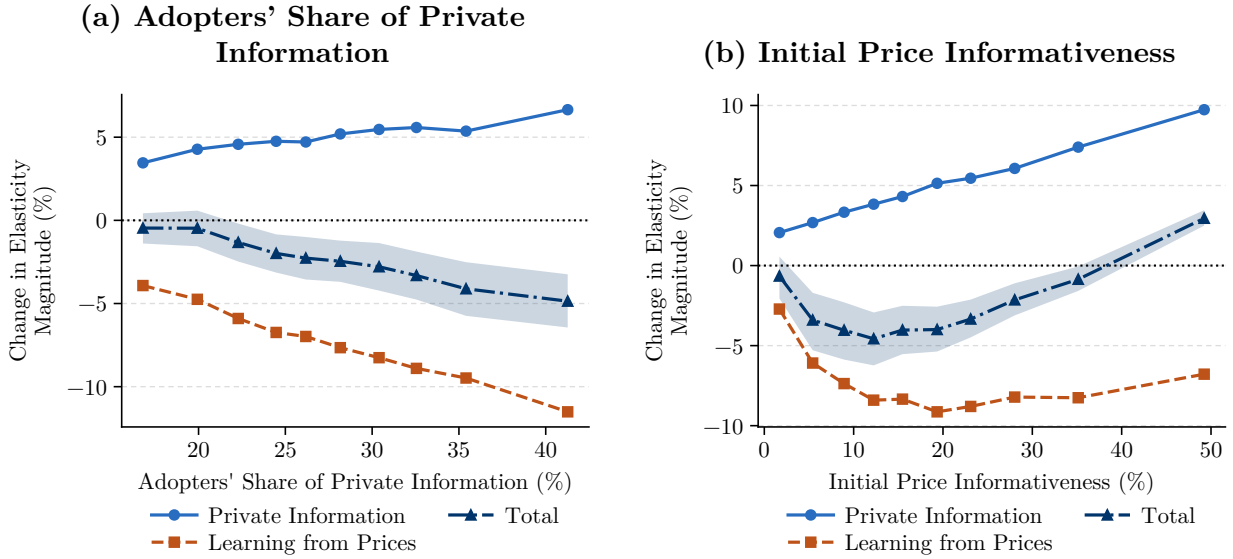


Figure 8: Private Information, Learning from Prices, and Aggregate Demand Elasticity

Notes. Percentage changes in aggregate demand-elasticity magnitude after a productivity increase equal to its 2011–2024 increase ($\delta = 1$), evaluated at initial prices and holdings for informative stocks. Panel (a) sorts stocks by adopters' share of aggregate private information. Panel (b) sorts by initial price informativeness. Both panels give equal-weighted means within ten bins. Private Information holds the contribution of learning from prices to the demand slope fixed. Learning from Prices gives the additional effect of more informative prices. Total is their sum. Negative changes mean less elastic demand. Shading around Total gives 90 percent confidence intervals.

Figure 8(a) shows larger elasticity declines where adopters supply a larger share of aggregate private information. Across the lowest and highest information-share groups, the mean decline in elasticity magnitude rises from 0.5 to 4.9 percent. Both the direct private-information effect and the effect of learning from prices grow with adopters' share, but the effect of learning from prices grows more.

In Panel (b), the change in elasticity magnitude is non-monotonic in initial price informativeness. The direct private-information effect increases with initial price informativeness. More informative prices make investors' payoff forecasts move more with prices, weakening the price response they would have without private information. Private information therefore accounts for a larger share of aggregate elasticity magnitude. The reduction in elasticity magnitude from learning from prices is strongest at intermediate price informativeness. When initial price informativeness is low, prices reveal little information, so the increase in learning from prices has a limited effect on demand elasticity. When initial price

informativeness is high, the price response investors would have without private information is already weak, leaving less scope for further learning from prices to reduce aggregate elasticity magnitude. The largest decline in elasticity magnitude occurs at intermediate initial price informativeness. The largest net decline is 4.6 percent in the group with average initial price informativeness of 12.2 percent. In the group with the most informative prices, the direct private-information effect dominates and demand becomes 3.0 percent more elastic on average.

Equity prices rise by 2.9 percent value-weighted and 1.4 percent equal-weighted. Appendix G gives the price calculation. Doubling and tripling the technology advance raise value-weighted prices by 5.5 and 7.9 percent.

8 Conclusion

Asset managers with larger shares of data-science labor show greater increases in stock-picking ability after technology breakthroughs. I measure investors' private information from holdings' forecasting power for future payoffs beyond public information and prices. Using U.S. institutional holdings and employment records, I estimate information production across investors and a demand system with heterogeneity in private information and learning from prices.

A counterfactual data technology advance equal to the estimated data scientist productivity growth over more than one decade raises average price informativeness by 3.1 percentage points from 11.8 percent. Price informativeness increases more for small stocks because adopters supply a larger share of the private information about these firms. Data technology adopters' better forecasts raise their demand, and their demand makes prices more informative. This erodes other investors' informational advantage and reduces the expected payoff from trading on their existing private signals. They shift portfolio weight toward stocks with less information from adopters, increasing market segmentation. Learning from prices reduces their payoff uncertainty and partly offsets their aggregate holdings losses.

The data technology advance also changes aggregate demand elasticity. Adopters' better private information reduces uncertainty about payoffs and makes their demand more elastic. However, their trades increase price informativeness and make all other investors learn more from prices. This makes their demand less elastic. Aggregate demand elasticity is an endogenous equilibrium outcome. For each stock, the distribution of ownership across investors with different data scientist teams shapes the gains in private information and how informative prices become. In the counterfactual, learning from prices dominates the direct effect of adopters' new information for many stocks, making aggregate demand less elastic.

References

Abis, Simona and Laura Veldkamp, "The Changing Economics of Knowledge Production," *Review of Financial Studies*, 2024, *37* (1), 89–118.

Admati, Anat R., "A Noisy Rational Expectations Equilibrium for Multi-Asset Securities Markets," *Econometrica*, 1985, *53* (3), 629–657.

Agarwal, Vikas, Wei Jiang, Yuehua Tang, and Baozhong Yang, "Uncovering Hedge Fund Skill from the Portfolio Holdings They Hide," *Journal of Finance*, 2013, *68* (2), 739–783.

Aragon, George O., Sanghyun (Hugh) Kim, and Vikram K. Nanda, "Will AI Replace or Enhance Human Intelligence in Asset Management?," 2026. Working paper, SSRN 5224162.

Azimi, Mehran and Anup Agrawal, "Is Positive Sentiment in Corporate Annual Reports Informative? Evidence from Deep Learning," *Review of Asset Pricing Studies*, 2021, *11* (4), 762–805.

Bai, Jennie, Thomas Philippon, and Alexi Savov, "Have Financial Markets Become More Informative?," *Journal of Financial Economics*, 2016, *122* (3), 625–654.

Baker, Malcolm, Lubomir Litov, Jessica A. Wachter, and Jeffrey Wurgler, "Can Mutual Fund Managers Pick Stocks? Evidence from Their Trades Prior to Earnings Announcements," *Journal of Financial and Quantitative Analysis*, 2010, *45* (5), 1111–1131.

Bastianello, Francesca and Paul Fontanier, "Expectations and Learning from Prices," *Review of Economic Studies*, 2025, *92* (3), 1341–1374.

Begenau, Juliane, Maryam Farboodi, and Laura Veldkamp, "Big Data in Finance and the Growth of Large Firms," *Journal of Monetary Economics*, 2018, *97*, 71–87.

Bond, Philip, Alex Edmans, and Itay Goldstein, "The Real Effects of Financial Markets," *Annual Review of Financial Economics*, 2012, *4*, 339–360.

Bonelli, Maxime and Thierry Foucault, "Alternative Data, Firms' Ownership and Stock Price Fragility," 2026. Working paper, SSRN 7348301.

_ **and** _ , "Does Big Data Devalue Traditional Expertise? Evidence from Active Funds," *Journal of Finance*, forthcoming.

Bretscher, Lorenzo, Lukas Schmid, Ishita Sen, and Varun Sharma, "Institutional Corporate Bond Pricing," *Review of Financial Studies*, 2026, *39* (3), 605–660.

Breugem, Matthijs and Adrian Buss, "Institutional Investors and Information Acquisition: Implications for Asset Prices and Informational Efficiency," *Review of Financial Studies*, 2019, *32* (6), 2260–2301.

Carhart, Mark M., "On Persistence in Mutual Fund Performance," *Journal of Finance*, 1997, *52* (1), 57–82.

Cen, Ling, Bing Han, Yanru Han, and Chanik Jo, "Data Scientists on Wall Street," 2025. Working paper, SSRN 5055695.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, "The Effect of Minimum Wages on Low-Wage Jobs," *Quarterly Journal of Economics*, 2019, *134* (3), 1405–1454.

Chaudhary, Manav, Julie Zhiyu Fu, and Haonan Zhou, "Anatomy of the Treasury Market: Who Moves Yields?," 2026. Working paper, SSRN 5021055.

Chaudhry, Aditya and Carter Davis, "The Origins of the Factor Zoo: Investors Weakly Substitute Across Stocks," 2026. Fisher College of Business Working Paper 2026-03-008.

_ **and Jiacui Li**, "Endogenous Elasticities: Price Multipliers Are Smaller for Larger Demand Shocks," 2026. Working paper, SSRN 5325371.

Chen, Mike and Weili Zhou, "On Machine Learning Applications in Investments," in Larry Cao, ed., *Handbook of Artificial Intelligence and Big Data Applications in Investments*, CFA Institute Research Foundation, 2023, pp. 2–19.

Chen, Qi, Itay Goldstein, and Wei Jiang, "Price Informativeness and Investment Sensitivity to Stock Price," *Review of Financial Studies*, 2007, *20* (3), 619–650.

Chen, Yong, Bryan Kelly, and Wei Wu, "Sophisticated Investors and Market Efficiency: Evidence from a Natural Experiment," *Journal of Financial Economics*, 2020, *138* (2), 316–341.

Cohen, Lauren, Christopher Malloy, and Quoc Nguyen, "Lazy Prices," *Journal of Finance*, 2020, *75* (3), 1371–1415.

Corhay, Alexandre, Kejia Hu, Jun E. Li, Jincheng Tong, and Chi-Yang Tsou, "Data, Markups, and Asset Prices," 2025. Working paper, SSRN 5093509.

Cremers, K. J. Martijn and Antti Petajisto, "How Active Is Your Fund Manager? A New Measure That Predicts Performance," *Review of Financial Studies*, 2009, *22* (9), 3329–3365.

Cui, Kai and Jonathan Shahrabani, "Data Science for Active and Long-Term Fundamental Investing," in Larry Cao, ed., *Handbook of Artificial Intelligence and Big Data Applications in Investments*, CFA Institute Research Foundation, 2023, pp. 27–30.

Dávila, Eduardo and Cecilia Parlatore, "Identifying Price Informativeness," *Review of Financial Studies*, 2026, *39* (5), 1267–1309.

Davis, Carter, "The Elasticity of Quantitative Investment," *Review of Financial Studies*, 2025, *38* (10), 2845–2886.

Dou, Winston Wei, Leonid Kogan, and Wei Wu, "Common Fund Flows: Flow Hedging and Factor Pricing," *Journal of Finance*, forthcoming.

Dugast, Jérôme and Thierry Foucault, "Data Abundance and Asset Price Informativeness," *Journal of Financial Economics*, 2018, *130* (2), 367–391.

_ **and** _ , "Equilibrium Data Mining and Data Abundance," *Journal of Finance*, 2025, *80* (1), 211–258.

Duncan, Otis Dudley and Beverly Duncan, "A Methodological Analysis of Segregation Indexes," *American Sociological Review*, 1955, *20* (2), 210–217.

Easley, David and Maureen O'Hara, "Information and the Cost of Capital," *Journal of Finance*, 2004, *59* (4), 1553–1583.

Farboodi, Maryam, Adrien Matray, Laura Veldkamp, and Venky Venkateswaran, "Where Has All the Data Gone?," *Review of Financial Studies*, 2022, *35* (7), 3101–3138.

_ **and Laura Veldkamp**, "Long-Run Growth of Financial Data Technology," *American Economic Review*, 2020, *110* (8), 2485–2523.

_ **, Dhruv Singal, Laura Veldkamp, and Venky Venkateswaran**, "Valuing Financial Data," *Review of Financial Studies*, 2025, *38* (3), 938–980.

Friberg, Richard, Itay Goldstein, and Kristine W. Hankins, "Corporate Responses to Stock Price Fragility," *Journal of Financial Economics*, 2024, *153*, 103795.

Froot, Kenneth, Namho Kang, Gideon Ozik, and Ronnie Sadka, "What Do Measures of Real-Time Corporate Sales Say about Earnings Surprises and Post-Announcement Returns?," *Journal of Financial Economics*, 2017, *125* (1), 143–162.

Gabaix, Xavier and Ralph S. J. Koijen, "Granular Instrumental Variables," *Journal of Political Economy*, 2024, *132* (7), 2274–2303.

Gao, Meng and Jiekun Huang, "Informing the Market: The Effect of Modern Information Technologies on Information Production," *Review of Financial Studies*, 2020, *33* (4), 1367–1411.

Gârleanu, Nicolae and Lasse Heje Pedersen, "Dynamic Trading with Predictable Returns and Transaction Costs," *Journal of Finance*, 2013, *68* (6), 2309–2340.

_ **and** _ , "Efficiently Inefficient Markets for Assets and Asset Management," *Journal of Finance*, 2018, *73* (4), 1663–1712.

Goldstein, Itay, "Information in Financial Markets and Its Real Effects," *Review of Finance*, 2023, *27* (1), 1–32.

_ **and Liyan Yang**, "Information Diversity and Complementarities in Trading and Information Acquisition," *Journal of Finance*, 2015, *70* (4), 1723–1765.

_ , **Chester S. Spatt, and Mao Ye**, "Big Data in Finance," *Review of Financial Studies*, 2021, *34* (7), 3213–3225.

_ , **Chong Huang, and Liyan Yang**, "Fragility of Financial Markets," *Annual Review of Financial Economics*, 2025, *17*, 27–48.

_ , **Shijie Yang, and Luo Zuo**, "The Real Effects of Modern Information Technologies: Evidence from the EDGAR Implementation," *Journal of Accounting Research*, 2023, *61* (5), 1699–1733.

Green, T. Clifton and Shaojun Zhang, "Alternative Data in Active Asset Management," *Journal of Portfolio Management*, 2024, *51* (2), 149–161.

Griffin, John M. and Jin Xu, "How Smart Are the Smart Guys? A Unique View from Hedge Fund Stock Holdings," *Review of Financial Studies*, 2009, *22* (7), 2531–2570.

Grossman, Sanford J. and Joseph E. Stiglitz, "On the Impossibility of Informationally Efficient Markets," *American Economic Review*, 1980, *70* (3), 393–408.

Haddad, Valentin and Tyler Muir, "Market Macrostructure: Institutions and Asset Prices," *Annual Review of Financial Economics*, 2025, *17*, 133–150.

_ , **Paul Huebner, and Erik Loualiche**, "How Competitive Is the Stock Market? Theory, Evidence from Portfolios, and Implications for the Rise of Passive Investing," *American Economic Review*, 2025, *115* (3), 975–1018.

_ , **Zhiguo He, Paul Huebner, Péter Kondor, and Erik Loualiche**, "Causal Inference for Asset Pricing," 2026. NBER Working Paper 35413.

Hellwig, Martin F., "On the Aggregation of Information in Competitive Markets," *Journal of Economic Theory*, 1980, *22* (3), 477–498.

Huang, Allen H., Hui Wang, and Yi Yang, "FinBERT: A Large Language Model for Extracting Information from Financial Text," *Contemporary Accounting Research*, 2023, *40* (2), 806–841.

Huebner, Paul, "The Making of Momentum: A Demand-System Perspective," 2023. Working paper, SSRN 4395945.

Jansen, Kristy A. E., Wenhao Li, and Lukas Schmid, "Granular Treasury Demand with Arbitrageurs," 2024. NBER Working Paper 33243.

Jensen, Theis Ingerslev, Bryan Kelly, and Lasse Heje Pedersen, "Is There a Replication Crisis in Finance?," *Journal of Finance*, 2023, *78* (5), 2465–2518.

Kacperczyk, Marcin and Amit Seru, "Fund Manager Use of Public Information: New Evidence on Managerial Skills," *Journal of Finance*, 2007, *62* (2), 485–528.

_ , **Clemens Sialm, and Lu Zheng**, "On the Industry Concentration of Actively Managed Equity Mutual Funds," *Journal of Finance*, 2005, *60* (4), 1983–2011.

_ , _ , **and** _ , "Unobserved Actions of Mutual Funds," *Review of Financial Studies*, 2008, *21* (6), 2379–2416.

_ , **Jaromir Nosal, and Savitar Sundaresan**, "Market Power and Price Informativeness," *Review of Economic Studies*, 2025, *92* (3), 1955–1986.

_ , **Stijn Van Nieuwerburgh, and Laura Veldkamp**, "Time-Varying Fund Manager Skill," *Journal of Finance*, 2014, *69* (4), 1455–1484.

_ , _ , **and** _ , "A Rational Theory of Mutual Funds' Attention Allocation," *Econometrica*, 2016, *84* (2), 571–626.

Kang, Jung Koo, Lorien Stice-Lawrence, and Yu Ting Forester Wong, "The Firm Next Door: Using Satellite Images to Study Local Information Advantage," *Journal of Accounting Research*, 2021, *59* (2), 713–750.

Katona, Zsolt, Marcus O. Painter, Panos N. Patatoukas, and Jean Zeng, "On the Capital Market Consequences of Big Data: Evidence from Outer Space," *Journal of Financial and Quantitative Analysis*, 2025, *60* (2), 551–579.

Koijen, Ralph S. J. and Motohiro Yogo, "A Demand System Approach to Asset Pricing," *Journal of Political Economy*, 2019, *127* (4), 1475–1515.

_ **and** _ , "Exchange Rates and Asset Prices in a Global Demand System," *Journal of Political Economy*, 2026, *134* (8), 2441–2487.

_ , **Robert J. Richmond, and Motohiro Yogo**, "Which Investors Matter for Equity Valuations and Expected Returns?," *Review of Economic Studies*, 2024, *91* (4), 2387–2424.

Li, Feng, "The Information Content of Forward-Looking Statements in Corporate Filings—A Naïve Bayesian Machine Learning Approach," *Journal of Accounting Research*, 2010, *48* (5), 1049–1102.

Li, Yizhang, Stanislav Sokolinski, and Andrea Tamoni, "Which Investors Drive Anomaly Returns and How?," *Journal of Financial Economics*, 2026, *179*, 104257.

Loughran, Tim and Bill McDonald, "When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks," *Journal of Finance*, 2011, *66* (1), 35–65.

Sheng, Jinfei, Zheng Sun, Baozhong Yang, and Alan L. Zhang, "Generative AI and Asset Management," *Review of Financial Studies*, forthcoming.

Van Nieuwerburgh, Stijn and Laura Veldkamp, "Information Immobility and the Home Bias Puzzle," *Journal of Finance*, 2009, *64* (3), 1187–1215.

_ **and** _ , "Information Acquisition and Under-Diversification," *Review of Economic Studies*, 2010, *77* (2), 779–805.

Veldkamp, Laura, *Information Choice in Macroeconomics and Finance*, Princeton, NJ: Princeton University Press, 2011.

Wing, Coady, Seth M. Freedman, and Alex Hollingsworth, "Stacked Difference-in-Differences," 2024. NBER Working Paper 32054.

Zhu, Christina, "Big Data as a Governance Mechanism," *Review of Financial Studies*, 2019, *32* (5), 2021–2061.

Appendix

A Data Construction and the Breakthrough Registry

A.1 Holdings and Institution Types

I use the holdings, prices, and fundamentals described in Section 2.2 and stock characteristics from Jensen et al. (2023). Each 13F filer receives one fixed investor type based on its available regulatory history and the business behind its own 13F portfolio. Starting with Bushee's entity types and Thomson's codes, I use fund links and Form ADV to distinguish mutual funds and hedge funds from other advisers. Seven rules then use regulatory identifiers and the 13F filer's business to refine the classification. Asset managers include investment advisers, hedge funds, mutual funds, and public pension and sovereign funds. Banks, insurers, and broker-dealers form the comparison group. Online Appendix OA1.1 gives the source hierarchy, rules, and counts. Appendix F uses the alternative classification of Koijen and Yogo (2019) plus hedge funds.

A.2 Employment Measurement

Revelio Labs' 2024 release provides 56,785,614 U.S. positions at matched employers and companies in their corporate families for measuring employment over 2009Q1–2024Q4. Identifiers, names, and independent reviews link 12,080 13F filers to 11,033 Revelio companies. The reviews use no holdings, performance, or employee counts. A worker counts once per investor-quarter.

Job titles determine functions. DS labor includes data scientists working on statistical forecasts, machine learning, text, computer vision, and alternative-data signals, as well as all quantitative researchers. Generic analytics and financial-data analysis fall outside the definition. 13 institutions sharing an employer divide its staff in proportion to AUM. The matched employment panel includes all 13F institutions linked to Revelio in the corresponding quarter. Online Appendix OA1.5 describes the samples for employment comparisons and estimation, and Online Appendix OA1.2 gives the matching, title, and counting procedures. Online Appendix Table OA1 reports employment trends by institution type.

Breakthrough waves and stock groups. A candidate must introduce a new algorithm or model, appear in Epoch AI's notable-model database, and rank in the top 0.01 percent of its field and year by citations. Models made public within eight quarters of a wave's first model belong to that wave. The wave is dated by its highest-ranked model. The event falls

in the first calendar quarter ending at least two months after that model became public with code or trained models. Table A1 summarizes the seven events. Online Appendix OA2.5 lists all fifteen models and documents their dates.

Store-based retailers have SIC codes 5200–5999, excluding 5961, 5962, and 5963 (mail-order, vending-machine, and direct-selling businesses). The group includes restaurants. Each stock's latest nonmissing SIC in the cleaned stock panel fixes its classification across events. SIC comes from JKP, with CCM/CRSP codes filling missing values. Online Appendix OA1.4 describes the text-heavy, complex-text, and satellite-covered stock groups.

Table A1: Data-Technology Breakthrough Waves

Technology	Release	Public date	Event quarter	Window
Language processing	Word2Vec	2013-07-29	2013Q3	$[-6, 6]$
	BERT	2018-11-02	2019Q1	$[-6, 6]$
Computer vision	AlexNet	2012-10-13	2012Q4	$[-6, 6]$
	ResNet	2016-02-03	2016Q2	$[-6, 6]$
	Mask R-CNN	2018-01-22	2018Q1	$[-6, 6]$
	ViT	2020-10-22	2020Q4	$[-6, 6]$
	SAM	2023-04-05	2023Q2	$[-6, 6]$

Notes. The table lists the seven events in Section 2.4. Public date is the day the method became public with code or trained models (Online Appendix OA2.5). The event quarter is the first calendar quarter ending at least two months later. Models released within eight quarters of a wave's first model belong to that wave. Its most prominent model determines the wave's date. Windows are measured in quarters relative to the event quarter, which also begins the post period. Online Appendix Table OA3 lists all fifteen models.

A.3 Data Technology and Fundamental Forecasting

Data technology changes the cost of forecasting firms' payoffs. Investors combine alternative data with financial statements and disclosures, and DS teams develop methods to turn these inputs into investment signals (Green and Zhang, 2024).

Language models extract information from disclosures and earnings calls. Filing tone is informative about unexpected earnings (Loughran and McDonald, 2011), changes in filings predict earnings and profitability (Cohen et al., 2020), and FinBERT improves information extraction from calls (Huang et al., 2023). The NLP analysis therefore uses firms with extensive disclosures, measured as in Section 2.4.

Satellite images make customer traffic observable before retailers announce earnings. Orbital Insight processes parking-lot images into car counts (Kang et al., 2021). Parking-lot fill rates predict same-store sales and announcement returns (Katona et al., 2025). Real-time sales measures also explain earnings surprises (Froot et al., 2017). These applications motivate the store-based retailers and restaurants used in the CV analysis.

Asset managers report that their DS teams help forecast firms' earnings and profitability (Chen and Zhou, 2023; Cui and Shahrabani, 2023). These applications produce information about firm payoffs, the object of private signals in the model, and complement the extraction of other investors' information from prices studied by Farboodi and Veldkamp (2020).

Online Appendix OA2 documents the institutional applications and improvements in models and computing costs. Its breakthrough registry gives public-release dates and the grouping of fifteen models into seven events.

B Stock-Picking Ability around Breakthroughs: Specification and Estimates

Datasets. For event s, I follow asset managers with an employment observation in the four preceding quarters, using the sample in Online Appendix OA1.5. Investors who have not hired any DS staff form the comparison group. Investor-events observed in only one quarter are absorbed by the investor-by-event effects and dropped. Tables A2 and A3 report the resulting sample sizes.

Windows and overlap. Each window runs from six quarters before the event to six after and ends before the next event of the same technology and the sample endpoint in 2024Q4. The two NLP events are 22 quarters apart, with separate windows. Consecutive CV events are 7 to 14 quarters apart. Later pre-event quarters can therefore fall in an earlier event's post-event window: six quarters for Mask R-CNN, two for ViT, and three for SAM. The AlexNet and ResNet windows are separate. An investor-quarter in two windows enters both datasets with the corresponding investor-by-event effects. Clustering by investor accounts for dependence between these entries.

Pooled specification. The pooled regression replaces the quarter indicators of equation (2) with the indicator $\text{Post}_{s,t} = \mathbf{1}\{t \geq t_s\}$ for the event quarter and the six quarters after it:

$$\text{SP}_{i,s,t} = \beta \, \text{Post}_{s,t} \times \text{DS}_{i,s} + \alpha_{i,s} + \lambda_{s,t} + u_{i,s,t}. \tag{A1}$$

The coefficient β compares stock-picking ability on exposed stocks in the six pre-event quarters with the seven post-event quarters, per standard deviation of the pre-event DS labor share. The event-study coefficients compare each quarter with quarter -1. In columns (3) and (5) of Tables A2 and A3, $\text{DS}_{i,s}$ is an indicator for employing DS labor in the four pre-event quarters. Column (2) adds pre-breakthrough shares of six other types of employ-

ment, each standardized within event and interacted with $\text{Post}_{s,t}$: investment staff, data engineering, administration and operations, finance and accounting, human resources, and IT support (Online Appendix OA1.2).

Weights. Following Wing et al. (2024), DS employers receive weight one and event s's comparison investors receive $Q_s = (N_s^T/N^T)/(N_s^C/N^C)$. Here N_s^T and N_s^C count DS employers and comparison investors in event s, and N^T and N^C sum them across the technology's events. This gives every event the same ratio of comparison investors to DS employers. Weights are recomputed for each column's sample and also used in Figure 2.

Estimates. In column (1) of Tables A2 and A3, a one-standard-deviation higher pre-event DS labor share is associated with a larger post-event increase in stock-picking ability: 0.906×10^{-3} (s.e. 0.279×10^{-3}) for NLP and 0.545×10^{-3} (s.e. 0.112×10^{-3}) for CV. Adding the pre-event shares of six other types of employment in column (2) gives estimates of 0.822×10^{-3} for NLP and 0.555×10^{-3} for CV. The estimates are also positive when DS employment is measured by an indicator in column (3). Columns (4) and (5) use complex-text stocks for NLP and satellite-covered retailers for CV. The estimates for both measures of DS employment are larger in these samples.

C Proofs and Further Theory

Expectations condition on date-0 primitives and information allocations, and $\mathbb{E}_0$ makes this conditioning explicit. I use the precisions and portfolio adjustments in Section 3. Write $\eta_i(n) = \varepsilon(n) + \bar{u}_i(n)$, where $\bar{u}_i(n)$ has variance $1/\tau_i(n)$ when $\tau_i(n) > 0$. Terms involving private signals are zero when $\tau_i(n) = 0$. Let $B = [\beta, \phi]$, with row $v_n^\top = (\beta(n), \phi(n))$, and let $N_i = |\mathcal{N}_i|$. Without a mandate, use $B = \beta$, $v_n = \beta(n)$ and the scalar G_i in Proposition 1. Investor-specific matrices run over $\mathcal{N}_i$, with zeros elsewhere when aggregating portfolios.

Assumption C.1 (Primitives)**.** Each stock has finite positive aggregate risk tolerance and finite aggregate private precision. Private signal errors average out across investors: $\int \gamma_i^{-1} \tau_i(n) \bar{u}_i(n)\, di = 0$. For results as stock universes grow, information counts and exposure loadings are uniformly bounded, and payoff precisions and noise-trading variances are bounded above and away from zero. Each exposure is represented across the investor's universe: the smallest eigenvalue of $N_i^{-1} \sum_{n \in \mathcal{N}_i} v_n v_n^\top$ stays bounded away from zero.

The final condition requires broad exposure to the factor and, when present, the mandate attribute. It is used for the approximation and error bounds below. The distributional assumptions and strict concavity of portfolio choice are stated in Section 3.

Table A2: Stock-Picking Ability around Natural-Language-Processing Breakthroughs

Dep. Var.:	SP on Text-Heavy Stocks			SP on Complex-Text Stocks	
	(1)	(2)	(3)	(4)	(5)
Different Measures of DS Labor:	Share of DS Labor	Share of DS Labor (w. Other Staff)	Employs DS Labor (0/1)	Share of DS Labor	Employs DS Labor (0/1)
Post × DS Labor	0.906***	0.822**	3.113**	2.125***	9.388***
	(0.279)	(0.325)	(1.275)	(0.325)	(1.101)
Other Staff Controls		Y			
WFH Control Weights	Y	Y	Y	Y	Y
Investor × Event FE	Y	Y	Y	Y	Y
Event × Quarter FE	Y	Y	Y	Y	Y
Joint p, Other Staff		0.20			
Mean of Dep. Var.	−0.31	−0.31	−0.31	0.39	0.39
N(Observations)	19,971	19,971	19,971	27,456	27,456
N(Investors)	1,419	1,419	1,419	1,814	1,814

Notes. The table reports coefficients on Post × DS Labor in equation (A1) across the Word2Vec and BERT events. Coefficients, standard errors and dependent-variable means are multiplied by 10^3. The dependent variable is stock-picking ability on text-heavy stocks in columns (1)–(3) and complex-text stocks in columns (4)–(5). Text-heavy stocks are in the top decile of pre-event filing length. Complex-text stocks are in the top decile of the share of Loughran–McDonald complexity words in 10-K filings. Both measures are adjusted for size and industry (Online Appendix OA1.4), and investor-quarters require at least ten such stocks. Column (2) adds pre-event employment shares for investment staff, data engineering, administration and operations, finance and accounting, human resources, and IT support, each standardized within event and interacted with Post. Online Appendix Table OA2 reports all six coefficients. Other definitions follow Table A3. DS employers have weight one, and comparison investors are reweighted to give every event the same ratio of comparison investors to DS employers (Wing et al., 2024). The pre-event coefficients of the event study are jointly insignificant in columns (1)–(3) and jointly significant at the 1 percent level in columns (4) and (5). Standard errors clustered by investor are in parentheses. *, **, and *** denote 10, 5, and 1 percent significance.

Table A3: Stock-Picking Ability around Computer-Vision Breakthroughs

Dep. Var.:	SP on Store-Based Retailers			SP on Satellite-Covered Retailers	
	(1)	(2)	(3)	(4)	(5)
Different Measures of DS Labor:	Share of DS Labor	Share of DS Labor (w. Other Staff)	Employs DS Labor (0/1)	Share of DS Labor	Employs DS Labor (0/1)
Post × DS Labor	0.545***	0.555***	1.827***	1.984***	8.538***
	(0.112)	(0.125)	(0.432)	(0.272)	(0.968)
Other Staff Controls		Y			
WFH Control Weights	Y	Y	Y	Y	Y
Investor × Event FE	Y	Y	Y	Y	Y
Event × Quarter FE	Y	Y	Y	Y	Y
Joint p, Other Staff		0.17			
Mean of Dep. Var.	4.07	4.07	4.07	13.39	13.39
N(Observations)	48,889	48,889	48,889	87,335	87,335
N(Investors)	1,939	1,939	1,939	3,216	3,216

Notes. The table reports coefficients on Post × DS Labor in equation (A1) across the five computer-vision events. Coefficients, standard errors and the dependent-variable mean are multiplied by 10^3. The dependent variable, SP, is stock-picking ability, defined in equation (1), on store-based retailers in columns (1)–(3) and on the satellite-covered retailers of Katona et al. (2025) in columns (4)–(5). Investor-quarters require at least ten such stocks, or three for the satellite-covered retailers. Post marks the event quarter and the six quarters after it. The window is $[-6, 6]$. In the row label, DS Labor refers to the measure named in the column header. Share of DS Labor is pre-event DS labor per 100 employees, standardized within event. Employs DS Labor indicates DS staff in the four pre-event quarters. Column (2) adds pre-event employment shares for investment staff, data engineering, administration and operations, finance and accounting, human resources, and IT support, each standardized within event and interacted with Post. Joint p reports the p-value for a joint test of these six coefficients, and Online Appendix Table OA2 reports them individually. Comparison investors have no DS staff in the four pre-event quarters. WFH Control Weights follow Wing et al. (2024): DS employers have weight one, and comparison investors are reweighted to give every event the same ratio of comparison investors to DS employers (Appendix B). Means are unweighted. The five pre-event coefficients of the corresponding event study (quarters −6 to −2) are jointly insignificant in columns (1) and (2) and jointly significant at the 1 percent level in columns (3)–(5). Standard errors clustered by investor are in parentheses. *, **, and *** denote 10, 5, and 1 percent significance.

C.1 Beliefs, Demand, and Prices

Proof of Lemma 1. Prices reveal $\eta_p(n)$ under the conjecture. Given $\varepsilon(n)$, private and price signals have independent normal errors. Updating adds their precisions to the prior precision and weights each signal by its share of the sum, giving equation (11). Independence across stocks gives diagonal posterior covariance. The independence of F from signals and prices leaves its posterior at $N(0,1)$. □

Proof of Proposition 1. Conditional payoff covariance is $\hat{\Omega}_i+\beta\beta^\top$. Apart from initial wealth, the objective is $\gamma_i q_i^\top m_i - \gamma_i^2 q_i^\top M_i q_i/2$. Strict concavity gives the unique optimum $q_i = \gamma_i^{-1} M_i^{-1} m_i$.

To write this condition stock by stock, set $g_i^\beta = \gamma_i \beta^\top q_i$ and $g_i^\phi = -\delta_i^\phi \phi^\top q_i/\gamma_i$. The first-order condition becomes

$$q_i = \gamma_i^{-1}\hat{\Omega}_i^{-1}(m_i - \beta g_i^\beta - \phi g_i^\phi).$$

Multiplying by $\beta^\top$ and $\phi^\top$ and substituting the definitions of the two exposures gives the system G_i in the proposition. At $\delta_i^\phi = 0$, $g_i^\phi = 0$ and only the factor equation remains. □

Proof of Proposition 2. Substitute beliefs into demand and aggregate over eligible investors. Use mean exposure adjustments in market clearing. Private signal errors average out, giving

$$\mathcal{T}(n)[\mu(n) - R_f p(n)] + \kappa(n)\varepsilon(n) + z(n) + \bar{\gamma}^{-1}\tau_p(n)\eta_p(n) - \beta(n)H_\beta(n) - \phi(n)H_\phi(n) = 1.$$

For $\kappa > 0$, $\kappa\varepsilon + z = \kappa\eta_p$ and

$$\mathcal{T} d_p = \frac{[\bar{\gamma}^{-1}\tau_\varepsilon + \bar{\gamma}^{-1}\tau_p + \kappa]\kappa}{\kappa + \bar{\gamma}^{-1}\tau_p} = \bar{\gamma}^{-1}\tau_\varepsilon d_p + \kappa = \mathcal{T}_p.$$

Solving for price gives equation (13). Its positive loading on η_p verifies the conjecture and the precision $\tau_p = \kappa^2/\sigma_z^2$. At $\kappa = 0$, $\tau_p = 0$, $d_p = 1$, and price varies with noise trades alone.

The mean price and portfolio adjustments are also uniquely determined. Taking expectations of optimal demand and imposing mean market clearing gives

$$\left(\int \gamma_i^{-1} M_i^{-1}\, di\right)\left(\mu - R_f \mathbb{E}_0[p]\right) = \mathbf{1}.$$

Assume this aggregate matrix has finite entries. It is positive definite because each eligible investor's objective is strictly concave and each stock has holders with positive risk tolerance. This equation determines the mean excess payoffs, which in turn determine the mean adjustments through Proposition 1. □

Accuracy of benchmark market clearing. The benchmark replaces realized aggregate exposure adjustments by their means. To assess the resulting quantity error, write $\mathbf{g}_i = (g_i^\beta, g_i^\phi)^\top$ and $\bar{\mathbf{g}}_i = \mathbb{E}_0[\mathbf{g}_i]$, dropping the mandate component when absent. At benchmark prices, expected excess payoffs depend on independent stock-level shocks, so

$$\mathbf{g}_i - \bar{\mathbf{g}}_i = G_i^{-1} \sum_{n \in \mathcal{N}_i} \hat{\tau}_i(n) v_n \Big(m_i(n) - \mathbb{E}_0[m_i(n)] \Big),$$

$$\mathrm{Cov}_0(\mathbf{g}_i) = G_i^{-1} \left[\sum_{n \in \mathcal{N}_i} \hat{\tau}_i(n)^2 \mathrm{Var}_0(m_i(n)) v_n v_n^\top \right] G_i^{-1}.$$

Suppose $\delta_i^\phi \leq 0$ and posterior precisions and $\mathrm{Var}_0(m_i(n))$ are uniformly bounded. The exposure condition in Assumption C.1 implies $G_i^{-1} = O(N_i^{-1})$, while the sum in brackets is $O(N_i)$. Thus $\mathbb{E}_0\|\mathbf{g}_i - \bar{\mathbf{g}}_i\|^2 = O(N_i^{-1})$: broader portfolios reduce fluctuations in exposure adjustments.

At the benchmark price, the clearing error is exactly

$$\int q_i(n)\, di + z(n) - 1 = -\int \gamma_i^{-1} \hat{\tau}_i(n) v_n^\top (\mathbf{g}_i - \bar{\mathbf{g}}_i)\, di.$$

It has mean zero. Its root mean square is bounded by the integral of $\gamma_i^{-1}\hat{\tau}_i(n)\|v_n\|\{\mathbb{E}_0\|\mathbf{g}_i - \bar{\mathbf{g}}_i\|^2\}^{1/2}$. Hence the quantity error vanishes if the bounds hold uniformly across holders, their minimum portfolio breadth grows, and aggregate risk tolerance stays bounded. A positive lower bound on participating risk tolerance keeps $\mathcal{T}_p(n)$ away from zero and bounds the benchmark variances of prices and expected excess payoffs.

Corollary C.1 (The own-price slope)**.** *Under the benchmark price function, investor i's own-price slope is*

$$\frac{\partial q_i(n)}{\partial p(n)} = -R_f \gamma_i^{-1} [\tau_i(n) + \tau_\varepsilon(n) d_p(n)][1 - h_i(n)], \qquad h_i(n) = \hat{\tau}_i(n) v_n^\top G_i^{-1} v_n.$$

For $\delta_i^\phi \leq 0$, bounded posterior precisions and the exposure condition in Assumption C.1 give $h_i(n) = O(N_i^{-1})$. It is zero when the stock has no factor or mandate exposure.

Proof of Proposition 3 and Corollary C.1. Hold private signals, information allocations and other prices fixed. At $\kappa(n) > 0$, a unit rise in $R_f p(n)$ raises the inferred price signal by $\mathcal{T}_p(n)/\kappa(n)$. Using $\tau_p \mathcal{T}_p/\kappa = \tau_p + \tau_\varepsilon(1 - d_p)$ in posterior beliefs gives

$$\frac{\partial m_i(n)}{\partial p(n')} = -\mathbf{1}\{n = n'\} R_f \frac{\tau_i(n) + \tau_\varepsilon(n) d_p(n)}{\hat{\tau}_i(n)}.$$

The same formula holds at $\kappa = 0$, where prices convey no information. Differentiating demand and its exposure adjustments then yields

$$\frac{\partial q_i(n)}{\partial p(n')} = -R_f \gamma_i^{-1} \Big[\mathbf{1}\{n = n'\} - \hat{\tau}_i(n) v_n^\top G_i^{-1} v_{n'} \Big] [\tau_i(n') + \tau_\varepsilon(n') d_p(n')].$$

For comparable stocks, $\hat{\tau}_i(n')v_{n'} = \hat{\tau}_i(n)v_n$. Subtracting their responses to $p(n)$ cancels the exposure terms and gives Proposition 3. Integrating the remaining coefficients gives $\kappa + \bar{\gamma}^{-1}\tau_\varepsilon d_p = \mathcal{T}_p$. The diagonal gives the own-price slope, and $G_i^{-1} = O(N_i^{-1})$ gives its exposure bound. Appendix F develops the empirical comparison. □

C.2 Information Gains and Investor Demand

Corollary C.2 (Crowding out over a finite increase)**.** *Let $\beta(n) = \phi(n) = 0$ and let κ rise to $\kappa' > \kappa$, fixing σ_z, τ_ε, risk tolerances and investor i's private precision. Suppressing the stock index, her expected holding falls if and only if*

$$\kappa\kappa' < \sigma_z^2(\tau_\varepsilon + \tau_i) + \tau_i \bar{\gamma}^{-1}(\kappa + \kappa').$$

For $\kappa' = (1+\Delta_\kappa)\kappa$ with $\kappa > 0$, the condition $\tau^{\mathrm{R}} < 1/(2+\Delta_\kappa)$ is sufficient for every investor whose private precision is unchanged.

Proof of Propositions 4 and 5 and Corollary C.2. Mean market clearing gives $\mathbb{E}[m_i] = 1/\mathcal{T}$ and $\mathbb{E}[q_i] = \gamma_i^{-1}\hat{\tau}_i/\mathcal{T}$. At fixed τ_i,

$$\frac{\partial \mathbb{E}[q_i]}{\partial \kappa} = \frac{\gamma_i^{-1}}{\mathcal{T}^2} \left[\frac{2\kappa}{\sigma_z^2}(\kappa - \bar{\gamma}^{-1}\tau_i) - (\tilde{\tau} + \tau_i) \right].$$

The bracket decreases in τ_i and equals $\tau_p - \tau_\varepsilon$ at zero. Thus $\tau^{\mathrm{R}} < 1/2$ makes it negative for every investor whose private precision is unchanged.

For a finite change, the holdings difference has the sign of

$$(\tilde{\tau}' + \tau_i)\mathcal{T} - (\tilde{\tau} + \tau_i)\mathcal{T}' = \frac{\kappa' - \kappa}{\sigma_z^2} \Big[\kappa\kappa' - \sigma_z^2(\tau_\varepsilon + \tau_i) - \tau_i \bar{\gamma}^{-1}(\kappa + \kappa') \Big].$$

At $\tau_i = 0$, this expression is negative exactly when $\kappa\kappa' < \sigma_z^2\tau_\varepsilon$. Substitution of $\kappa' = (1+\Delta_\kappa)\kappa$ gives $\tau^{\mathrm{R}} < 1/(2+\Delta_\kappa)$. The bracket decreases further with τ_i. Finally, $\mathbb{E}[q_i']/\mathbb{E}[q_i] = [(\tilde{\tau}' + \tau_i)/(\tilde{\tau} + \tau_i)]\mathcal{T}/\mathcal{T}'$ decreases in τ_i when $\tilde{\tau}' > \tilde{\tau}$. Better-informed investors whose private information is unchanged therefore experience larger proportional losses. □

Proposition C.1 (Information rents and expected demand)**.** *In the benchmark economy without mandates, define the stock-selection position by removing the common factor premium from expected excess payoff:*

$$q_i^{\text{sel}} = \gamma_i^{-1}\hat{\tau}_i\left(m_i - \frac{\beta H_\beta}{\mathcal{T}}\right), \qquad q_i = q_i^{\text{sel}} + \gamma_i^{-1}\hat{\tau}_i\beta\left(\frac{H_\beta}{\mathcal{T}} - g_i^\beta\right).$$

The stock index is suppressed. Let supply be $S > 0$, and fix eligible holders and their risk tolerances. Writing $\bar{q}_i = \mathbb{E}[q_i^{\text{sel}}]$ and $\Pi_i = \mathbb{E}[q_i^{\text{sel}}\tilde{\varepsilon}]$ gives

$$\Pi_i = \gamma_i^{-1}\frac{\tau_i}{\tilde{\tau}}, \qquad \frac{\bar{q}_i}{S} = \frac{\gamma_i^{-1} + \Pi_i}{\bar{\gamma}^{-1} + \int \Pi_j\, dj} = \frac{\gamma_i^{-1}\hat{\tau}_i}{\mathcal{T}}. \tag{A2}$$

Between two information allocations,

$$\frac{\bar{q}_i'}{\bar{q}_i} = \frac{\hat{\tau}_i'}{\hat{\tau}_i}\frac{\mathcal{T}}{\mathcal{T}'}. \tag{A3}$$

Holding private precisions fixed, greater public precision reduces the rent from a given private signal and changes expected stock-selection demand by

$$\left.\frac{\partial \bar{q}_i}{\partial \tilde{\tau}}\right|_{\{\tau_j\}} = \frac{\gamma_i^{-1}S(\kappa - \bar{\gamma}^{-1}\tau_i)}{\mathcal{T}^2}. \tag{A4}$$

It raises demand below the risk-tolerance-weighted mean private precision $\kappa/\bar{\gamma}^{-1}$ and lowers demand above it.

Learning from prices reduces the advantage of private information and improves risk bearing most for relatively uninformed investors. Appendix G uses the stock-selection response to calculate changes in holdings while keeping mean factor contributions fixed.

Proof. Mean clearing gives $\mathbb{E}[m_i] = (S + \beta H_\beta)/\mathcal{T}$, so $\bar{q}_i = \gamma_i^{-1}\hat{\tau}_i S/\mathcal{T}$. The part of q_i^{sel} that covaries with residual payoff news is $\gamma_i^{-1}\tau_i\tilde{\varepsilon}$. Since $\text{Var}(\tilde{\varepsilon}) = 1/\tilde{\tau}$, its covariance is $\Pi_i = \gamma_i^{-1}\tau_i/\tilde{\tau}$. Divide the expected-holdings numerator and denominator by $\tilde{\tau}$ to obtain equation (A2). Take the ratio across allocations to obtain equation (A3). Differentiating, using $\mathcal{T} - \bar{\gamma}^{-1}\hat{\tau}_i = \kappa - \bar{\gamma}^{-1}\tau_i$, proves equation (A4). For $\tau_i > 0$, $\partial\Pi_i/\partial\tilde{\tau} = -\gamma_i^{-1}\tau_i/\tilde{\tau}^2 < 0$. □

Corollary C.3 (Aggregate elasticity after a finite increase)**.** *Under the conditions of Proposition 6, let $\kappa' = (1 + \Delta_\kappa)\kappa$ with $\Delta_\kappa > 0$. Aggregate demand becomes less elastic at a fixed positive price and quantity if and only if*

$$\kappa < \frac{\bar{\gamma}^{-1}\tau_\varepsilon d_p(1 - d_p)}{1 + \Delta_\kappa(1 - d_p)}.$$

The threshold approaches the local condition as $\Delta_\kappa \to 0$.

Proof of Proposition 6 and Corollary C.3. From $d_p = (1 + \bar{\gamma}^{-1}\kappa/\sigma_z^2)^{-1}$,

$$\frac{\partial \mathcal{T}_p}{\partial \kappa} = 1 - \frac{(\bar{\gamma}^{-1})^2 \tau_\varepsilon}{\sigma_z^2} d_p^2 = 1 - \frac{\bar{\gamma}^{-1}\tau_\varepsilon d_p(1-d_p)}{\kappa}.$$

The second expression applies at $\kappa > 0$ and gives the local condition. For a finite increase,

$$d_p' = \frac{d_p}{1 + \Delta_\kappa(1-d_p)}, \qquad \mathcal{T}_p' - \mathcal{T}_p = \Delta_\kappa \left[\kappa - \frac{\bar{\gamma}^{-1}\tau_\varepsilon d_p(1-d_p)}{1+\Delta_\kappa(1-d_p)}\right].$$

The first term is the increase in demand sensitivity from private information, and the second is the reduction from stronger learning from prices. Since the slope is $-R_f\mathcal{T}_p$, the sign proves both elasticity results. □

Corollary C.4 (The increase in price informativeness)**.** *Let* $\kappa > 0$ *rise to* $(1+\Delta_\kappa)\kappa$*, keeping* σ_z *and* τ_ε *fixed. Then*

$$\tau^{\mathrm{R}\prime} = \frac{(1+\Delta_\kappa)^2 \tau^{\mathrm{R}}}{1 - \tau^{\mathrm{R}} + (1+\Delta_\kappa)^2 \tau^{\mathrm{R}}}.$$

For a given $\Delta_\kappa > 0$*, the increase is largest at initial price informativeness* $\tau^{\mathrm{R}} = 1/(2+\Delta_\kappa)$*. Its maximum is* $\Delta_\kappa/(2+\Delta_\kappa)$*. For a small change,* $d\tau^{\mathrm{R}} = 2\tau^{\mathrm{R}}(1-\tau^{\mathrm{R}})\, d\ln\kappa$.

Proof. Substitute $\tau_p' = (1+\Delta_\kappa)^2\tau_p$ into $\tau^{\mathrm{R}\prime} = \tau_p'/(\tau_\varepsilon + \tau_p')$. The derivative of the increase with respect to initial τ^{R} has the sign of $1 - 2\tau^{\mathrm{R}} - [(1+\Delta_\kappa)^2 - 1](\tau^{\mathrm{R}})^2$. Its unique root in $(0,1)$ is $1/(2+\Delta_\kappa)$, and substitution gives the maximum. Differentiating $\tau^{\mathrm{R}} = \kappa^2/(\kappa^2 + \sigma_z^2\tau_\varepsilon)$ gives the local formula. □

C.3 Multiple Factors

Proposition C.2 (Multiple factors)**.** *Let* $F \sim N(0, I)$ *have a fixed number of factors, with vector loadings* $\beta(n)$*, and suppose investors have no mandates. Under the benchmark price function, beliefs about factors stay at their prior. With* $G_i = I + \beta^\top \hat{\Omega}_i^{-1}\beta$*, demand is*

$$q_i(n) = \gamma_i^{-1}\hat{\tau}_i(n)[m_i(n) - \beta(n)^\top g_i^\beta], \qquad g_i^\beta = G_i^{-1}\beta^\top\hat{\Omega}_i^{-1}m_i.$$

The factor term in prices becomes $\beta(n)^\top H_\beta(n)$*, with* $H_\beta(n) = \int \gamma_i^{-1}\hat{\tau}_i(n)\mathbb{E}_0[g_i^\beta]\, di$*. Aggregate private information* $\kappa(n)$ *and price precision* $\tau_p(n)$ *keep their definitions. The information identity and feasible correction in Appendix D apply with* $v_n = \beta(n)$ *and*

$$h_i(n) = \hat{\tau}_i(n)\beta(n)^\top G_i^{-1}\beta(n) \in [0,1).$$

Bounded posterior precisions and the exposure condition in Assumption C.1, applied with $v_n = \beta(n)$*, give* $h_i(n) = O(N_i^{-1})$*. Comparable stocks in Proposition 3 satisfy* $\hat{\tau}_i(n')\beta(n') = \hat{\tau}_i(n)\beta(n)$*. The crowding-out and aggregate-elasticity results apply where* $\beta(n) = 0$*.*

Proof. The first-order condition in the proof of Proposition 1 applies with vector factor exposures. Multiplication by $\beta^\top$ gives G_i and the displayed demand. Benchmark clearing aggregates private-signal coefficients into the same κ. Replacing factor loadings by vectors changes the mean exposure term. For the information identity, $\mathbb{E}[g_i^\beta \tilde{\varepsilon}(n)] = G_i^{-1}\beta(n)\tau_i(n)/\tilde{\tau}(n)$, which gives the stated $h_i(n)$. Positive definite covariance gives $0 \leq h_i(n) < 1$, and broad factor exposure gives its order. Replacing posterior by public precisions yields Corollary D.2's correction. The relative-demand proof uses the same cancellation with vector loadings. □

D Measuring Information: Estimation and Properties

D.1 The Information Identity and Exposure Correction

The information measure uses the covariance of holdings with payoff news left unrevealed by public information. This covariance separates private information from trades based on public beliefs. Portfolio exposure adjustments change the response to private news and give the correction below. Notation follows Appendix C.

Corollary D.1 (Zero information and public controls)**.** *In the benchmark economy,* $k_i(n) = 0$ *implies* $\mathbb{E}[q_i(n)\tilde{\varepsilon}(n)] = 0$*, including factor and mandate adjustments. Removing any public-information component from holdings leaves their population covariance with* $\tilde{\varepsilon}(n)$ *unchanged. In any equilibrium, an investor with* $K_i = 0$ *has a zero moment when payoff news is residualized on the full public information set, including the price vector.*

Proof of Proposition 7 and Corollary D.1. Write $\tilde{m}(n) = \mu(n) + \mathbb{E}[\varepsilon(n) \mid p] - R_f p(n)$ for the expected excess payoff based on public information. Posterior beliefs give

$$\hat{\tau}_i(n)m_i(n) = \hat{\tau}_i(n)\tilde{m}(n) + \tau_i(n)\tilde{\varepsilon}(n) + \tau_i(n)\bar{u}_i(n).$$

Residual payoff news has variance $1/\tilde{\tau}(n)$ and is independent of public information, individual signal errors and other stocks' residual news. Thus only stock n contributes to the covariance of the exposure adjustment with its residual news:

$$\mathbb{E}\left[\begin{pmatrix} g_i^\beta \\ g_i^\phi \end{pmatrix} \tilde{\varepsilon}(n)\right] = G_i^{-1} v_n \frac{\tau_i(n)}{\tilde{\tau}(n)}.$$

Substituting into demand yields

$$\begin{aligned}\mathbb{E}[q_i(n)\tilde{\varepsilon}(n)] &= \gamma_i^{-1}\frac{\tau_i(n)}{\tilde{\tau}(n)}\left[1-\hat{\tau}_i(n)v_n^\top G_i^{-1}v_n\right]\\ &= \gamma_i^{-1}k_i(n)[1-\tau^{\mathrm{R}}(n)][1-h_i(n)].\end{aligned}$$

Since $1-h_i(n) = [M_i^{-1}]_{nn}/\hat{\tau}_i(n) > 0$, division by the two adjustment factors and summation prove Proposition 7. For $\delta_i^\phi \le 0$, G_i is positive definite and $0 \le h_i(n) < 1$.

The same formula gives zero when $k_i(n) = 0$. The conditional mean $\mathbb{E}[\tilde{\varepsilon}(n) \mid p] = 0$ also gives zero covariance with every public variable by iterated expectations. This proves invariance to public controls and, when all holdings are based on public information, the zero moment in any equilibrium. □

Corollary D.2 (A feasible exposure correction)**.** *Suppose $\delta_i^\phi \le 0$. Omitting the exposure correction gives*

$$[1-\max_n h_i(n)]\gamma_i^{-1}K_i \le \sum_n \frac{\mathbb{E}[q_i(n)\tilde{\varepsilon}(n)]}{1-\tau^{\mathrm{R}}(n)} \le \gamma_i^{-1}K_i.$$

A correction using public precision is

$$\hat{h}(n) = \tilde{\tau}(n)v_n^\top\left[\sum_{n'\in\mathcal{N}_i}\tilde{\tau}(n')v_{n'}v_{n'}^\top\right]^{-1}v_n.$$

Under Assumption C.1, bounded public precisions and fixed exposure preferences, using $\hat{h}(n)$ in place of $h_i(n)$ gives relative error $O(\max_n k_i(n)/N_i + N_i^{-2})$ for $K_i > 0$.

Proof. The bounds follow from summing $\gamma_i^{-1}k_i(n)[1-h_i(n)]$. For the approximation, denote the matrix inside the inverse by $\hat{G}$. Then

$$h_i(n) - \hat{h}(n) = \tau_i(n)v_n^\top G_i^{-1}v_n + \tilde{\tau}(n)v_n^\top(G_i^{-1}-\hat{G}^{-1})v_n.$$

Broad exposure across stocks bounds both inverses by $O(N_i^{-1})$. The first term is $O(k_i(n)/N_i)$. For the second, use $G_i^{-1}-\hat{G}^{-1} = -G_i^{-1}(G_i-\hat{G})\hat{G}^{-1}$. The difference $G_i-\hat{G}$ consists of the fixed exposure-preference matrix and $\sum_{n'}\tau_i(n')v_{n'}v_{n'}^\top$. Their contributions are $O(N_i^{-2})$ and $O(\sum_{n'}k_i(n')/N_i^2)$. Since $\hat{h}(n) = O(N_i^{-1})$, dividing each moment by $1-\hat{h}(n)$ and summing gives the stated relative error. □

Without a mandate, the feasible correction is simply $\tilde{\tau}(n)\beta(n)^2/\sum_{n'}\tilde{\tau}(n')\beta(n')^2$. Proposition C.2 extends it to multiple factor loadings, and Online Appendix Proposition OA3.1 covers a hard exposure target.

D.2 Estimation Specification

Earnings forecasts. I forecast the change in operating earnings from the same fiscal quarter a year earlier, per dollar of book equity at the holdings date. Earnings are announced in the following quarter. I use the latest fiscal period when several reports map to one holdings quarter and winsorize the change at that quarter's 1st and 99th percentiles.

The baseline forecast uses stock characteristics, analyst-consensus revisions for the same fiscal period, log one plus analyst coverage, and forecast dispersion. Revisions compare quarterly forecast snapshots. The augmented forecast adds price, price–characteristic interactions, and twelve-industry indicators. Its error is the residual earnings surprise. I assume the forecast improvement comes from price and its characteristic interactions.

Variances and weights. Models of squared forecast errors give prior precision $\hat{\tau}_\varepsilon(n)$ and public precision $\hat{\tau}_\varepsilon(n) + \hat{\tau}_p(n)$. The baseline variance model is pooled, and the augmented model is fitted within quarter. I use a common conversion from earnings innovations to payoff innovations across stocks and quarters. It multiplies information estimates by the same constant and cancels from the variance ratio.

Within each size-quintile and market-to-book-tercile bin, I match the median correction $1/[1-\hat{\tau}^{\mathrm{R}}(n)]$ to the degrees-of-freedom-adjusted ratio of baseline to augmented mean squared errors. Fitted variances supply within-bin variation. I bound corrections between one and 100 and public precision between one hundredth and one hundred times its quarterly median. The lower correction bound assigns zero price informativeness when the fitted variance does not fall.

Exposure and position controls. I compute $\hat{h}(n)$ using public precision and market betas estimated from 60 months of returns, and cap it at 0.5. Position controls include every earnings-forecast regressor, together with price and characteristic terms scaled by precision and the correction for price informativeness. Write P for the least-squares projection on these controls over $\mathcal{N}_{it}$. The estimator uses residual positions $(I - P)Q_{it}$ and divides each stock's contribution by $\max\{1 - P_{nn}, 0.05\}$ to correct for its weight in the fitted position.

An observed mandate attribute enters the exposure correction alongside market beta. If the attribute is a linear combination of the characteristics, its public-precision-weighted term also enters the position controls. These controls remove the part of forecast errors explained by the included variables. A common rescaling of both variance models cancels from the correction for price informativeness, while a common rescaling of public precision leaves $\hat{h}(n)$ unchanged. Group statistics use quarterly time-series standard errors because institutions share earnings surprises and fitted forecasts.

Persistence and size. Online Appendix Table OA4 uses twelve-quarter formation averages and follow-up windows separated by a four-quarter gap (Carhart, 1997). Within quarter and group, I residualize standardized information on the log market value and log stock count of positions entering the estimate, then sort on and follow the residual. I also sort investors on trailing size and breadth and track their standardized information.

D.3 Statistical Properties

The sampling results use full positions, known forecast functions and precisions, and non-binding estimation bounds. I normalize the common conversion from earnings to payoffs to one. Information is allocated before signals over $\mathcal{N}_{it}$. Assume $\delta_i^{\phi} \leq 0$, the large-universe conditions in Assumption C.1, bounded posterior precisions and bounded corrections for price informativeness, and $\max_n P_{nn} = O(N_{it}^{-1})$, where $N_{it} = |\mathcal{N}_{it}|$. All moments condition on public information and the allocation. Time subscripts on model objects are suppressed within the investor-quarter. These conditions keep any one stock from dominating the exposure or position adjustment.

Sampling variation has three sources: realized payoff news, private signal noise, and public demand left in residual positions. The proof first calculates their leading contributions and then bounds the dependence introduced by fitting portfolio controls.

Proof of Propositions 8 and 9. Book-equity positions times earnings surprises per dollar of book equity equal ownership shares times dollar earnings surprises. In these units, the controls and precision corrections are fixed, and $(\tilde{\varepsilon}(n), \bar{u}_i(n))$ are independent across stocks.

Let $a = (a_{nn'})$ describe how direct private-information trades enter residual positions:

$$a = (I - P)\Big[I - \operatorname{diag}(\hat{\tau}_i) B G_i^{-1} B^{\top}\Big].$$

The exposure matrix has entries $O(N_{it}^{-1})$. The projection identities $|P_{nn'}| \leq \sqrt{P_{nn} P_{n'n'}}$ and $\sum_{n'} P_{nn'}^2 = P_{nn}$ imply $a_{nn} = 1 + O(N_{it}^{-1})$, $a_{nn'} = O(N_{it}^{-1})$ for $n' \neq n$, and $\hat{h}(n) = O(N_{it}^{-1})$.

Write the adjusted public-demand component and the leading contribution from stock n as

$$e_{it}^{\text{mis}}(n) = \frac{\gamma_{it}^{-1}[a \operatorname{diag}(\hat{\tau}_i)\tilde{m}]_n}{[1 - \hat{h}(n)][1 - P_{nn}]},$$
$$Y_{it}(n) = \frac{\{\gamma_{it}^{-1}\tau_{it}(n)[\tilde{\varepsilon}(n) + \bar{u}_i(n)] + e_{it}^{\text{mis}}(n)\}\tilde{\varepsilon}(n)}{1 - \tau^{\text{R}}(n)}.$$

Here $\tilde{m}$ is the public expected excess payoff defined above. Conditional on public information, the $Y_{it}(n)$ are independent with mean $\gamma_{it}^{-1} k_{it}(n)$. The exact own-stock multiplier differs from

one by $O(N_{it}^{-1})$, and cross-stock private terms have mean zero. Therefore $\mathbb{E}[\hat{\theta}_{it}] - \theta_{it} = O(\theta_{it}/N_{it})$.

Normal second and fourth moments give

$$V_{it}(n) = \mathrm{Var}[Y_{it}(n)] = 2\gamma_{it}^{-2}k_{it}(n)^2 + \frac{\gamma_{it}^{-2}k_{it}(n)}{1-\tau^{\mathrm{R}}(n)} + \frac{e_{it}^{\mathrm{mis}}(n)^2}{\tau_\varepsilon(n)[1-\tau^{\mathrm{R}}(n)]}.$$

The first two terms are payoff-news and signal-noise variation. Define $V_{it} = \sum_n V_{it}(n)$ and let $\sigma^2_{\mathrm{mis},it}$ be the sum of the last term. Dividing by $\theta_{it}^2 = \gamma_{it}^{-2}K_{it}^2$ gives the concentration, signal-noise and residual-public-demand terms in Proposition 8.

To justify this leading variance, write

$$\hat{\theta}_{it} - \mathbb{E}[\hat{\theta}_{it}] = \sum_n \{Y_{it}(n) - \mathbb{E}[Y_{it}(n)]\} + \mathcal{R}_{it}.$$

Cross-stock products have coefficients of order $\gamma_{it}^{-1}\tau_{it}(n')/N_{it}$. Gaussian independence and bounded counts and precisions give $\mathbb{E}[\mathcal{R}_{it}^2] = O(\gamma_{it}^{-2}K_{it}/N_{it})$. The cross-stock products are uncorrelated with the leading terms. The own-stock approximation changes private-component variance by a relative $O(N_{it}^{-1})$. Since $V_{it} \geq \gamma_{it}^{-2}K_{it}$,

$$\mathrm{Var}(\hat{\theta}_{it}) = V_{it}[1 + O(N_{it}^{-1})], \qquad \frac{\mathcal{R}_{it}}{\sqrt{V_{it}}} \xrightarrow{p} 0.$$

At $K_{it} = 0$, the remainder and bias are zero. Otherwise, bounded counts give $K_{it} = O(N_{it})$, so

$$\frac{|\mathbb{E}[\hat{\theta}_{it}] - \theta_{it}|}{\sqrt{V_{it}}} = O\left(\frac{\sqrt{K_{it}}}{N_{it}}\right) \to 0.$$

Each centered $Y_{it}(n)$ is a linear or quadratic function of normal variables, with third absolute moment bounded by a constant times $V_{it}(n)^{3/2}$. Thus the ratio of summed third absolute moments to $V_{it}^{3/2}$ is bounded by a constant times $[\max_n V_{it}(n)/V_{it}]^{1/2}$, which tends to zero under Proposition 9. The negligible remainder and bias give its stated limit.

If $\mathrm{HHI}_{it} \to 0$, $K_{it} \to \infty$, and $\sigma^2_{\mathrm{mis},it}/\theta^2_{it} \to 0$, the relative variance tends to zero. Together with the bias bound, this gives $\hat{\theta}_{it}/\theta_{it} \xrightarrow{p} 1$. □

Empirical normalization. I sum the squared contributions of the twelve Fama–French industries, $\hat{V}_{it} = \sum_b \hat{\theta}^2_{it,b}$, where $\hat{\theta}_{it,b}$ sums stock contributions within industry b. Standardized information is $\hat{\theta}_{it}/\sqrt{\hat{V}_{it}}$. This scale accounts for common fluctuations within industries and also reflects the concentration of information across them. Online Appendix OA4.1 gives its relation to sampling variance and the bound $|\hat{\theta}_{it}|/\sqrt{\hat{V}_{it}} \leq \sqrt{12}$.

Lemma D.1 (Scale invariance)**.** *Fix forecasts, controls, the lagged-holdings set and industries. Multiplying every position by a positive constant scales* $\hat{\theta}_{it}$ *and* $\sqrt{\hat{V}_{it}}$ *by that constant, leaving standardized information unchanged. At fixed prices and information, an investor without a mandate has the same scaling when her risk tolerance changes. Under Assumption F.1, scaling reference AUM at fixed relative risk aversion also leaves information per dollar of AUM unchanged.*

Proof. Residualization and equation (22) are linear in positions, while $\hat{V}_{it}$ is quadratic. Without mandates, M_i and m_i are independent of γ_i, so $q_i = \gamma_i^{-1} M_i^{-1} m_i$ scales with risk tolerance. The AUM result follows from Assumption F.1. □

Lemma D.2 (Long positions)**.** *Let* $q_i^+(n) = \max\{q_i(n), 0\}$ *be the reported long position. Then*

$$\mathbb{E}[q_i^+(n)\tilde{\varepsilon}(n)] = \Pr(q_i(n) > 0)\,\mathbb{E}[q_i(n)\tilde{\varepsilon}(n)].$$

For a universe fixed before signals, censoring reduces each expected information contribution by its holding probability. The expected estimate from long positions is a lower bound on θ_{it} *up to the exposure and position-projection approximations.*

Proof. Joint normality implies $\mathrm{Cov}(\max\{X, 0\}, Y) = \Pr(X > 0)\mathrm{Cov}(X, Y)$. Apply this identity with $X = q_i(n)$ and $Y = \tilde{\varepsilon}(n)$, whose mean is zero. In residualized positions, the own-stock coefficient is $1 - P_{nn}$, which the estimator divides out. Cross-stock terms have the order of the exposure adjustment. The leading contribution is therefore the full-position information contribution times its holding probability. □

E Information Production: Identification

The production regression measures research output through long positions. With known forecast functions and precisions, Lemma D.2 gives, up to the exposure approximation,

$$\mathbb{E}[\hat{\theta}_{it}] = \gamma_{it}^{-1} K_{it}^+, \qquad K_{it}^+ = \sum_{n \in \mathcal{N}_{it}} \Pr(q_{it}(n) > 0)\, k_{it}(n).$$

Assumption E.1 (Stable investor scale)**.** Risk tolerance and the share of research output captured by measured long positions are constant within each investor over the production sample.

In equation (23), investor fixed effects ω_i absorb persistent efficiency, risk tolerance, the share of research output captured by long positions and the units used to measure payoffs. The error u_{it} combines sampling error, productivity shocks and unmeasured input changes.

Identification uses changes in employment within investors after accounting for other staffing and common annual changes. I assume these changes are uncorrelated with u_{it} in the estimation sample and that its mean is stable across years. Investors observed in multiple years connect the annual productivity estimates. I set 2011 productivity to one and estimate all coefficients jointly by least squares with investor fixed effects.

Sample and timing. The production panel spans 2011Q1–2024Q4 and uses workforce histories beginning in 2009Q1. Active status uses full-sample mean quarterly active share. The cutoff of 0.553 is fixed before applying the production screens. Employment comes from the preceding calendar quarter. Each observation needs observed positive DS employment in at least four of the preceding eight calendar quarters. I apply the positive-outcome and positive-input screens in Section 5.1 and require at least two eligible observations per investor. The outcome is raw quarterly information.

The DS labor elasticity is 0.143 (standard error 0.081). The coefficients on log employment of investment analysts and portfolio managers and log employment outside that group and DS are 0.252 (0.156) and 0.230 (0.198). The 2024-to-2011 productivity ratio is 3.28, with a 90 percent confidence interval of $[2.29, 4.69]$.

F Demand Estimation: Details

Positions and flows. In book-equity units, $Q_{is}(n) = \mathrm{BE}_s(n)q_{is}(n)$, where $q_{is}(n)$ is investor i's fraction of outstanding shares. Price is market-to-book, $\mathrm{MB}_s(n) = p_s(n)/\mathrm{BE}_s(n)$ (Koijen et al., 2024). Payoff and signal precisions below use payoffs per dollar of book equity. AUM scales positions and information by portfolio value.

Assumption F.1 (Reference tolerance in a quarterly comparison)**.** For the comparison of quarters $t-1$ and t, hold reference risk tolerance fixed and write

$$\gamma_i^{-1} = \frac{\mathrm{AUM}_{i,t-1}}{\gamma_i^{\mathrm{R}}}.$$

The local relative-risk-aversion scale γ_i^{R} may vary across comparisons as assets change.

For $s = t-1, t$, the level schedules are

$$\frac{Q_{is}(n)}{\mathrm{AUM}_{i,t-1}} = -\frac{R_f}{\gamma_i^{\mathrm{R}}}[\tau_{is}(n) + \tau_{\varepsilon,s}(n)d_{p,s}(n)]\mathrm{MB}_s(n) + C_{is}(n) + u_{is}(n),$$
$$u_{is}(n) = \frac{\tau_{is}(n)}{\gamma_i^{\mathrm{R}}}\eta_{is}(n).$$

Here C_{is} collects expected-payoff and exposure-management terms, while u_{is} is demand from the private signal. Empirical residual demand can also include stock-specific tastes.

I value both quarters' shares at lagged book equity per split-adjusted share. With $\rho_t(n)$ the ratio of this value in $t-1$ to its value in t,

$$\Delta Q_{it}(n) = \frac{\rho_t(n)Q_{it}(n) - Q_{i,t-1}(n)}{\text{AUM}_{i,t-1}}, \qquad \Delta\text{MB}_t(n) = \text{MB}_{t-1}(n)r_t(n),$$

where r_t is the ex-dividend return. The earnings innovation accrues to book equity, so the price change excludes dividends. A nontrading holder has zero flow under issuance, splits and dividends.

Separately, estimation holds the relative demand slope fixed in lagged book-equity units:

$$\rho_t(n)^2[\tau_{it}(n) + \tau_{\varepsilon,t}(n)d_{p,t}(n)] = \tau_{i,t-1}(n) + \tau_{\varepsilon,t-1}(n)d_{p,t-1}(n).$$

The two powers of ρ_t convert quantity and price. Expected-payoff and residual-demand terms require only the quantity conversion.

First difference. Suppose both quarter-end positions lie on the level schedules above. Under Assumption F.1 and the fixed-slope restriction,

$$\begin{aligned}\Delta Q_{it}(n) = &-\frac{R_f}{\gamma_i^{\text{R}}}[\tau_{i,t-1}(n) + \tau_{\varepsilon,t-1}(n)d_{p,t-1}(n)]\Delta\text{MB}_t(n)\\ &+ \rho_t(n)C_{it}(n) - C_{i,t-1}(n) + \Delta u_{it}(n),\\ \Delta u_{it}(n) = &\,\rho_t(n)u_{it}(n) - u_{i,t-1}(n).\end{aligned}$$

Substitute the level schedules into ΔQ_{it} and use $\text{MB}_t = \rho_t(\text{MB}_{t-1} + \Delta\text{MB}_t)$. The fixed-slope restriction cancels the terms in MB_{t-1}, leaving the stated price change and differences in C and u.

Substitution conditional on information. In the share units of Section 3, let $v_n = (\beta(n), \phi(n))^\top$ and let G_i be the matrix in Proposition 1. Without a mandate, keep only the factor component. The proof of Proposition 3 in Appendix C gives

$$\begin{aligned}\frac{\partial q_i(n)}{\partial p(m)} = -R_f\gamma_i^{-1}&\Big[\mathbf{1}\{n=m\} - \hat{\tau}_i(n)v_n^\top G_i^{-1}v_m\Big]\\ &\times\Big[\tau_i(m) + \tau_\varepsilon(m)d_p(m)\Big].\end{aligned} \tag{A5}$$

The exposure adjustment depends on the stock through its precision-weighted loadings, $\hat{\tau}_i(n)v_n$ (Haddad et al., 2026). Matching two distinct stocks on these loadings gives

$$\frac{\partial[q_i(n) - q_i(n')]}{\partial p(n)} = -R_f \gamma_i^{-1}[\tau_i(n) + \tau_\varepsilon(n) d_p(n)],$$

the relative slope in equation (16). Since $\hat{\tau}_i = \tau_\varepsilon + \tau_p + \tau_i$, matching depends on information from the prior, prices and private signals as well as exposures.

At fixed book equity and AUM, conversion to the demand units gives

$$\frac{\partial[Q_i(n)/\text{AUM}_i]}{\partial \text{MB}(m)} = \frac{\text{BE}(n)\text{BE}(m)}{\text{AUM}_i} \frac{\partial q_i(n)}{\partial p(m)}.$$

The row loadings are proportional to $\text{BE}(n)\hat{\tau}_i(n)v_n$. The factor controls therefore interact exposures with public and measured private information, with coefficients free by institution-quarter.

Controls and information inputs. I control for lagged characteristics and their ρ-adjusted changes, changes in public earnings forecasts adjusted for market-to-book and industry, and changes in relative price signal precision. Factor-loading interactions with information allow portfolio-exposure adjustments to vary with posterior precision. These controls have institution-quarter-specific coefficients. Online Appendix OA4.2 lists the terms.

For each slope-index term except private information, additional controls multiply its variable by payoff precision and expected payoff divided by R_f, separately at each date. Expected payoff is measured by the lagged twelve-quarter average of operating profitability. The current control also includes ρ_t. These controls have separate coefficients common across institutions and quarters and serve as their own instruments.

The prediction $\bar{k}_{is}$ targets $\hat{\theta}_{is}/\text{AUM}_{is}$, the quarterly information estimate scaled by AUM, winsorized within quarter at the 5th and 95th percentiles. Weighted least squares explains its deviation from the quarterly mean using seven type indicators, lagged log AUM, a historical score and a missing-score indicator. At date s, the score divides the mean target through $s-4$, using at least four observations, by its Newey–West standard error with four lags.

Weights equal the number of demand observations. I divide institutions into five groups and fit each group's prediction coefficients using the other four. I add the fitted deviation to the average of weighted quarterly means and impose a positive floor. These coefficients and means use the full panel. Demand in t uses the $t-1$ prediction, whose score stops at $t-5$. The gap also excludes the $t-4$ estimate that shares an earnings figure with t through seasonal differencing. Under uniform allocation, $\bar{k}_{i,t-1}(n) = \bar{k}_{i,t-1}/N_{i,t-1}$. I winsorize this

prediction within quarter at the 1st and 99th percentiles across investor–stock observations, then standardize across those observations.

Measuring relative price signal precision. Forecasting gives $\hat{\tau}_p = \hat{\tau}_\varepsilon \hat{\tau}^{\mathrm{R}}/(1-\hat{\tau}^{\mathrm{R}})$. Measured aggregate information $\hat{\kappa}$ is payoff precision times the sum of holders' terms in equation (22). For informative stock-quarters with positive $\hat{\kappa}$, the identity $\kappa = \sqrt{\tau_p}\sigma_z$ gives a noise-trade scale. I regress its log on log turnover and log dollar volume per dollar of book equity, with quarter effects, and use fitted $\hat{\sigma}_z$ throughout. The ratio $\sqrt{\hat{\tau}_p}/\hat{\sigma}_z$ equals aggregate information divided by noise-trading variance and is zero for uninformative prices.

For informative prices, ℓ is the log ratio minus its quarterly median across investor–stock observations. The between-quarter term is that median minus the pooled median. Every observation has equal weight. The reference slope sets both terms to zero.

The GIV. Within each stock-quarter, I project flows on type-by-trading-activity indicators, the holder variables governing price and expected-payoff terms, and predicted payoff-news loadings at both dates. Trading activity groups investors into terciles of their median residual-flow variance across quarters. Projection weights are inverse residual variances estimated leaving the current quarter out. Holder variables include log age and predicted information per stock.

Under uniform allocation, the payoff-news loading for a position dated s uses predicted information in its original units and is proportional to

$$\bar{k}_{j,s-1}(n)\tau_{\varepsilon,s-1}(n), \qquad s = t-1, t.$$

The current loading also receives the quantity conversion ρ_t. Including both loading vectors removes common payoff news from the differenced flow.

I exclude investor i from the projection and sum other holders' residuals with lagged-AUM weights, as in equation (26). I use observations for which the projection can be estimated after excluding that investor. Decompose private-signal demand as

$$u_{is}(n) = \frac{\tau_{is}(n)}{\gamma_i^{\mathrm{R}}}\varepsilon_s(n) + u_{is}^{\perp}(n), \qquad \Delta u_{it}^{\perp}(n) = \rho_t(n)u_{it}^{\perp}(n) - u_{i,t-1}^{\perp}(n),$$

where $u^{\perp}$ contains signal noise and tastes. Conditional on the projection design and weights, exclusion requires

$$\mathbb{E}\Big[\Delta u_{it}^{\perp}(n) \mid \Delta u_{jt}^{\perp}(n), \varepsilon_t(n), \varepsilon_{t-1}(n), \psi_{it}(n), X_{it}(n)\Big] = 0, \qquad j \neq i, \tag{A6}$$

where ψ_{it} stacks the slope-index variables.

Proposition F.1 (The GIV)**.** *Conditional on the projection design and weights, suppose Assumption F.1 and equation (A6) hold. Each holder-specific component has zero conditional mean given both payoff innovations. The true payoff-news loadings at both dates are linear combinations of the projection variables. Remaining characteristic-based demand from other holders is uncorrelated with investor i's residual demand. The resulting linear moments satisfy*

$$\mathbb{E}[Z_{it}(n)\Delta u_{it}(n) \mid \psi_{it}(n), X_{it}(n)] = 0.$$

It is relevant if other holders' lagged-AUM-weighted price covariances exceed the component removed by projection.

Proof. Let P_{-i} be the projection matrix for other investors, a_{-i} be their lagged AUM vector and r_{-i} contain their holder-specific shocks and remaining characteristic-based demand. The modeled price and expected-payoff responses and both payoff-news loadings are linear combinations of the projection variables for every coefficient value. Hence $Z_i = a_{-i}^\top(I - P_{-i})r_{-i}$. The conditional restrictions make this combination uncorrelated with Δu_i. Its price covariance is

$$\mathrm{Cov}(Z_i, \Delta\mathrm{MB}) = a_{-i}^\top \mathrm{Cov}(r_{-i}, \Delta\mathrm{MB}) - a_{-i}^\top P_{-i}\mathrm{Cov}(r_{-i}, \Delta\mathrm{MB}),$$

conditional on the design and weights. The relevance condition makes the difference positive. □

Holding net issuance and outside demand fixed, market clearing gives

$$\Delta\mathrm{MB}_t(n) \approx \frac{\sum_j \mathrm{AUM}_{j,t-1}\Delta u_{jt}(n)}{R_f \mathcal{T}_{p,t}(n)} + \text{other terms}, \tag{A7}$$

where $-R_f\mathcal{T}_{p,t}$ is the aggregate institutional slope and the other terms collect controls and fixed supply. Other investors hold private signals fixed and infer payoff news from price movements whose source is uncertain (Haddad et al., 2026, Appendix B.6.1). Appendix H gives the signal-extraction conditions and the restrictions under which effective tolerances incorporate investors' own price impact.

Estimation. I remove institution-quarter and common controls from the interacted GIVs and price regressors, giving each observation equal weight. The moment conditions in equation (27) require the resulting instruments to be uncorrelated with residual demand. Identification also requires the instruments to predict distinct variation in the price interactions. I

estimate the slope coefficients by linear GMM. Inverse-variance weights enter the construction of the GIV.

Standard errors. I use a covariance approximation based on equation (A.7) of Chaudhary et al. (2026). It assumes independent investor–stock–quarter shocks and accounts for each investor's shock entering other investors' instruments. Shock variances are estimated by investor-quarter, leaving the current quarter out. The calculation holds the information inputs fixed.

Elasticities and profiles. Position elasticity is fitted slope times market-to-book divided by book-equity holdings per dollar of AUM. I average with book-equity weights within investor-quarter and report medians across investor-quarters. Figure 5 evaluates equation (25) at the 10th, 50th and 90th percentiles of predicted information among informed investors. Type is investment adviser. Age, stock inputs and normalized holdings are fixed at their medians. The between-quarter term is zero, and price and quantity stay fixed along each profile. Online Appendix OA4.2 gives the construction of these profiles.

The discount implied by the estimates. Let $\bar{d}$ denote the discount at the median informative stock. With aggregate risk tolerance common across stocks in a quarter, equation (14) gives $d_p(\ell) = 1/[1 + e^{\ell}(1 - \bar{d})/\bar{d}]$. Let $\bar{\zeta}$ be the AUM-weighted uninformed slope index at $\ell = 0$.

Lemma F.1 (The discount implied by relative price signal precision)**.** *If ζ_ℓ equals the derivative at $\ell = 0$ of this average slope index, then $\bar{d} = 1 + \zeta_\ell/\bar{\zeta}$.*

Proof. The average uninformed slope index is $\bar{\zeta}\, d_p(\ell)/\bar{d}$. Its derivative at zero is $-\bar{\zeta}(1 - \bar{d})$ because $d_p'(\ell) = -d_p(\ell)[1 - d_p(\ell)]$. Equating this derivative to ζ_ℓ gives the result. □

The uninformed average sets private information to zero. The confidence set for $\bar{d}$ inverts a test using the full coefficient covariance in Table 3.

G The Counterfactual: Inputs and Solution

The experiment starts from $t = 2024Q4$ and uses information inputs dated $t - 1$. Table A4 lists the inputs.

Table A4: Inputs of the Counterfactual

Input	Value	Source
Panel (a) Estimated		
Demand Slopes	By type, age and information	Table 3
Private-Information Component	0.16	Appendix G
Price-Inference Discount	0.39	Section 6
Annual DS Productivity	2011–2024, normalized in 2011	Figure 4
Panel (b) Measured		
Predicted Information	Institution level, 2024Q3	Section 6
Payoff Precision	Stock level, 2024Q3	Section 4
Price Informativeness	Stock level, 2024Q4	Section 4
Institutional Holdings	13F and CRSP, 2024Q4	Section 2
Adopters	172 institutions, 10.7% of holdings	Section 7
Panel (c) Fixed Inputs and Assumptions		
Proportional Information Gain	$1 - \widehat{A}_{2011}/\widehat{A}_{2024}$	Section 7
Size of the Technology Advance	Equation (28)	Section 7
Placement of New Information	Proportional to current information	Section 7
Non-13F Investors	No information, price-inelastic	Appendix G
Variance of Noise Trades	Solved from price informativeness	Appendix G

Notes. Inputs are for the 2024Q4 experiment in Section 7.1. Price-Inference Discount is the reference estimate. The information and risk-tolerance scales match its implied mean discount and the sum of fitted stock slopes (Online Appendix OA4.3). Private-Information Component is $\kappa/\mathcal{T}_p$, aggregate private precision weighted by risk tolerance divided by the aggregate demand-slope coefficient with learning from prices. The technology advance repeats the estimated 2011–2024 increase in annual productivity, raising adopters' private information by 69.5 percent at fixed employment and risk tolerance. Information and payoff-precision inputs use 2024Q3, and current payoff precision determines posterior uncertainty. Appendix G gives the allocation and treatment of missing lags. Prices with zero measured informativeness stay uninformative. Demand-coefficient draws hold measured information, production inputs and proportional information gains fixed.

Information and its allocation. Adopters are asset managers with DS employees at $t-1$ and full-sample mean active share at or above the production cutoff of 0.553. Let $\theta_{i,t}$ be total predicted information based on the lagged portfolio. I form the per-stock input $\theta_{i,t}/N_{i,t-1}$ by multiplying lagged AUM by $\bar{k}_{i,t-1}(n)$ after the within-quarter winsorization used in demand estimation and before standardization.

With common scale $c > 0$, private precision satisfies

$$\gamma_i^{-1}\tau_{i,t}(n) = c\frac{\theta_{i,t}}{N_{i,t-1}}\tau_{\varepsilon,t-1}(n).$$

Information is equal across positions in units of lagged prior precision. Current precision $\tau_{\varepsilon,t}(n)$ determines posterior uncertainty. It also replaces a missing or nonpositive lagged precision in the assignment. Equation (28) multiplies each adopter's private precision by $1+\delta(1-\widehat{A}_{2011}/\widehat{A}_{2024})$. Employment, investor efficiency, risk tolerance and priors stay fixed, as does other institutions' private information. Below I suppress t and use a prime for values after the technology advance.

Demand and market clearing. Positions $Q_i(n)$ and supply are in book-equity dollars. Payoffs and prices are per dollar of book equity, as in Appendix F. Each stock's institutional supply $S(n)$ equals its initial aggregate holdings. Outside holders have zero private information and fixed demand, and institutions trade competitively with zero mandate penalties. Proposition C.1 gives the mean stock-selection position

$$\bar{Q}_i(n) = \mathbb{E}[Q_i^{\text{sel}}(n)] = \frac{\gamma_i^{-1}\hat{\tau}_i(n)S(n)}{\mathcal{T}(n)}.$$

I hold each investor's mean factor-management position $\gamma_i^{-1}\hat{\tau}_i(n)\beta(n)[H_\beta(n)/\mathcal{T}(n)-\bar{g}_i^\beta]$ and the common factor premium $\beta(n)H_\beta(n)/\mathcal{T}(n)$ at their initial values. The counterfactual solves stock-selection demand conditional on these fixed factor components. Changes in full expected holdings then equal changes in stock-selection holdings, $\Delta\mathbb{E}[Q_i] = \Delta\bar{Q}_i$. This restriction limits cross-stock factor rebalancing, consistent with the weak substitution estimated by Chaudhry and Davis (2026).

Risk tolerance is proportional to lagged AUM times fitted type-and-age tolerance. I use current AUM when its lag is unusable and impose a small positive floor on fitted tolerance. I choose the two common scales jointly to match the sum of fitted stock slopes and the holder-weighted mean discount implied by Lemma F.1. Online Appendix OA4.3 gives these targets. Effective tolerances stay fixed under Assumption H.1.

Beliefs. For stocks with informative prices, aggregate assigned information $\kappa(n)$ and price precision imply noise variance

$$\sigma_z^2(n) = \frac{\kappa(n)^2}{\tau_p(n)}.$$

I hold this variance fixed. For stocks whose measured price precision is zero, I take the limit $\sigma_z^2 \to \infty$. This gives $\tau_p = \tau_p' = 0$ and $d_p = d_p' = 1$. For informative stocks, Proposition 2 gives $\tau_p'(n) = [1+\Delta_\kappa(n)]^2\tau_p(n)$, where $\Delta_\kappa(n) = \kappa'(n)/\kappa(n) - 1$. Updated posterior precisions determine $\mathcal{T}'$, discounts and demand slopes.

Ownership decomposition. To isolate the effect of private information, first update private precision while keeping information from prices fixed. The intermediate stock-selection position is

$$Q_i^{\mathrm{D}}(n) = \frac{\gamma_i^{-1} S(n)[\tau_\varepsilon(n) + \tau_p(n) + \tau_i'(n)]}{\bar{\gamma}^{-1}[\tau_\varepsilon(n) + \tau_p(n)] + \kappa'(n)}. \tag{A8}$$

Updating price information as well gives the final position $\bar{Q}_i' = \gamma_i^{-1}\hat{\tau}_i' S/\mathcal{T}'$. At fixed private precision, learning from prices raises model holdings for investors below the risk-tolerance-weighted average private precision and lowers them for investors above it (Proposition C.1).

For observed book-equity positions $Q_i^{\mathrm{obs}}(n)$ and $S(n) = \sum_j Q_j^{\mathrm{obs}}(n) > 0$, map any model allocation x to

$$\mathcal{H}_n(x)_i = S(n)\frac{[Q_i^{\mathrm{obs}}(n) + x_i - \bar{Q}_i(n)]_+}{\sum_j [Q_j^{\mathrm{obs}}(n) + x_j - \bar{Q}_j(n)]_+}, \qquad [y]_+ = \max\{y, 0\}. \tag{A9}$$

Every model allocation x sums to $S(n)$, so the denominator is at least $S(n)$. The rule adds model changes to observed positions, imposes long-only holdings and keeps each stock's institutional supply fixed. At the baseline, $\mathcal{H}_n(\bar{Q}) = Q^{\mathrm{obs}}$.

Let $Q' = \mathcal{H}(\bar{Q}')$ be the reported positions. Their change decomposes as

$$Q_i' - Q_i^{\mathrm{obs}} = \underbrace{\mathcal{H}(Q^{\mathrm{D}})_i - Q_i^{\mathrm{obs}}}_{\text{private information}} + \underbrace{Q_i' - \mathcal{H}(Q^{\mathrm{D}})_i}_{\text{learning from prices}}. \tag{A10}$$

Table 7 sums each contribution over a group's investors and stocks and divides by its initial positions, multiplied by 100. Online Appendix Table OA5 reverses the order and averages the two orders for $\delta = 1, 2, 3$. All contributions include the long-only adjustment and rescaling.

[Insert Online Appendix Table OA5 here]

Prices and elasticities. Holding the factor premium fixed, the stock-selection component of expected excess payoff falls by the fraction $1 - \mathcal{T}/\mathcal{T}'$. I calibrate each stock's price response

using its estimated demand slope:

$$\frac{\Delta \mathbb{E}[p]}{p} = \left(1 - \frac{\mathcal{T}}{\mathcal{T}'}\right) \frac{d_p S}{R_f \widehat{\mathcal{T}}_p p}. \tag{A11}$$

Here p is initial market-to-book. The slope input $\widehat{\mathcal{T}}_p$ combines the fitted type-and-age index at zero private information with the fitted private-information component. Learning from prices enters through the calibrated d_p. The model supplies the proportional change $\mathcal{T}/\mathcal{T}'$ and the fitted baseline slope sets the price scale. The estimated elasticity magnitude is $R_f \widehat{\mathcal{T}}_p p/S$. Dividing by d_p gives its magnitude at fixed payoff forecasts. Posterior precision scales that magnitude by $\mathcal{T}'/\mathcal{T}$ after the anticipated technology advance. Reported elasticity changes allow learning from prices and equal the proportional changes in $\mathcal{T}_p$, evaluated at initial prices and holdings.

Coverage, information shares and crowding out. For adopters $\mathcal{A}$ and initial holders $\mathcal{I}(n)$, let $\kappa_{\mathcal{A}}(n) = \sum_{i \in \mathcal{A} \cap \mathcal{I}(n)} \gamma_i^{-1} \tau_i(n)$ be adopters' private information, weighted by risk tolerance. Total information is $\mathcal{T}(n) = \sum_{i \in \mathcal{I}(n)} \gamma_i^{-1} \hat{\tau}_i(n)$, the weighted sum of holders' posterior precisions. It includes prior information, private information and information learned from prices. The sorting variables are

$$\text{Coverage}(n) = \frac{\kappa_{\mathcal{A}}(n)}{\mathcal{T}(n)}, \qquad \text{Information share}(n) = \frac{\kappa_{\mathcal{A}}(n)}{\kappa(n)}.$$

Adopters' coverage is the proportion of total information that comes from adopters' private information. Information share is the proportion of all private information contributed by adopters. Table 6 and Figure 7(a) use approximately equal-count coverage quintiles. Panel (b) takes medians of non-adopters' percentage changes in total positions within quintiles of winsorized $\bar{k}_{i,t-1}(n)$. Each institution enters once, and both position totals cover its initially held stocks.

Elasticity contributions. In Table 7(b), the direct change in magnitude is $100(\kappa' - \kappa)/\mathcal{T}_p$ and the total is $100(\mathcal{T}_p'/\mathcal{T}_p - 1)$. The direct component holds d_p at its initial value. The panel takes separate medians across informative stocks and defines the contribution of learning from prices as their difference. Figure 8 averages each stock's direct change, its change due to learning from prices, and its total change within fixed equal-count bins of initial information share or price informativeness. Contributions add within each stock and bin.

Confidence intervals. I draw 200 joint demand-coefficient vectors using Appendix F's covariance and recalibrate both scales within each draw. Measured information, production estimates, the information-gain rule, recipients and information allocation stay fixed. Stock and investor bins stay fixed at the point estimate across draws and sizes of the technology advance. I calculate each statistic within each draw, then take its 5th and 95th percentiles. This includes the share of stocks with declining elasticity magnitude and the total-effect bands in Figure 8.

H Finite Investors and Price Impact

Price impact reduces the positions investors take on their information. With finitely many investors, prices also reveal part of each investor's own signal.

H.1 Portfolio Choice with Price Impact

Consider one stock without factor or mandate exposure. Let p be its public reference price and $1/\hat{\tau}_i$ investor i's posterior payoff variance. A marginal execution price $p + \lambda_i^K x$ at quantity x, with $\lambda_i^K \geq 0$, adds cost $\lambda_i^K q_i^2/2$ at the trading date. Beliefs and the execution-cost coefficient are given when she chooses her position.

Proposition H.1 (Price impact and effective risk tolerance)**.** *With expected excess payoff $m_i = \mathbb{E}[f \mid \mathcal{I}_i] - R_f p$, optimal demand is*

$$q_i^\lambda = \frac{m_i}{\gamma_i/\hat{\tau}_i + R_f \lambda_i^K} = \left(\gamma_i^{\text{eff}}\right)^{-1} \hat{\tau}_i m_i, \qquad \gamma_i^{\text{eff}} = \gamma_i + R_f \lambda_i^K \hat{\tau}_i. \tag{A12}$$

Proof. The objective, divided by risk aversion, is $m_i q_i - (\gamma_i/\hat{\tau}_i + R_f\lambda_i^K) q_i^2/2$. It is strictly concave, and its first-order condition gives the result. □

For a portfolio, write $\Sigma_i = \hat{\Omega}_i + \beta\beta^\top$ for posterior payoff covariance and $q_i^\top \Lambda_i^K q_i/2$ for execution cost. The first-order condition is

$$\left(\gamma_i \Sigma_i + R_f \Lambda_i^K - \frac{\delta_i^\phi}{\gamma_i} \phi\phi^\top\right) q_i = m_i.$$

Costs proportional to posterior covariance allow the same substitution:

$$R_f \Lambda_i^K = (\gamma_i^{\text{eff}} - \gamma_i)\Sigma_i, \qquad \delta_i^{\phi,\text{eff}} = \delta_i^\phi \frac{\gamma_i^{\text{eff}}}{\gamma_i}. \tag{A13}$$

With these costs and the adjusted mandate coefficient, demand has the form in Proposition 1. Covariance-proportional costs can arise from compensation for liquidity suppliers' inventory risk (Gârleanu and Pedersen, 2013, Assumption 1). The restriction here uses the investor's posterior covariance.

H.2 Learning from Prices with Finite Investors

In the same one-stock setting, let J investors have signals $\eta_j = \varepsilon + \bar{u}_j$, with independent normal errors of variance $1/\tau_j$. These errors are independent of the payoff innovation and noise trades. Suppose the public price reveals

$$y = \sum_{j=1}^{J} a_j^p \eta_j + z, \qquad \tau_p = \frac{(\sum_j a_j^p)^2}{\sigma_z^2 + \sum_j (a_j^p)^2/\tau_j}. \tag{A14}$$

The price coefficients are deterministic and $\sigma_z^2 > 0$. Investors without a signal have $a_j^p = 0$ and contribute zero to both sums.

For investor i with a private signal, subtracting $a_i^p \eta_i$ from y removes her own contribution to price. The remaining statistic has noise independent of her signal error, conditional on ε. Its precision and her posterior precision are

$$\tau_{p,-i} = \frac{(\sum_{j\neq i} a_j^p)^2}{\sigma_z^2 + \sum_{j\neq i} (a_j^p)^2/\tau_j}, \qquad \hat{\tau}_i = \tau_\varepsilon + \tau_i + \tau_{p,-i}. \tag{A15}$$

Public precision is $\tilde{\tau} = \tau_\varepsilon + \tau_p$. The difference between private and public posterior precision measures the information the investor adds beyond the price.

Lemma H.1 (Information beyond the price)**.** *Given the affine price function and the portfolio choice in Proposition H.1,*

$$\tau_i^{\text{inc}} = \hat{\tau}_i - \tilde{\tau} \geq 0, \qquad \mathbb{E}\Big[q_i^\lambda \Big(\varepsilon - \mathbb{E}[\varepsilon \mid p]\Big)\Big] = \Big(\gamma_i^{\text{eff}}\Big)^{-1} \frac{\tau_i^{\text{inc}}}{\tilde{\tau}}. \tag{A16}$$

Proof. The public component of demand has zero covariance with $\varepsilon - \mathbb{E}[\varepsilon \mid p]$. The covariance of the investor's additional forecast with this news is

$$\begin{aligned} &\mathbb{E}\Big[\Big(\mathbb{E}[\varepsilon \mid \eta_i, p] - \mathbb{E}[\varepsilon \mid p]\Big)\Big(\varepsilon - \mathbb{E}[\varepsilon \mid p]\Big)\Big] \\ &\quad = \mathbb{E}\Big[\Big(\mathbb{E}[\varepsilon \mid \eta_i, p] - \mathbb{E}[\varepsilon \mid p]\Big)^2\Big] = \frac{1}{\tilde{\tau}} - \frac{1}{\hat{\tau}_i} \geq 0. \end{aligned}$$

The first equality follows from iterated expectations, and the second is the reduction in conditional variance under normal updating. Multiplication by the demand coefficient $(\gamma_i^{\text{eff}})^{-1}\hat{\tau}_i$

gives the result. □

The moment is zero when the private signal adds no information beyond price. For an investor with a signal, $\tau_i^{\text{inc}} = \tau_i + \tau_{p,-i} - \tau_p$. The continuum identity follows when aggregate signal noise becomes negligible: $\sum_j (a_j^p)^2/\tau_j = o(\sigma_z^2)$, $a_i^p \to 0$, and $\sum_j a_j^p \to \kappa > 0$, with noise variance bounded away from zero. Then $\tau_{p,-i} - \tau_p \to 0$. Relative recovery also requires this difference to be $o(\tau_i)$. It follows, for example, when τ_i stays bounded away from zero. Matching the benchmark price function requires the aggregate coefficient to converge to $\kappa = \int (\gamma_j^{\text{eff}})^{-1} \tau_j \, dj$.

H.3 Estimation and Counterfactuals

Assumption H.1 (Stable effective risk tolerance)**.** At the counterfactual date, preference and effective risk aversion are fixed as information changes. In the scalar problem, execution costs satisfy

$$R_f \lambda_i^K(n) \hat{\tau}_i(n) = \gamma_i^{\text{eff}} - \gamma_i \geq 0 \tag{A17}$$

across stocks and counterfactual states. In the full portfolio problem, costs satisfy equation (A13). Appendix G applies the scalar condition to stock-selection positions.

Under benchmark inference or the large-market limits above, effective tolerance replaces preference tolerance. Production requires a stable investor scale (Assumption E.1). Demand fixes reference tolerance within each quarterly comparison (Assumption F.1).